FOREWORD

This is the second iteration of preparing our own courseware material, after successful completion of a similar task undertaken a few years ago. These contents have been carefully prepared and should serve as excellent auxiliary material for both instructors and students.

The Special Academic Group, Autonomy (SAGA) was formed for the sole purpose of preparing courseware contents primarily in the first-year theory subjects; a few second-year subjects were also included. The subjects for the first year were - Basic Electrical Engineering, Basic Electronics Engineering, Computer Programming, Data Structures and Algorithms, Engineering Mathematics I and II, Engineering Physics, Engineering Chemistry, Constitution of India, Environmental Science and Engineering and Communicative and Technical English. For the second year, the subjects for which courseware material was prepared were Analog Electronic Circuits, Digital Systems Design, Circuit Theory and Measurements and Instruments.
Faculty members from all the departments contributed to the task. They were, in no particular order, Nalini Singh, Bimal Meher, Saumyaranjan Dash, Mukti Routray, Susmita Biswal, Manasa Dash, Bipin Tripathy, Sibasankar Nayak, Janmejay Senapati, Subrat Sahu, Pradeep Moharana, Rupambika Pattanaik, Dhananjay Tripathy, Jagadish Patra, Sachin Das, Deepak Ranjan Nayak, Amulya Roul, Bodhisattva Dash, Sanghamitra Das, Gyana Ranjan Biswal, Nibedita Swain and Rajan Mishra.

The entire group worked diligently to successfully complete the task which included a peer review of the material. I take this opportunity to thank all the members of the SAGA group for a job well done.

I sincerely hope that this courseware material comes in handy and is utilized to the fullest extent. These are readily available additional resources prepared in accordance with the Silicon autonomy syllabus, to complement textbooks and classroom lectures. If there are any errors, I would be grateful if they are brought to my notice so that we can correct them in subsequent versions.

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Module – 1: WAVES

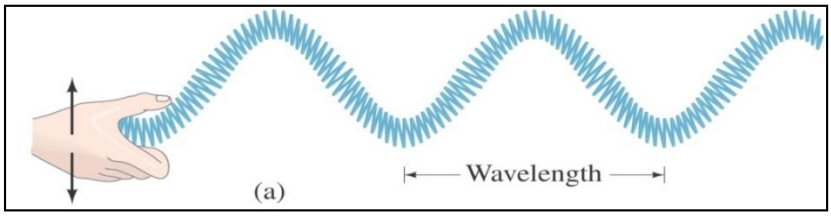
1.0 Wave: *a **wave** can be described as a disturbance that travels through a medium, transporting energy from one location to another location without transporting matter. As a disturbance moves through a medium from one particle to its adjacent particle, energy is being transported from one end of the medium to the other. The general name of this disturbance or wave is called progressive wave.*

Types of waves:

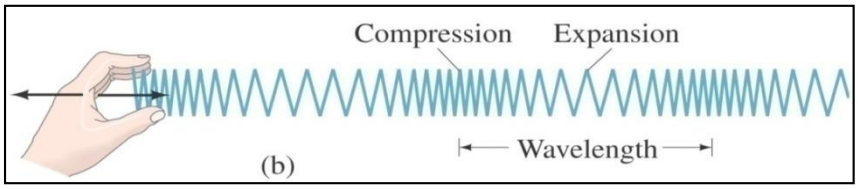
Waves are of two types

- Transverse wave
- Longitudinal wave

If the disturbance takes place perpendicular to the direction of propagation of wave, the wave is called transverse wave.



If the disturbance takes place along the direction of propagation of wave, then it is called longitudinal wave.



Wave as a periodic variation in space and time

- A wave is a disturbance that travels through a medium from one location to another.
- A wave is the motion of a disturbance that moves through a medium in such a manner that at any position, the displacement of the particle of the medium is a function of time, and at any instant, the displacement of the particles of the medium is a function of the position at that point.
- The disturbance that is propagated is called a progressive or a travelling wave.
- During its propagation, energy is transferred through the medium

- At a given instant, the wave periodically varies with position and at a given position, wave varies periodically with time.

Example:

- In sound wave, the disturbance is a pressure variation in a medium.
- In light wave, the disturbance is the variation of the strength of electric and magnetic fields.

1.1 Wave in one dimension:

$$\psi(x,t) = A\sin(kx \pm wt) = A\sin[k(x \pm vt)]$$

If we consider the initial phase factor Ø

$$\psi(x,t) = A\, sin(\, kx \pm wt + Ø)$$

Wave Characteristics:

- *Amplitude (A)* – Maximum displacement of particle of the medium from its equilibrium point. The bigger the amplitude, the more energy the wave carries.
- *Wavelength (λ)* – Distance between two consecutive crests (max positive displacement) or between two consecutive trough (max negative displacement).
- *Wave vector or propagation vector (k)* – 2π / λ
- *Period (T)* –The time required for one complete oscillation. It is the reciprocal of frequency: T = 1 / f.
- *Frequency (f)* – The number of oscillations/vibrations per second. The SI unit for frequency is the Hertz (Hz).
- *Wave speed (v)* – How fast the wave is moving (the disturbance). Speed depends on the medium. It is expressed as v = λ f.

1.2 Wave equation in differential form:

Let's take the equation, $\psi(x,t) = A\, sin(\, kx - wt + \phi)$

$$\frac{\partial\psi}{\partial t} = -wAcos(kx - wt + \phi)$$

$$\frac{\partial^2\psi}{\partial t^2} = -w^2Asin(kx - wt + \phi) = -w^2\psi$$

Again,

$$\frac{\partial\psi}{\partial x} = kAcos(kx - wt + \phi)$$

$$\frac{\partial^2\psi}{\partial x^2} = -k^2Asin(kx - wt + \phi) == -k^2\psi$$

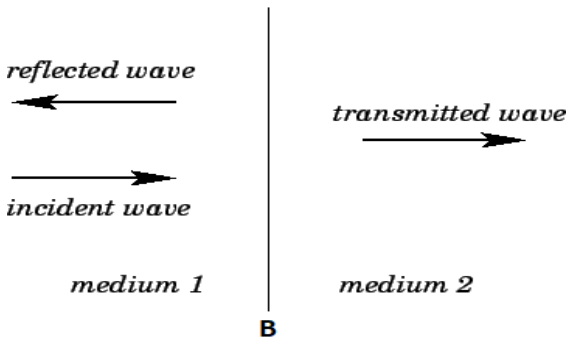
$$= \frac{k^2}{w^2}\frac{\partial^2\psi}{\partial t^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

So, $$\frac{\partial^2\psi}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

It is the general wave equation, in second order differential form with **wave velocity/ phase velocity** $v = \frac{w}{k}$**.**

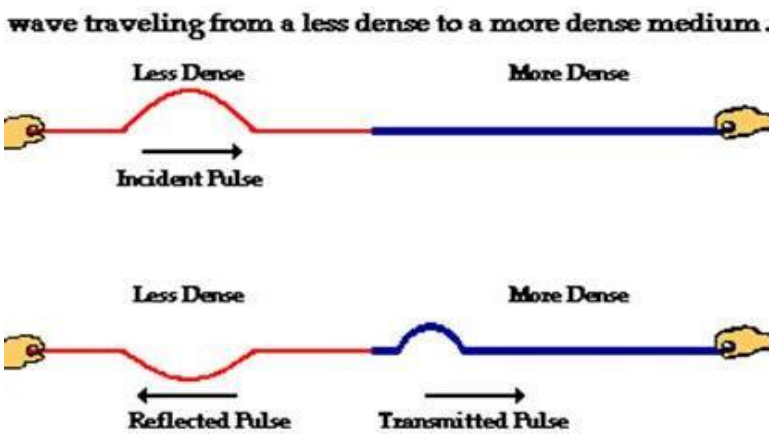
1.3 Reflection and transmission of a transverse wave at boundary of two media

When a wave strikes an obstacle, or comes to the end of the medium in which it is traveling, part of it is reflected and a part is transmitted



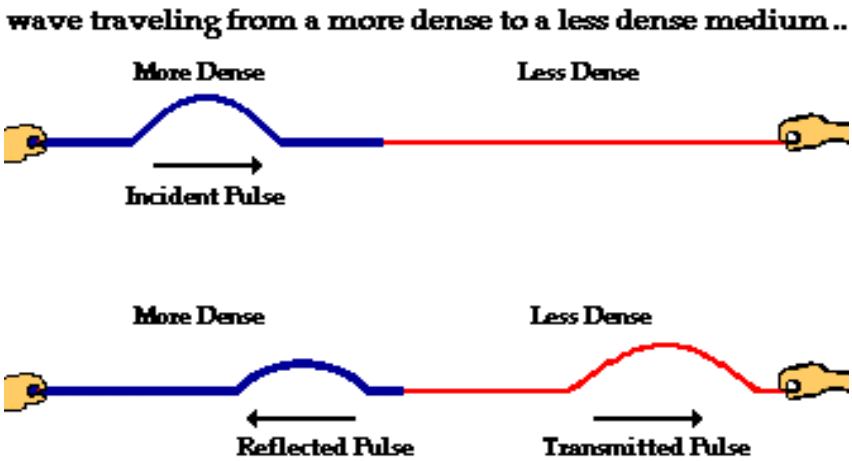
- ✓ Incident wave – the wave that strikes the boundary
- ✓ Reflected wave – the wave that is returned as a result of reflection
- ✓ Transmitted wave – the wave that is propagated to the second medium

1.3.1 Rarer to Denser medium:



Parameters	Reflected Component	Transmitted Component
Amplitude	Decreases	Decreases
Velocity	No change	Decreases
Frequency	No Change	No change
Wavelength	No change	Decreases
Phase	Changes by π	No change

1.3.2 Denser to Rarer medium



Parameters	Reflected Component	Transmitted Component
Amplitude	Decreases	Decreases
Velocity	No change	Increases
Frequency	No Change	No change
Wavelength	No change	Increases
Phase	No change	No change

Superposition

2.1 Principle of Superposition:

The principle of superposition states that the resultant displacement of a particle of a medium acted upon by two or more waves simultaneously is the algebraic sum of the displacement of the same particle due to individual waves.

Let the displacement of the particle due to individual waves are ψ_1 and ψ_2

Then the resultant displacement is

$\boldsymbol{\Psi} = \boldsymbol{\Psi_1} + \boldsymbol{\Psi_2}$ (If both are in same direction)

and $\boldsymbol{\Psi} = \boldsymbol{\Psi_1} - \boldsymbol{\Psi_2}$ (If both are in opposite direction)

2.2 Types of Superposition:

- Coherent Superposition
- Incoherent Superposition

Coherent Superposition

- Phase difference remains constant.
- The resultant intensity differs from sum of the individual intensity.
- $I \neq I_1 + I_2$.
- It gives an interference (fringe) pattern.

Incoherent Superposition

- Phase difference changes frequently and randomly.
- The intensity of the resultant wave is equal to the sum of the intensities of the component waves.
- $I = I_1 + I_2$.
- Cannot form interference pattern because their intensity is constant.

2.3 Two beam Superposition:

Let's take two beams or two waves having different amplitude, same frequency and different phases superpose

$$\psi_1 = A_1 \sin(kx - wt + \phi_1)$$

$$\psi_2 = A_2 \sin(kx - wt + \phi_2)$$

The resultant wave is $\psi = \psi_1 + \psi_2$

$$\psi = A_1 \sin(kx - wt + \phi_1) + A_2 \sin(kx - wt + \phi_2)$$

$$= A_1\,[\sin(kx - wt)cos\phi_1 + \cos(kx - wt)sin\phi_1]$$
$$+ A_2\,[\sin(kx - wt)cos\phi_2 + \cos(kx - wt)sin\phi_2]$$
$$= \sin(kx - wt)[A_1cos\phi_1 + A_2cos\phi_2] + \cos(kx - wt)\left[A_1sin\phi_1 + A_2sin\phi_2\right]$$

Let $\quad Acos\phi = [A_1cos\phi_1 + A_2cos\phi_2]$

$$Asin\phi = \left[A_1sin\phi_1 + A_2sin\phi_2\right]$$

Then $\quad \psi = \sin(kx - wt)Acos\phi + \cos(kx - wt)Asin\phi$

$$= A\sin(kx - wt + \phi)$$

So the resultant wave is of same nature as that of the incident wave.

Where A is the resultant amplitude and ϕ is the resultant phase.

So $\quad A^2 = (Asin\phi)^2 + (Acos\phi)^2$

$$= (A_1sin\phi_1 + A_2sin\phi_2)^2 + (A_1cos\phi_1 + A_2cos\phi_2)^2$$
$$= {A_1}^2 + {A_2}^2 + 2A_1A_2\cos(\phi_1 - \phi_2)$$

The third term of above equation is the **interference term**. The resultant intensity depends upon the interference term.

2.3.1 Coherent superposition:

In this case phase difference remains constant, so $(\phi_1 - \phi_2)$ is constant

Case I

If $(\phi_1 - \phi_2) = 2n\pi$, $\quad$ where $n\ is\ an\ integere = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots\ldots$

Then $\cos(\phi_1 - \phi_2) = 1$, $\quad A^2 = (A_1 + A_2)^2$

So intensity $I = I_{max} = (A_1 + A_2)^2$

Special case

If $\quad A_1 = A_2 = a$,

Then $I_{max} = 4a^2$, that means the resultant intensity (intensity of bright fringe) is four times the intensity of the individual wave.

Case II

If $(\phi_1 - \phi_2) = (2n + 1)\pi$, $\quad$ where $n\ is\ an\ integere = 0, \pm1, \pm2, \pm3, \ldots\ldots\ldots\ldots$

Then $\cos(\phi_1 - \phi_2) = -1$, $\quad A^2 = (A_1 - A_2)^2$

So intensity $I = I_{min} = (A_1 - A_2)^2$

Special case

If $\quad A_1 = A_2 = a$,

Then $I_{min} = 0$, that means the resultant intensity is zero. Dark fringe is formed.

So, in case of coherent superposition alternate bright and dark fringes appear.

2.3.2 Incoherent superposition:

In this case phase difference changes frequently and randomly, so any particular value cannot be taken. So, time average value of $\cos(Ø_1 - Ø_2)$ is considered.

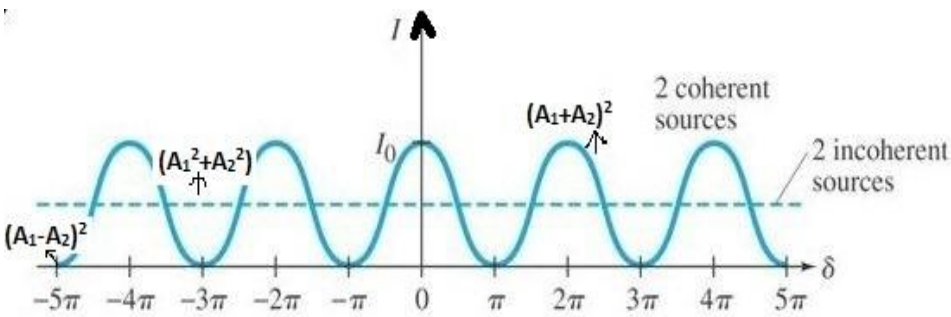
But$< \cos(\phi_1 - \phi_2) >= 0$, as its value varies in between +1 to -1.

So, $A^2 = {A_1}^2 + {A_2}^2$

Special case

If $A_1 = A_2 = a$, then $A^2 = 2a^2$, which is constant always.

So, no fringe pattern is observed.



Intensity distribution curve

2.4 Multiple beams Superposition:

Let's take a number of beams or waves having different amplitude, same frequency and different phases superpose

$$\psi_1 = A_1 \sin(kx - wt + \phi_1)$$

$$\psi_2 = A_2 \sin(kx - wt + \phi_2)$$

$$\psi_3 = A_3 \sin(kx - wt + \phi_3)$$

$$\psi_n = A_n \sin(kx - wt + \phi_n)$$

The resultant wave is $\psi = \psi_1 + \psi_2 + \psi_3 + \ldots\ldots\ldots + \psi_n$

$$\psi = \sum_{i=1}^{n} \psi_i = \sum_{i=1}^{n} A_i \, sin(kx - wt + \phi_i)$$

Where $A_i$and $Ø_i$ represents amplitude and phase of the ith component wave.

By comparison with two beam superposition the resultant wave is

$$\Psi = Asin(kx - wt + \emptyset)$$

where $A\ sin\emptyset = [A_1 sin\phi_1 + A_2 sin\phi_2 + A_3 sin\phi_3 + \cdots + A_n sin\phi_n]$=$\sum_{i=1}^{n} A_i\ sin\phi_i$

and $A\ cos\emptyset = [A_1 cos\phi_1 + A_2 cos\phi_2 + A_3 cos\phi_3 + \cdots + A_n cos\phi_n]$=$\sum_{i=1}^{n} A_i\ cos\phi_i$

So $A^2 = (\sum_{i=1}^{n} A_i\ sin\emptyset_i)^2 + (\sum_{i=1}^{n} A_i\ cos\phi_i)^2$

$$=\sum_{i=1}^{n} A_i^2 + 2\sum_{\substack{i,j=1 \\ i\neq j}}^{n} A_i A_j \cos(\phi_i - \phi_j)$$

Discussion:

2.4.1 Coherent superposition:

$(\phi_i - \phi_j)$ remains constant

$(\phi_i - \phi_j)$ =2nπ where $n = 0, \pm 1, \pm 2, \pm 3, \ldots \ldots \ldots \ldots$

So, $\cos(\phi_i - \phi_j)$ =1

Then $A^2 = =\sum_{i=1}^{n} A_i^2 + 2\sum_{\substack{i,j=1 \\ i\neq j}}^{n} A_i A_j$

Example Let n=3 so $A^2 = A_1^2 + A_2^2 + A_3^2 + 2A_1A_2 + 2A_2A_3 + 2A_1A_3$

$=(A_1 + A_2 + A_3)^2$

Special Case: If $A_1 = A_2 = A_3 = a$

Then $A^2 = (3a)^2$=$9a^2$

So, in general

$\boldsymbol{I_{coh} = (na)^2}$ where 'n' is the number of beams superpose.

2.4.2. Incoherent superposition:

$(\phi_i - \phi_j)$ changes frequently and randomly

So $< \cos\ (\phi_i - \phi_j)\ >= 0$

Then A^2 =$\sum_{i=1}^{n} A_i^2$

Example Let n=3 so $A^2 = A_1^2 + A_2^2 + A_3^2$

Then $A^2 = 3a^2$

So, in general

$\boldsymbol{I_{incoh} = na^2}$ where 'n' is the number of beams superpose.

$$\boldsymbol{I_{coh} = nI_{incoh}}$$

2.5. Superposition of multiple waves with a constant and successive phase difference:

Consider '**N**' number of waves having **same amplitudes, frequencies but phase varies with a constant amount** (**Φ, say**) and are represented as follows;

$\psi_1 = Asin(kx - \omega t)$

$\psi_2 = Asin(kx - \omega t + \phi)$

$\psi_3 = Asin(kx - \omega t + 2\phi)$

$\psi_N = Asin(kx - \omega t + [N-1]\phi)$

Phase difference between successive waves is Ø. The resultant superposed wave can be written as;

$\psi = \psi_1 + \psi_2 + \cdots + \psi_N$

In order to get the resultant wave in a convenient way we can express the wave function in exponential form

$\psi_1 = Ae^{i(kx-\omega t)}$

$\psi_2 = Ae^{i(kx-\omega t+\phi)}$

$\psi_3 = Ae^{i(kx-\omega t+2\phi)}$

Similarly, $\psi_N = Ae^{i(kx-\omega t+(N-1)\phi)}$

So the resultant superposed wave can be written as;

$$\psi = \psi_1 + \psi_2 + \cdots + \psi_N$$

$\psi(x,t) = Ae^{i(kx-\omega t)}\{1 + e^{i\phi} + e^{i2\phi} + e^{i3\phi} + \cdots + e^{i(N-1)\phi}\}$.....(i)

Using the Algebraic Formulation;

$1 + x + x^2 + x^3 + \cdots + x^m = \frac{1-x^{m+1}}{1-x}$

Now, equation (i) can be written as;

$\psi = Ae^{i(kx-\omega t)}\left\{\frac{1-e^{iN\phi}}{1-e^{i\phi}}\right\} = Ae^{i(kx-\omega t)}\left\{\frac{e^{iN\phi}-1}{e^{i\phi}-1}\right\}$

With suitable arrangement of the exponent power terms;

$\psi = Ae^{i(kx-\omega t)}\frac{e^{i(N/2)\phi}[e^{i(N/2)\phi} - e^{-i(N/2)\phi}]}{e^{i(\phi/2)}[e^{i(\phi/2)} - e^{-i(\phi/2)}]}$

$$\psi = Ae^{i\left(kx-\omega t+\frac{N-1}{2}\phi\right)} \frac{\left(2isin\frac{N\phi}{2}\right)}{2isin\frac{\phi}{2}} = \quad Ae^{i\left(kx-\omega t+\frac{N-1}{2}\phi\right)} \frac{sin\frac{N\phi}{2}}{sin\frac{\phi}{2}} \text{.....(ii)}$$

If we represent the resultant wave function as $\psi = Re^{i(kx-wt+\theta)}$.....(iii)

where R------- Resultant amplitude

$\theta \cdots\cdots\cdots Resultant\ phase$

Then comparing equations (ii) and (iii), we have

$$R = A\frac{sin\frac{N\phi}{2}}{sin\frac{\phi}{2}} \quad \text{and} \quad \theta = \frac{N-1}{2}\phi$$

We will apply this resultant amplitude expression in diffraction phenomena like, single slit and multiple slit (diffraction grating).

3.1 Interference:

We have already discussed that in case of coherent superposition, there is redistribution of light intensity leading to interference.

Definition:

*When two light waves from different coherent sources meet together (superpose), then the resultant amplitude (or intensity) in the region of superposition is different than that of the amplitude (or intensity) of individual waves. This modification in the distribution of intensity (light energy) due to superposition of two or more light waves is called "**Interference of light**.*

Examples

- ✓ One of the best examples of interference is demonstrated by the light reflected from a film of oil floating on water.
- ✓ The thin film of a soap bubble, which reflects a spectrum of beautiful colors when illuminated by natural or artificial light sources.

Interference is of two types:

- ❖ **Constructive interference**
- ❖ **Destructive interference**

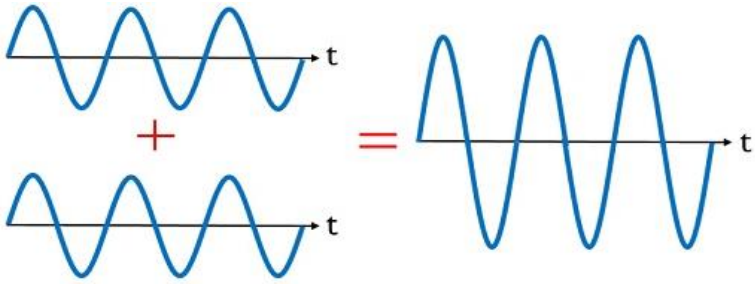
Constructive interference *occurs when the wave amplitudes reinforce each other, building a wave of even greater amplitude. It occurs when the crests of one wave overlap with the crests of the other wave, and troughs of one wave with the troughs of the other wave, causing an increase in wave amplitude/intensity.*

Or the resultant amplitude is the sum of the amplitudes of the waves

Condition:

Path difference = nλ or 2n ($\frac{\lambda}{2}$)

Phase difference = 2nπ, n is an integer = $0, \pm 1, \pm 2, \pm 3, \ldots\ldots\ldots\ldots$



Constructive Interference

Destructive interference *occurs when the wave amplitudes oppose each other, resulting in waves of reduced amplitude. It occurs when the crests of one wave overlap with the troughs of the other wave, causing a decrease in wave amplitude/intensity.*

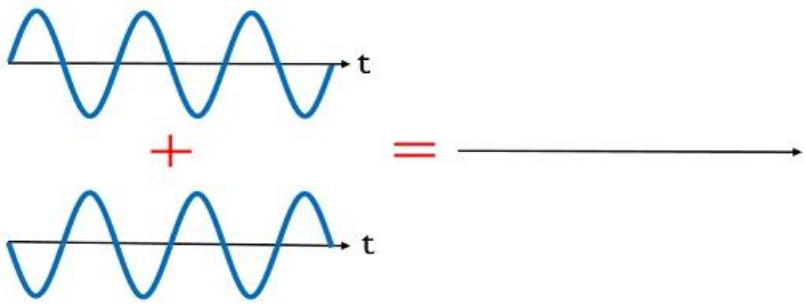
Or *if we take two waves then the resultant amplitude is the difference of two amplitudes, then that is called destructive interference*.

Condition:

Path difference = (2n + 1) $\frac{\lambda}{2}$

Phase difference = (2n + 1)π, n is an integer = $0, \pm 1, \pm 2, \pm 3, \ldots\ldots\ldots\ldots$

(Phase difference = $\frac{2\pi}{\lambda}$ x path difference)



Destructive Interference

3.2 Methods of obtaining interference:

i. **Division of Amplitude**: *In this method, the amplitude of the incident beam is divided into two or more parts either by partial reflection or refraction. Thus, we have coherent beams produced by division of amplitude. These beams travel different optical paths and are finally united together to produce interference. Examples are Newton's rings, Michelson interferometer.*

ii. **Division of Wave front**: *Under this category, the coherent sources are obtained by dividing the wave front, originating from a common source, by employing mirrors, biprisms or lenses. These beams travel unequal optical path and finally reunite to produce interference. Examples are Fresnel' biprism, Lloyds single mirror.*

3.2.1 Condition for Interference of light:

To obtain a permanent or stationary interference pattern the conditions are classified into following three categories

- *Condition for sustained interference*
- *Condition for clear observation*
- *Condition for good contrast between maxima and minima*

3.2.2 Condition for Sustained Interference Pattern:

- *The two interfering wave should be coherent i.e., both light waves are in same phase or maintain constant phase difference between them.*
- *The source should have same frequency.*
- *Both the wave must be in same set of polarization.*

3.2.3 Condition for clear observation of fringes

- ✓ *The distance between the coherent sources should small.*
- ✓ *The distance between the source and screen must be large.*
- ✓ *The background should be dark.*

3.2.4 Condition for Good Contrast between maxima and minima

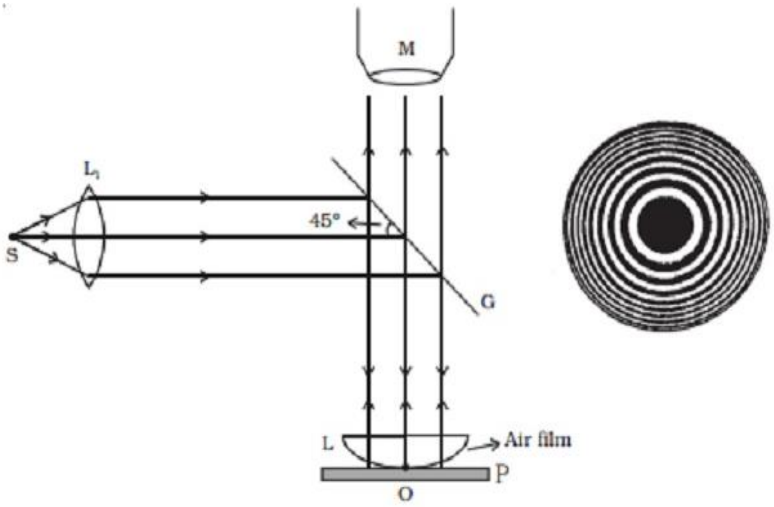
- ✓ *The amplitude of the both interfering wave be the equal or very nearly equal.*
- ✓ *The sources must be narrow.*
- ✓ *The sources must be monochromatic.*

Newton's Rings (Fringes due to Division of Amplitude)

4.1: Description:

When a plane-convex lens of large focal length is placed as a plane glass plate, a thin film of air is formed between the plano-convex lens and plane glass plate. The thickness of the air film is very small at the point of contact and gradually increased from the centre upwards. If a monochromatic light is allowed to fall normally on this film, a set of alternate dark and bright fringes will be seen in the film. The fringes are concentric circle. These circles or rings are called Newton's rings. Normal incidence of monochromatic light on the plano-convex lens will produces bright and dark concentric rings around the point of contact between the lens and the glass plate. The phenomenon first explained by Newton so called Newton's rings.

Experimental arrangement:



S – Source, producing monochromatic light like, sodium light

L_1 - Convergent lens, used to produce parallel rays

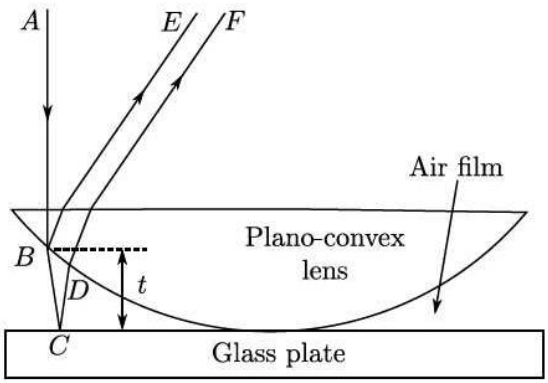
G - A glass plate, making an angle 45^0 with the direction of incident light, so that vertically reflected downward reflected rays are produced

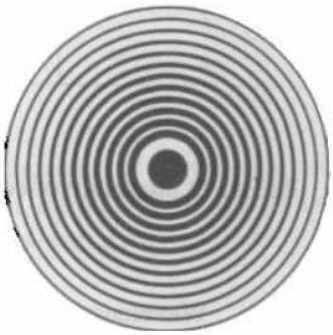
L - A plano-convex lens of large radius of curvature, placed on a plane glass plate P, with its convex surface facing upward

M - A travelling microscope, which is used to see the fringe patterns

4.1.1 Explanation of formation of Fringes:

Newton's rings are formed as a result of interference between light wave reflected from the upper and lower surfaces of the air film. Fig. below shows that from a single incident ray (A), two reflected rays (E&F) are produced by division of amplitude. These two reflected rays are coherence and superimpose to produce interference pattern.





Suppose, the radius of curvature of the plano-convex lens is R and the thickness of the air film is 't'. The ray E and F are coherent as they produced from same incident ray A. The effective path difference between the interfering rays is

$$\Delta = 2\mu t cos r + \frac{\lambda}{2}$$

For air, μ=1 and for normal incidence, r = 0

So effective path difference in this case is $\Delta = 2t + \frac{\lambda}{2}$ .....(1)

At the centre, $t = 0$

So effective path difference $\Delta = \frac{\lambda}{2}$, this is the condition for minimum intensity. So the centre is dark.

For the bright fringes $\Delta = \boldsymbol{n\lambda}$

So $2t + \frac{\lambda}{2} = \boldsymbol{n\lambda}$

Or, $2t = (2n - 1)\frac{\lambda}{2}$ ..... (2)

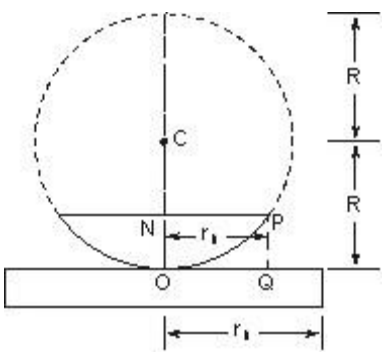
For the dark rings $\Delta = \boldsymbol{(2n + 1)\frac{\lambda}{2}}$

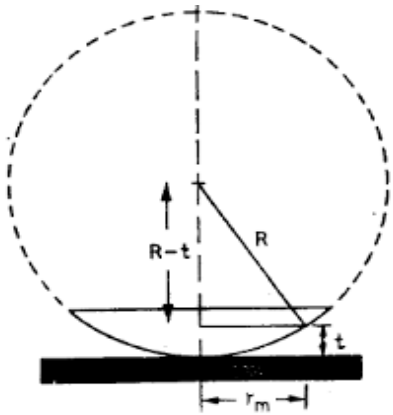
So $2t + \frac{\lambda}{2} = \boldsymbol{(2n + 1)\frac{\lambda}{2}}$ $\boldsymbol{n = 0, 1, 2, 3, 4}$ ....

Or, $2t = \boldsymbol{n\lambda}$ ..... (3)

It is clear that a bright or dark fringe of any order n depend upon the thickness of the air film. Since t is constant along a circle with its centre at the point of contact, the fringes are in the form of concentric circle.

4.1.2 Diameter of Bright Rings





Let r_n be the radius of newton rings corresponding to a point P where the firm thickness is t.

From the figure

$$(PN)^2 = ON \times NE = ON \times (2R - ON)$$

$$(PN)^2 = PQ \times (2R - PQ)$$

$$(PN)^2 = T(2R - t) = 2Rt - t^2$$

$$r_n^2 = 2Rt$$

$$2t = \frac{r_n^2}{R} \qquad \ldots\ldots\ldots\ldots (4)$$

Substituting the value of t in eqn. (2)

For bright rings

$$\frac{r_n^2}{R} = (2n-1)\frac{\lambda}{2}$$

$$r_n^2 = \frac{(2n-1)}{2}\lambda R$$

$$r_n = \sqrt{\frac{(2n-1)}{2}\lambda R} \qquad \ldots\ldots\ldots\ldots (5)$$

and $D_n = 2r_n$ where D is diameter of nth bright ring.

So from eqn. 5

$$D_n^2 = 2(2n-1)\lambda R$$

$$D_n = \sqrt{2\lambda R}\,\sqrt{(2n-1)}$$

$$D_n \propto \sqrt{(2n-1)}$$

Thus, the diameters of bright rings are proportional to the square root of the odd number

$$D_1 = \sqrt{1}\,, D_2 = \sqrt{3}\,, D_3 = \sqrt{5}$$

The separations between successive rings are 0.732:0.504:0.410.

4.1.3 Diameter of Dark Rings

From equation (3) and (4)

$$\frac{r_n^2}{R} = n\lambda$$

$$r_n = \sqrt{nR\lambda}$$

If D is the diameter of dark ring

$$D_n = 2r_n = 2\sqrt{nR\lambda} = \sqrt{4n\lambda R}$$

$$D_n \propto \sqrt{n}$$

Thus, the diameters of dark ring are proportional to the square root of natural number

$$D_1 = \sqrt{1}\,, D_2 = \sqrt{2}\,, D_3 = \sqrt{3}.....$$

4.1.4 Newton's rings by transmitted light:

The effective path difference between the interfering rays is

$$\Delta = 2\mu t cosr$$

For μ=1 and for normal incidence r=0

So effective path difference in this case is $\Delta = 2t$

At centre $t = 0$

So effective path difference $\Delta = 0$, this is the condition for maxima so center is bright.

For the bright fringes $\Delta = \boldsymbol{n\lambda}$

So $2t = \boldsymbol{n\lambda}$

So, for bright fringe $D_n^2 = 4Rn\boldsymbol{\lambda}$ **so** $D_n = \sqrt{4Rn\boldsymbol{\lambda}}$

For the dark rings $\Delta = (\boldsymbol{2n-1})\frac{\boldsymbol{\lambda}}{\mathbf{2}}$

So $2t = (\boldsymbol{2n-1})\frac{\boldsymbol{\lambda}}{\mathbf{2}}\boldsymbol{n} = \mathbf{1,2,3,4}$ ....

So $D_n^2 = 2R\boldsymbol{\lambda}(\boldsymbol{2n-1})$ **So** $D_n = \sqrt{\boldsymbol{2R\lambda(2n-1)}}$

Hence, the fringe patterns, in reflected and transmitted light are complimentary.

4.1.5 Application of Newton's rings:

4.1.5.1 Determination of wavelength of light:

By forming Newton's rings and measuring the radii of the rings formed, we can calculate the wavelength of the light used if the radius of curvature of the lens is known. Let R be the radius of curvature of the lens and λ is the wavelength of the light used. So the diameter of the nth dark ring can be written as

$$D_n^2 = 4\, n\, \lambda R \qquad (1)$$

Similarly, the diameter of the (n +m) th dark ring is

$$D_{(n+m)}^2 = 4\,(n+m)\,\lambda R \qquad (2)$$

Subtracting equation (1) from (2) we get

$$D_{(n+m)}^2 - D_n^2 = 4\,(n+m)\,\lambda R - 4\, n\, \lambda R = 4mR\lambda$$

Or, $\boldsymbol{\lambda} = \frac{\mathbf{D}_{(n+m)}^2 - \mathbf{D}_n^2}{4mR}$

Using the above relation, wavelength of the monochromatic light can be calculated

4.1.5.2 Determination of refractive index of a liquid using Newton's rings:

By forming Newton's rings and measuring the diameter of the rings formed, we can calculate the refractive index of the liquid.
In air film, the diameters of the nth and (n+m)th dark rings are measured with the help of travelling microscope.

$$D_{(n+m)}^2 - D_n^2 = 4\,(n+m)\,\lambda R - 4\, n\, \lambda R = 4mR\lambda \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad (3)$$

Now, the air film is replaced by liquid film (refractive index μ). In liquid film, the diameters of the same nth and (n+m)th dark rings are D'_n and $D'_{(n+m)}$ are measured with the help of travelling microscope.

$$D'^2_n = 4\, n\, \lambda R/\mu \quad \text{and} \quad D'^2_{(n+m)} = 4\,(n+m)\,\lambda R/\mu$$

$$\text{So, } D'^2_{(n+m)} - D'^2_n = 4mR\lambda/\mu \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (4)$$

Dividing equation (3) by (4)

$$\frac{\mathbf{D}_{(n+m)}^2 - \mathbf{D}_n^2}{\mathbf{D}'^2_{(n+m)} - \mathbf{D}'^2_n} = \boldsymbol{\mu}$$

Using the above relation μ can be calculated.

Numerical:

1. A wave along s string is given by the equation $y = 0.01\sin(50\pi t - 31.4x)\,m$ calculate the speed with which the wave travels.

   Sol: V=$\frac{w}{k} = \frac{50\pi}{31.4} = 5m/s$

2. Two coherent sources, whose intensity ratio is 9:4, produce interference fringes. Find the ratio of maximum to minimum intensity of the fringe system.

   Sol: Let I_1 and I_2 be the intensities, and a_1 and a_2 the amplitudes of the two interfering beams. We can write

   $\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2}$

   or, $\frac{a_1}{a_2} = \sqrt{\frac{I_1}{I_2}} = \frac{3}{2}$

   If a_1 =3x then a_2 = 2x

   Now, $\frac{I_{max}}{I_{min}} = \frac{(a_1+a_2)^2}{(a_1-a_2)^2} = \frac{(3x+2x)^2}{(3x-2x)^2} = \frac{25x^2}{x^2} = \frac{25}{1}$

3. Find the ratio of the intensity at the center of a bright fringe in an interference pattern to the intensity at a point one-quarter of the distance between two fringes from the center.

   Sol: Resultant intensity I = $2a^2$(1 + cos φ)

   Where a is the amplitude of the wave.

   At the center, φ = 0

   So, I_0 = $2a^2$(1 + cos φ) = $2a^2$(1 + cos 0) = $4a^2$

   At one-quarter of the distance between two fringes from the center, φ = π/2

   I_1 = $2a^2$(1 + cos φ) = $2a^2$(1 + cos π/2)= $2a^2$

   So, $I_0 : I_1$ = $4a^2 : 2a^2$ = 2 :1

4. In a two source interference experiment, let 'p' be a point on the screen, which is equidistant from both the sources. If one source is closed the intensity at 'p' is 0.02 watt/m^2 then what will be the intensity at 'p' when both sources are opened?

   Sol: 'P' is the point of central principal maxima

   I_1=0.02 watt/m^2

   Resultant intensity I= $4I_1$=0.08 watt/m^2

5. Twenty five sinusoidal waves of equal amplitude superpose incoherently to produce a resultant intensity of 0.5 watt/m^2. What would be the maximum possible resultant intensity if the superpose coherently?

   Sol: In case of incoherent superposition,

   $I_{in\text{-}coherent}$ = na^2 =nI, and in case of coherent superposition, $I_{coherent}$ = n^2a^2 =n^2I

So, $I_{coherent} = n\, I_{in\text{-}coherent} = 25 \times 0.5 = 12.5$ watt/m^2.

6. Newton's rings are observed normally in reflected light of wavelength 5893 Å. The diameter of the 10th dark ring is 0.005 m. Find the radius of curvature of the lens and the thickness of the film.

   Sol: $D_n^2 = 4\, n\, \lambda\, R$

   Here, $D_n = 0.005$ m, n=10, $\lambda = 5893 \times 10^{-9}$ m.

   Now, $R = \frac{D_n^2}{4\, n\, \lambda} = 1.06$ m = 106 cm

   If t be the thickness of the film corresponding to a ring of diameter D, then we have

   $2t = \frac{D_n^2}{4\, R} = 6 \times 10^{-6}$ m

   or, $t = 3 \times 10^{-6}$ m

7. If in a Newton's rings experiment, the air in the interspace is replaced by a liquid of refractive index 1.33, in what proportion would the diameter of a ring change?

   Sol: $\frac{(D_n^2)_{air}}{(D_n^2)_{liquid}} = \mu$

   or, $\frac{(D_n)_{liquid}}{(D_n)_{air}} = \sqrt{\frac{1}{\mu}} = 0.867$

   The rings are contracted to 0.867 their previous diameter.

8. In a Newton's rings experiment in air, the diameter of 10th dark ring is 0.272 cm and that of 15th dark ring is 0.555 cm. If the radius of curvature of the plano-convex lens is 200 cm, calculate the wavelength of the monochromatic light used.

   Sol: $D_n = 0.272$ cm

   $D_{n+p} = 0.555$ cm

   $n = 10$, n + m =15, m = 5

   $R = 200$ cm

   We know that $\lambda = \frac{D_{(n+m)}^2 - D_n^2}{4mR}$

   or, $\lambda = \frac{0.555^2 - 0.272^2}{4 \times 5 \times 200} = 5851$ Å.

9. In a Newton's rings experiment, the diameter of the 10th bright ring changes from 1.5 cm to 1.27 cm when a liquid is introduced between the plate and the lens. Calculate the refractive index of the liquid.

10. In a Newton's rings experiment, the diameter of the 5th ring was 0.336 cm and the diameter of the 15th ring was 0.59 cm. Find the radius of curvature of the plano convex lens if the wavelength of light is 5890 Å.

Questions: Short & medium answer type

1. Write the difference between coherent and incoherent superposition.
2. Discuss why two independent sources of light of the same wavelength cannot produce interference fringes
3. Do you expect any fringe pattern in case of incoherent superposition? Give justification to your answer.
4. Find the ratio of intensity at the centre of a bright fringe to the intensity at a point one quarter of the distance between two fringes from the centre.
5. Write down the condition for the destructive interference in terms of phase difference and in terms of path difference.
6. Write the conditions of sustained interference pattern.
7. Write the conditions for good contrast between dark and bright fringes.
8. What are the different methods of obtaining interference pattern? Give example of each.
9. Show the intensity distribution curve in an interference pattern.
10. Why the fringes in Newton's ring interference are circular?
11. In a Newton's ring experiment the center is bright, whether it is formed by reflected light or transmitted light. Justify your answer.
12. In a Newton's ring experiment the center is dark, whether it is formed by reflected light or transmitted light. Justify your answer.
13. Why the planoconvex lens has large radius of curvature in Newton's ring experiment?
14. In a Newton's ring experiment, the fringes near to the center are wider than the fringes away from the centre. Justify it why?
15. Write down the formula for the refractive index of a liquid determined by In a Newton's ring experiment.
16. Explain how refractive index of liquid can be determined by Newton's rings method.

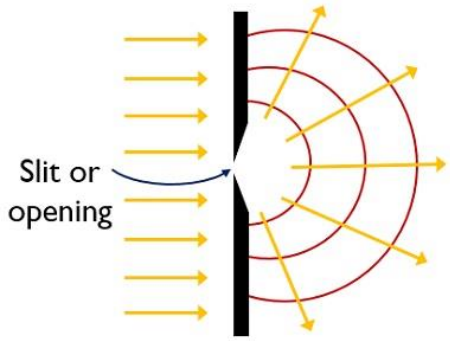
Long answer type

1. Show that the resultant wave due to the superposition of two harmonic waves of same frequency, different phases and travelling in same direction is also harmonic. Find the amplitude and phase of the resultant wave.
2. Show that coherent superposition leads to interference pattern, discuss it with two beams having different amplitude, different phases propagating in same direction.
3. Draw the schematic diagram for production of Newton's rings with reflected light. Explain how will u obtained two coherent waves.

4. With a suitable diagram, explain the formation of Newton's rings. Derive the expression for the diameter of Newton's rings formed by transmitted light. Explain why the fringes in Newton's rings interference are circular.
5. Explain the formation of fringes in a Newton's rings in reflected light with a suitable diagram. Prove that in reflected light (i) diameters of the dark rings are proportional to the square root of natural numbers, and (ii) diameters of the bright rings are proportional to the square root of odd natural numbers.

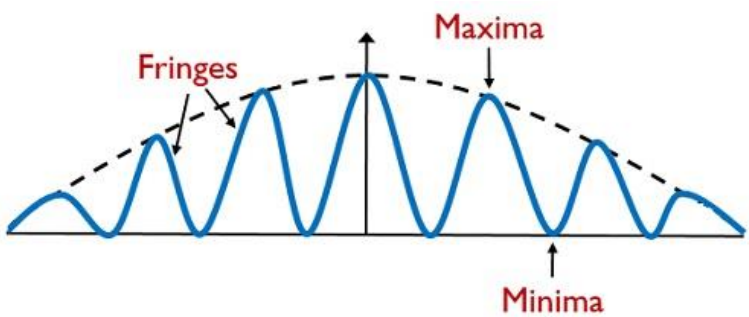
DIFFRACTION

5.1 **Diffraction** is the slight bending of light as it passes around the edge of an object. The amount of bending depends on the relative size of the wavelength of light to the size of the opening. If the opening is much larger than the light's wavelength, the bending **will** be almost unnoticeable.



Diffraction of light

Diffraction is the spreading out of **waves** as they pass through an aperture or around objects. It occurs significantly when the size of the aperture or obstacle is of similar linear dimensions to the wavelength of the incident wave. The essential condition for diffraction to occur is that the wavelength of **light** should be comparable to that of the size of the object/obstacle. It might also occur if the size of the object is less than the wavelength of **light**. It is due to the interaction among the secondary wavelets which are not blocked off by the obstacle.



Diffraction pattern

The most striking **examples of diffraction** are those that involve light; for **example**, the closely spaced tracks on a CD or DVD act as a **diffraction** grating to form the familiar rainbow pattern seen when looking at a disc.

Difference between Diffraction and Interference	
Interference	Diffraction
Interference may be defined as waves emerging from two different coherent sources, producing different wave fronts.	Diffraction on the other hand can be termed as secondary waves that emerge from the different parts of the same wave.
In interference the intensity of all the positions on maxima are equal.	In diffraction, there is a variance of the intensity of all maxima positions.
The width of the fringes in interference may be equal (bi-prism) or may not be equal (Newton's rings).	The widths of the fringes are not equal in diffraction.
It is absolutely dark in the region of minimum intensity, in the case of interference.	In the case of diffraction, the position of minima is not perfectly dark
If the number of sources are few such as two sources, then they are referred to as interference sources.	If the number of sources is many, that is more than two then it is referred to as diffraction sources.

Types of diffraction (Fresnel diffraction and Fraunhofer diffraction)

Characteristics	**Fresnel diffraction**	**Fraunhofer diffraction**
Observation distance	Either the source or screen or both are at finite distance from the obstacle	The source and the screen are effectively at infinite distance from the obstacle
Requirement of lens	Lens are not required	Lens (usually 02) are required
Wave fronts	Cylindrical or spherical wave fronts	Plane wave fronts
Diffraction pattern	Shape and intensity of diffraction pattern change as the waves propagate downstream of the scattering source	Shape and intensity remain constant

Movement of diffraction pattern	Move along the corresponding shift in the object	Remains in a fixed position

5.1.1 Diffraction due to Single Slit (Fraunhofer diffraction)

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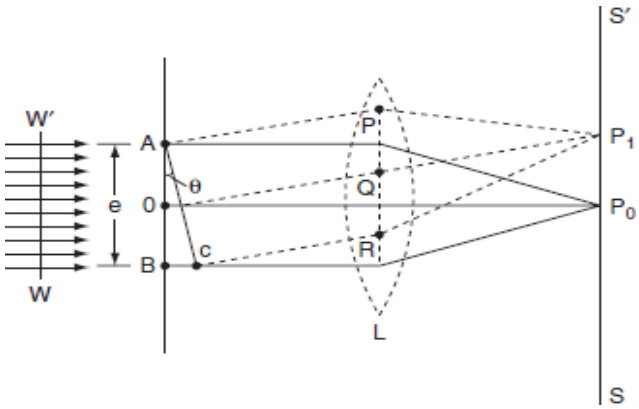


'AB' is a narrow slit of width 'e' perpendicular to the plane of the paper. Consider a plane wavefront 'ww' originated from a monochromatic source of wavelength 'λ' propagating normal to the slit be incident on it. After diffraction Let the diffracted rays are focused by using a convex lens and the screen placed at the focal plane of the lens.

According to Hygen's theory, each point of the wave front in the plane of the slit is a source of secondary wavelets. The secondary wavelets traveling normal to the slits are brought to focus by the lens at the point P_0 on the screen. So, the point P_0 must be the point of maxima called as central principal maxima. In order to consider all the secondary wavelets originated and propagated in different directions, consider the secondary wavelets in the direction of' θ' are brought to the focus by the lens to the point P_1 on the screen as shown in the figure. The intensity at the point P_1 is either minimum or maximum and depends upon the path difference between the secondary waves originating from the corresponding points of the wave front.

Theory:

In order to find out the intensity at P_1, draw a perpendicular *AC* on *BR*.

The path difference between secondary wavelets from *A* and *B* in direction θ is *BC i.e.*,

$$\Delta = BC = AB sin\boldsymbol{\theta} = esin\boldsymbol{\theta}$$

So, the phase difference,

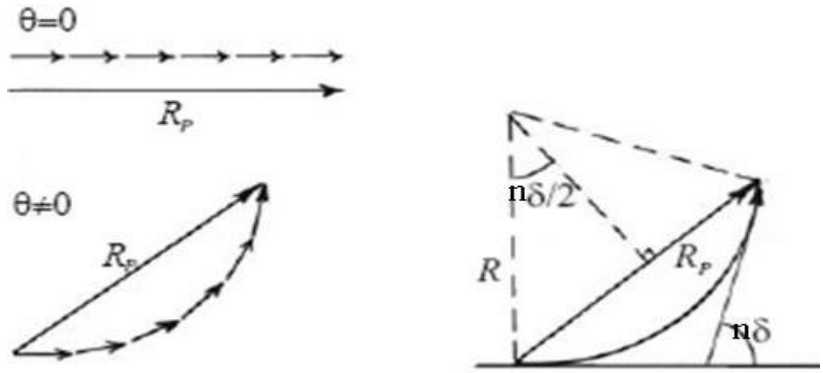
$$= \frac{2\pi}{\lambda} \times \Delta = \frac{2\pi}{\lambda}(esin\boldsymbol{\theta})$$

Let us consider that the width of the slit is divided into '*n*' equal parts and the amplitude of the wave from each part is *'a'*.

So, the phase difference between two consecutive points

$$\delta = \frac{1}{n}.\left\{\frac{2\pi}{\lambda}(esin\boldsymbol{\theta})\right\}$$ .....(2.32)

Then the *resultant amplitude R* is calculated by using the method of vector addition of amplitudes



The resultant amplitude of n number of waves having same amplitude 'a' and having common phase difference of 'δ' is

$$R = a\frac{sin(n\delta/2)}{sin(\delta/2)}$$ .....(2.33)

Substituting the value of δ in equation (2.33)

$$R = a\frac{\sin(\frac{n}{2}\left(\frac{2\pi}{n\lambda}\right)esin\theta)}{\sin(\frac{1}{2})\frac{1}{n}\frac{2\pi}{\lambda}esin\theta}$$

So, $$R = a\frac{\sin(\left(\frac{\pi}{\lambda}\right)esin\theta)}{\sin\frac{1\pi}{n\lambda}esin\theta}$$ .....(2.34)

Substituting $\alpha = \frac{\pi}{\lambda}.esin\boldsymbol{\theta}$ in equation 2.34

$$R = a\frac{sin\alpha}{sin(\alpha/n)}$$

As α/n is small value; $sin\alpha/n \rightarrow \alpha/n$

$$R = na\frac{sin\alpha}{\alpha}$$

and na = A

Therefore,

$$\boldsymbol{R = A\frac{sin\alpha}{\alpha}} \text{ .....(2.35)}$$

Therefore, the Intensity is given by

$$I^2 = R^2 = \boldsymbol{A^2\frac{sin^2\alpha}{\alpha^2}} \text{ .....(2.36)}$$

<u>Case (i): *Principal Maximum:*</u>

Eqn (2.35) takes maximum value for

α=0

$$\alpha = \frac{\pi}{\lambda}esin\theta = 0$$

$\Rightarrow sin\theta = 0$ or $\theta = 0$

The condition

The condition $\theta = 0$ means that this maximum is formed by the secondary wavelets which travel normally to the slit along OP_o and focus at P_o. This maximum is known as *"Principal maximum"*.

Intensity of Principal maxima

$$R_{max} = Lim_{\alpha\rightarrow 0}\frac{Asin\alpha}{\alpha} = ALim_{\alpha\rightarrow 0}\frac{sin\alpha}{\alpha}$$
$$R_{max} = A.1 = A$$

Therefore

$$I^2_{max} = R^2_{max} = A^2$$

Case (ii): *Minimum Intensity positions:*

For minimum intensity, *sin*α*=0.* The values of 'α' which satisfy $sin\alpha = 0$ are

$$\alpha = \pm\pi, \pm 2\pi, \pm 3\pi ..... \pm n\pi$$

$$\frac{\pi}{\lambda}.esin\theta = \pm n\pi$$

$$\boldsymbol{esin\theta = \pm n\lambda} \text{ where } n = 1, 2, 3, ... \quad ...(2.37)$$

In the above eq. (2.37) *n = 0* is not applicable because corresponds to principal maximum. Therefore, the positions according to eq. (2.37) are on either side of the principal maximum.

Case (iii): *Secondary maximum:*

In addition to principal maximum at α = 0, there are weak secondary maxima between minima positions.
The positions of these weak secondary maxima can be obtained with the rule of finding maxima and minima
of a given function in calculus. So, differentiating eq^n (2.35) and equating to zero, we have

$$\because A^2 \neq 0; sin\alpha \neq 0$$

Because $sin\alpha = 0$ correspond to minima positions

so, $\alpha cos\alpha - sin\alpha = 0$

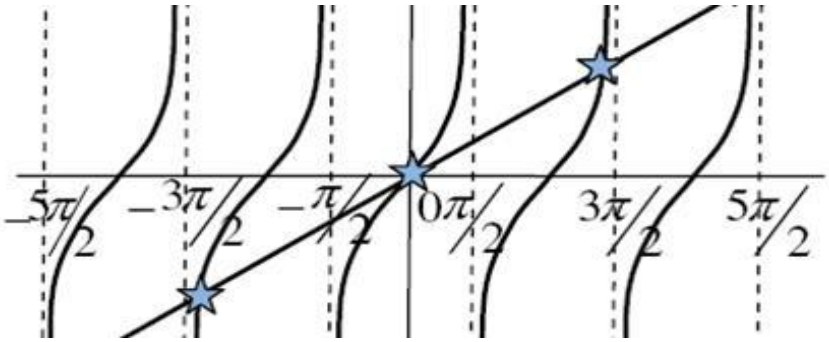
$$\Rightarrow \alpha = tan\alpha ..... (2.38)$$

The values of 'α' satisfying the eqn (2.38) are obtained graphically by plotting the curves $y = \alpha$ and $y = tan\alpha$

on the same graph. The points of intersection of the two curves gives the values of 'α' for which satisfy eqn (2.38).

The points of intersections are

$$\alpha = 0, \pm\frac{3\pi}{2}, \pm\frac{5\pi}{2}, \pm\frac{7\pi}{2} ..... \pm \frac{(2n+1)\pi}{2}$$



But $\alpha = 0$, gives principal maximum, substituting the values of 'α' in eqn(2.36), we get

$$I_1 = A^2 \left[\frac{sin3\pi/2}{3\pi/2}\right]^2 = \frac{A^2}{22}$$

$$I_2 = A^2 \left[\frac{sin5\pi/2}{5\pi/2}\right]^2 = \frac{A^2}{62}$$

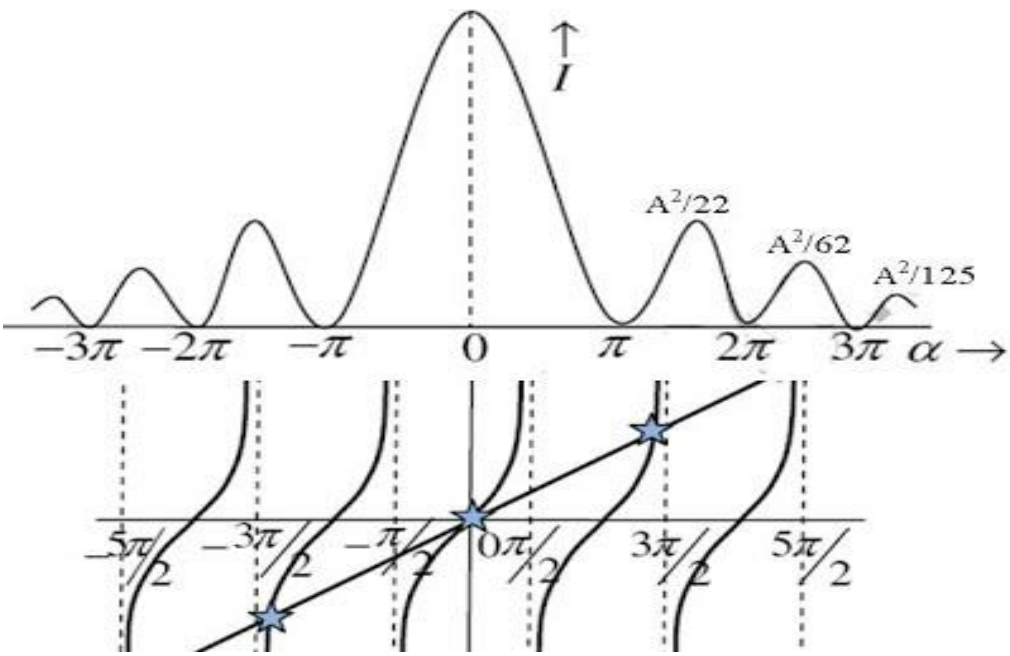
$$I_3 = A^2 \left[\frac{sin7\pi/2}{7\pi/2}\right]^2 = \frac{A^2}{125}$$

and so on.

From the above expressions, I_{max}, I_1, I_2, I_3... it is evident that most of the incident light is concentrated at the principal maximum.

5.1.2 Intensity distribution graph:

A graph showing the variation of intensity with 'α' is as shown in the adjacent figure

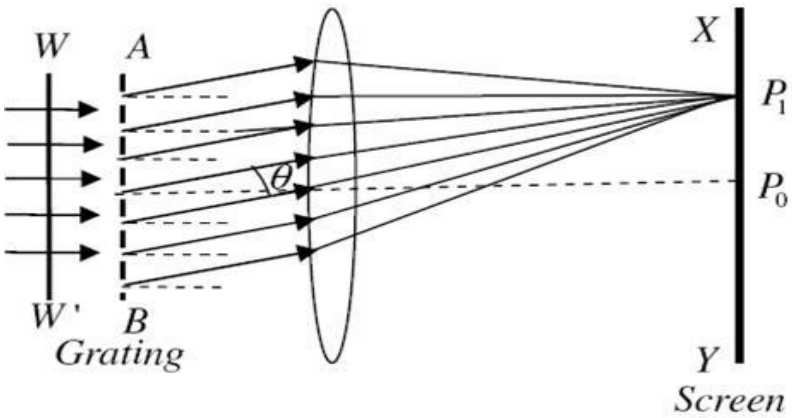


5.1.3 Plane diffraction grating [Diffraction due to N-Slits (Grating)]:

An arrangement consisting of large number of parallel slits of the same width and separated by equal opaque spaces is known as **Diffraction grating**.

Plane diffraction grating consists of a number of parallel and equidistant lines ruled on an optically **plane** and parallel glass plate by a fine diamond point. Each ruled line behaves as an opaque line while the transparent portion between two consecutive ruled lines behaves as a slit. If a is the width of transparency and b is width of opacity (width of ruling lines) then a+b is called grating element.

Theory: A section of a *plane transmission grating AB* placed perpendicular to the plane of the paper is as shown in the figure.



Let *'e'* be the width of each slit and *'d'* the width of each opaque space. Then *(e+d)* is known as grating element and *XY* is the screen. Suppose a parallel beam of monochromatic light of wavelength 'λ' be incident normally on the grating. By Huygen's principle, each of the slit sends secondary wavelets in all directions.

Now, the secondary wavelets travelling in the direction of incident light will focus at a point P_o on the screen. This point P_o will be a central maximum. Now consider the secondary waves travelling in a direction inclined at an angle 'θ' with the incident light will reach point P_1 in different phases. As a result dark and bright bands on both sides of central maximum are obtained. The intensity at point P_1 may be considered by applying the theory of Fraunhofer diffraction at a single slit. The wavelets proceeding from all points in a slit along their direction are equivalent to a single wave of amplitude $A\frac{sin\alpha}{\alpha}$ starting from the middle point of the slit, Where $\alpha = \frac{\pi}{\lambda}esin\theta$

If there are N slits, then we have N diffracted waves. The path difference between two consecutive slits is $(e+d)sin\theta$

Therefore, the phase difference

$$\delta = \frac{2\pi}{\lambda}(e+d)sin\theta = 2\beta \quad .....(2.39)$$

Hence the intensity in a direction can be found by finding the resultant amplitude of N vibrations each of amplitude $A\frac{sin\alpha}{\alpha}$ and a phase difference of 2β"

Since in the previous case

$$a = A\frac{sin\alpha}{\alpha}; n = N; \delta = 2\beta$$

Substituting these in equation $R = a\frac{sin(n\delta/2)}{sin(\delta/2)}$

The resultant amplitude on screen at P_1 becomes

$$R = \left(A\frac{sin\alpha}{\alpha}\right)\frac{sinN\beta}{sin\beta} .....(2.40)$$

Thus, Intensity at P_1 will be

$$I^2 = R^2 = \left(A\frac{sin\alpha}{\alpha}\right)^2\frac{sin^2N\beta}{sin^2\beta} \quad ...(2.41)$$

The $\left(A\frac{sin\alpha}{\alpha}\right)^2$ factor gives the distribution of Intensity due to a single slit while the factor

$\frac{sin^2N\beta}{sin^2\beta}$ gives the distribution of intensity as a combined effect of all the slits

5.1.4 Intensity Distribution:

<u>Case (i): Principal maxima:</u> The eqn (2.40) will

take a maximum value if

$$sin\beta = 0$$

$$\beta = \pm n\pi; n = 0, 1, 2, 3.....$$

$$\frac{\pi}{\lambda}(e+d)sin\theta = \pm n\pi$$

$$\boldsymbol{(e+d)sin\theta = \pm n\lambda} .....(2.42)$$

n = 0 corresponds to zero order maximum. For *n = 1,2,3,…* we obtain first, second, third,… principal

maxima respectively. The ± sign indicates that there are two principal maxima of the same order lying on either side of zero order maximum.

Case(ii): Minima Positions: The eq. (2.40) takes minimum value if $sinN\beta = 0$ but $sin\beta \neq 0$

$$\therefore N\beta = \pm m\pi$$

$$N(e+d)sin\theta = \pm m\lambda \quad \text{.....(2.43)}$$

$$N\frac{\pi}{\lambda}(e+d)sin\theta = \pm m\pi$$

where ***m*** has all integral values except *m = 0, N, 2N, …, nN,* because for these values $sin\beta$ becomes zero and we get principal maxima. Thus, *m = 1, 2, 3, …, (N-1).* Hence

$$N(e+d)sin\theta = \pm m\lambda \quad \text{where} \quad m = 1, 2, 3....(N-1), (N+1), ...(2N-1)....$$

gives the minima positions which are adjacent to the principal maxima.

Case(iii): Secondary maxima: As there are *(N-1)* minima between two adjacent principal maxima there must be *(N-2)* other maxima between two principal maxima. These are known as secondary maxima. To find their positions

$$\frac{\mathrm{dI}}{\mathrm{d}\beta} = 0$$

$$\because \frac{sin\alpha}{\alpha} \neq 0; sinN\beta \neq 0$$

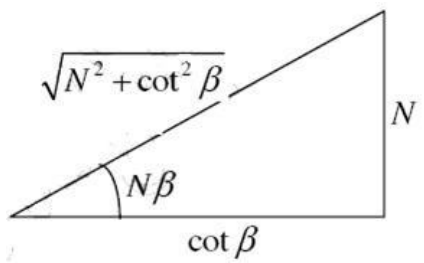
Only

$$[NcosN\beta sin\beta - SinN\beta cos\beta] = 0$$

$$\boldsymbol{Ntan\beta = tanN\beta} \text{.....(2.44)}$$

The roots of the above equation other than those for which $\beta = \pm n\pi$ give the positions of secondary maxima The eqn (2.44) can be written as

$$tanN\beta = \frac{N}{cot\beta}$$



From the triangle we have

$$sinN\beta = \frac{N}{\sqrt{N^2 + cot^2\beta}}$$

$$\therefore \frac{sin^2 N\beta}{sin^2\beta} = \frac{N^2}{(N^2 + cot^2\beta)sin^2\beta}$$

$$\therefore \frac{sin^2 N\beta}{sin^2\beta} = \frac{N^2}{N^2 sin^2\beta + (1 - sin^2\beta)}$$

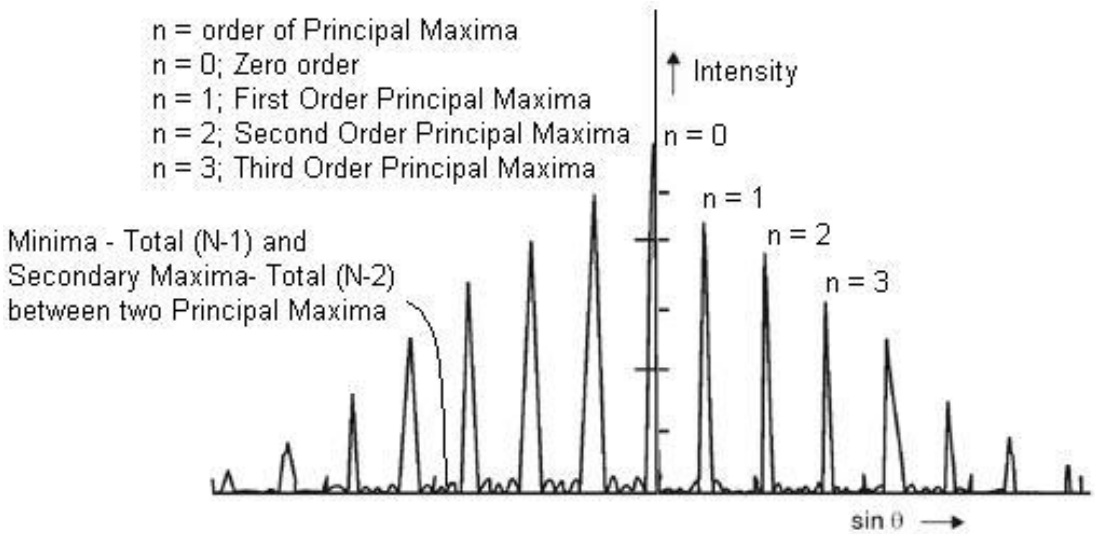
$$\therefore \frac{sin^2 N\beta}{sin^2\beta} = \frac{N^2}{1 + (N^2 - 1)sin^2\beta}$$

$$I = I_o \left(\frac{sin\alpha}{\alpha}\right)^2 \frac{N^2}{1 + (N^2 - 1)sin^2\beta}$$

Since intensity of principal maxima is proportional to N^2,

$$\frac{\text{Intensity of secondry maxima}}{\text{Intensity of Principal maxima}} = \frac{\frac{N^2}{1+(N^2-1)\sin^2\beta}}{N^2} = \frac{1}{1+(N^2-1)\sin^2\beta}$$

Hence if the value of N is larger, then the secondary maxima will be weaker and becomes negligible when N becomes infinity.



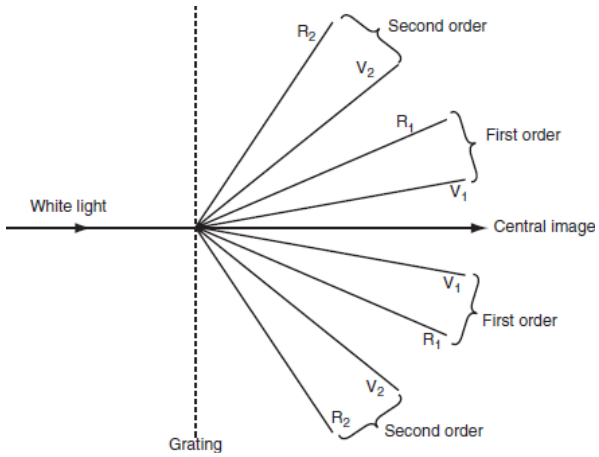
5.1.5 Formation of Spectra with Grating

The principle maxima in a grating are formed in direction given by $(\boldsymbol{e}+\boldsymbol{d})\boldsymbol{sin\theta}=\boldsymbol{n\lambda}$. **w**here $(\boldsymbol{e}+\boldsymbol{d})$ is the grating element, *'n'* the order of the maxima and 'λ' the wavelength of the incident light.

1) For a given wavelength 'λ' the angle of diffraction'θ' is different for principal maxima of different orders.

2) For white light and for a particular order *n*, the light of different wavelengths will be diffracted in different directions.

The longer the wavelength, greater is the angle of diffraction. So in each order, we will get the spectra having as many lines as the wavelength in the light source. At centre *(n = 0, zero order)* $\theta=0$ gives the maxima of all wavelengths. So here different wavelengths coincide to form the central image of the same color as that the light source. Similarly, the principal maxima of all wavelengths corresponding to *n = 1* will form the first order spectrum, the principal maxima of all wavelengths corresponding to *n = 2*, will form the second order spectrum and so on.



Important characteristics of grating spectra:

- ✓ Spectra of different orders are situated symmetrically on both sides of *zero order.*
- ✓ Spectral lines are almost straight and quite sharp.
- ✓ Spectral colors are in the order from Violet to Red.
- ✓ Spectral lines are more dispersed as we go to higher orders.
- ✓ Most of the incident intensity goes to *zero order* and rest is distributed among the other orders.

5.1.6 Maximum number of orders formed by a Grating:

The principal maxima in a grating satisfy the condition

$$(e+d)sin\theta = n\lambda$$

Or $n = \dfrac{(e+d)sin\theta}{\lambda}$

The maximum angle of diffraction is 90°, hence the maximum possible order is given by

$$n_{max} = \frac{(e+d)sin90^o}{\lambda} = \frac{(e+d)}{\lambda}$$

Ex: Consider a grating having grating element which is less than twice the wavelength of the incident light,

then $(e+d) < 2\lambda$

$$\therefore n_{max} < \frac{2\lambda}{\lambda} < 2$$

i.e., only the first order is possible.

5.1.7 Absent spectrum with a diffraction grating*:*

We know the intensity maxima at a point due to grating can be observed when that satisfies the maxima for single slit s well as for 'N' no. of slits.

But when the condition for principal maxima due to 'N' no. of slits and the condition for minima due to single slit are satisfied simultaneously for the same angle θ, then the principal maxima due to grating for that particular order will be missed or absent, which is called missing spectra.

We know that, in case of a grating the principal maxima are obtained in the directions given by

$$(e+d)sin\theta = n\lambda \quad n = 0,1,2,3,4 \ldots \ldots$$

Also, in case of a single slit, the minima are obtained in the directions given by

$$esin\theta = m\lambda \quad m = 1,2,3,4 \ldots \ldots ..$$

If both the conditions are satisfied simultaneously for same angle 'ϑ', a particular maximum of order ***n*** will be missing in the grating spectrum.

Dividing above equations we have

$$\frac{(e+d)}{e} = \frac{n}{m}$$

which is the ***condition of absent spectra***.

If the width of the ruling is equal to the width of the slit, $\because e = d \therefore \frac{(e+e)}{e} = 2 \Rightarrow n = 2m$
the second order spectrum will be missed.

5.1.8 Dispersive power of grating

Dispersive power of grating is defined as the ratio of the difference in the angle of diffraction of any two neighboring spectral lines to the difference in the wavelength between the two spectral lines.

It can also be defined as the difference in the angle of diffraction per unit change in wave length. The diffraction grating of the nth order principal maxima for wavelength given by equation

$$(e + d)sin\theta = n\lambda$$

Differentiating this equation, we have , $(e + d)cos\theta \, d\theta = n \, d\lambda$

(e+d) is constant and *'n'* is constant for given order.

So, $\frac{d\theta}{d\lambda}$ is the dispersive power of grating $\frac{d\theta}{d\lambda} = \frac{n}{(e+d)cos\theta}$

So dispersive power of grating $\frac{d\theta}{d\lambda}$ α *n* (order of spectrum)

α $\frac{1}{(e+d)}$

α $\frac{1}{cos\theta}$

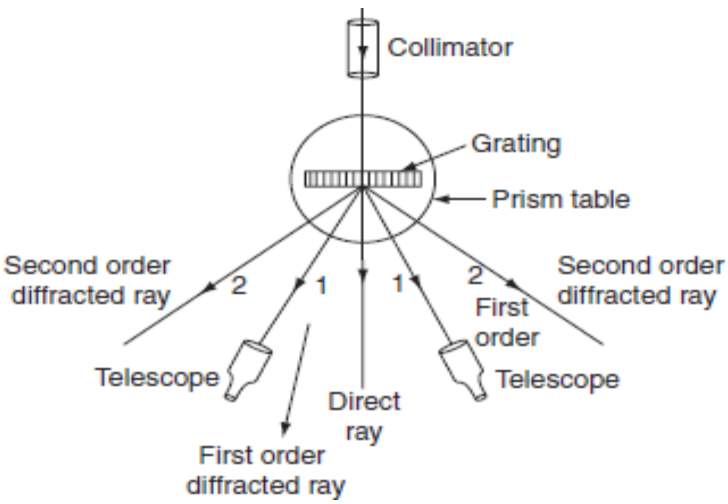
5.1.9 Determination of Wavelength Using Grating

In laboratory, the grating spectrum can be obtained by using a spectrometer.

Adjustments: Before performing the experiment, the following adjustments are made.

(1) The spectrometer is adjusted for parallel rays by ***Schuster's method***.

(2) The grating is adjusted for ***normal incidence***.



Measurement of θ***:*** When a white light is incident on the grating normally, the beam gets dispersed and in each order of the spectrum we can observe constituent wavelengths (i.e., VIBGYOR)

1) The telescope is now turned to get the first order spectrum in the field of view on left.

2) The cross-wire is adjusted on the line whose wavelength is to be determined (say on RED line)

3) Now, the readings of the two verniers are noted.

4) The telescope is then turned to the right side to receive the first order spectrum and repeat steps (2) & (3).

5) The difference between readings of the same vernier gives twice the angle of diffraction θ for that line in first order.

6) By substituting the values of 'ϑ', $(e+d)$ and *n* *in* $(e+d)sin\theta = n\lambda$ we can determine the wavelength of light.

The same procedure from step (1) through (6) is repeated for second order and even in higher orders.

Numerical:

1. In Fraunhoffer diffraction due to a narrow slit, a screen is placed 2 meters away from the lens to obtain the pattern. If the slit width is 0.2 mm and the first minima lie 5 mm on either side of the central maximum, find the wavelength of light.

   Sol: In the Fraunhoffer diffraction due to a single slit of width e, the direction of minima are given by
   $e \sin\theta = \pm m\lambda$

For first minima on either side of central maximum, m =1. Also, when θ is very small sin θ = θ.

Hence, e θ = λ

or, θ = λ/e = λ/0.02

θ is also given by 0.5/200

So, λ/0.02 = 0.5/200

or, λ = 5000 Å

2. A parallel beam of light is normally incident on a plane transmission grating having 4250 lines per cm and a second order spectral line is observed at an angle of 30^0. Calculate the wavelength of light.

Sol: We know that (e +d) sinθ = nλ, n is an integer

Here, (e +d) is the grating element, n is the order of the spectrum

There are 4250 lines per cm. therefore

(e +d) = $\frac{1}{4250}$ cm

Also, n =2, θ = 30^0

So, λ = $\frac{(e+d)\sin\theta}{n}$ = 5882 Å

3. In a grating spectrum, which spectral line in 4th order will overlap with 3rd order line of 5600 Å?

Sol: The grating equation is

(e +d) sinθ = nλ

If nth order of λ_1 coincides with (n+1)th order of λ_2, then

(e +d) sinθ = $n\lambda_1$ = (n+1) λ_2

or, $\lambda_2 = \frac{n\lambda_1}{n+1} = \frac{3\times5600}{4}$ = 4200 Å

4. What is the highest order spectrum which may be seen with light of wavelength 5000 Å by means of a grating with 3000 lines/cm?

Sol: The grating equation is

(e +d) sinθ = nλ

The maximum value of sinθ is 1. Therefore, the highest order visible with grating spectrum is given by

$n_{max} = \frac{e+d}{\lambda} = \frac{1}{3000\times5\times10^{-5}}$ =6.6

So, the highest order, which may be seen, is 6.

5. A plane transmission grating having 5000 lines per cm is being used under normal incidence of light. If the width of opaque parts be double than that of the transparent parts of the grating, then which orders of spectra will be absent?

   Sol: The condition for absent spectra is

   $\frac{e+d}{e} = \frac{n}{m}$ where m = 1,2,3.....

   If d = 2e, then

   n = 3m = 3, 6, 9, ...

   Hence the 3rd, 6th, 9th .....order spectra would be absent.

6. For a grating with grating element 18000 Å, obtain the dispersions in the first order spectrum around wavelength 5000 Å, assuming normal incidence.

   Sol: The expression for angular dispersion is given by

   $\frac{d\theta}{d\lambda} = \frac{n}{(e+d)cos\theta} = \frac{1}{\sqrt{\left(\frac{e+d}{n}\right)^2 - \lambda^2}}$

   $(e+d)$= 18000 Å, λ = 5000 Å, n=1

   Substituting, we get

   $\frac{d\theta}{d\lambda} = 5.78 \times 10^{-5}$ rad/ Å.

7. In a plane transmission grating, the angle of diffraction for the second order principal maxima for wavelength 5000 Å is 30^0. Calculate the number of lines in one cm of the grating surface.

8. A plane transmission grating produces an angular separation of 0.01 radian between two wavelengths observed at an angle of 30^0. If the mean value of the wavelength is 5000 Å and the spectrum is observed in the second order, calculate the difference in the two wavelengths.

Questions: Short & medium answer type

1. Distinguish between Fresnel's and Frounhofer's diffraction.
2. Distinguish between diffraction and interference.
3. Diffraction can occur if slit width is greater than wavelength of light. Justify.
4. Find the approximate expression for the secondary maxima in case of Fraunhofer's single slit diffraction pattern.
5. Explain the meaning of missing spectra in the diffraction pattern of a plane transmission grating. Obtain the expression for missing spectra.
6. In a plane diffraction grating, if the width of opaque space is equal to the slit width, find the orders of missing spectra.
7. Define dispersive power of grating. Write the factors on which it depends.

8. A glass piece and a student grating both are identical in size and shape is given to you. How can you distinguish the students grating from the glass piece without using any optical instrument?
9. Find the position of maxima, minima and secondary maxima in a single slit diffraction pattern.
10. Draw the intensity distribution curve in case of a single slit diffraction.
11. Why the secondary maxima are not visible in case of grating?
12. Write the characteristics of grating spectra.

Long answer type

1. Derive an expression for intensity incase of Fraunhofer's single slit diffraction with a neat diagram.
2. In Fraunhofer's diffraction due to single slit, obtain the conditions for principal maximum, secondary maxima and minima. Find its expression, and then show it graphically.
3. How grating spectra are formed? Write its important characteristics. Explain the meaning of missing spectra in the diffraction pattern of a plane transmission grating. Obtain the expression for missing spectra.
4. Define dispersive power of grating. Obtain the expression for it Write the factors on which it depends. Show that the dispersive power of a grating is $n \big/ \{(a+b)^2 - n^2\lambda^2\}^{\frac{1}{2}}$

   Where $(a+b)$ is the grating element and $'n'$ is the order of the spectrum.

Module – 2: ELECTROMAGNETISM

Scalar and Vector Function:

A scalar function f(x, y, z) specifies a magnitude but independent of directions. A vector function F(x, y, z) specifies a magnitude and a direction at every point (x, y, z) in some region of space. We can picture a vector function as a collection of arrows (Fig. 1), one for each point (x, y, z).

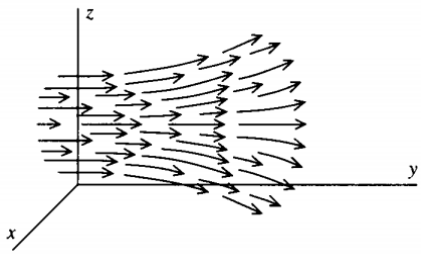


Fig. 1: Vector function

The direction of the arrow at any point is the direction specified by the vector function, and its length is proportional to the magnitude of the function. A vector function, like any vector, can be resolved into components as in fig. 2.

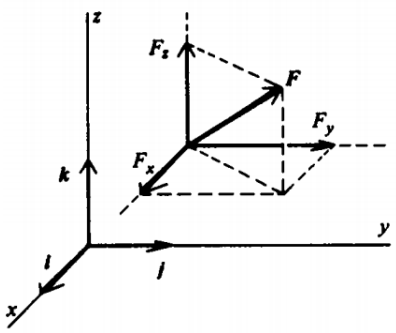


Fig. 2: Vector function components

Letting, **i**, **j** and **k** unit vectors along the x-, y-, and z-axes, respectively, we write

$$\vec{F}\,(x, y, z) = \hat{\imath}F_x + \hat{\jmath}F_y + \hat{k}F_z \qquad (1)$$

F_x, F_y and F_z are themselves scalar functions of x, y and z. The magnitude or the direction (or both) of the vector quantities may change with position and time. The various laws of electromagnetism describe how the different scalar and the vector functions change with position and time. The laws are expressed mathematically in terms of derivatives of the fields with respect to space coordinates and time. The derivatives w.r.t. space coordinates are expressed in terms of *gradient, divergence* and *curl* operators.

Gradient($\vec{\nabla}$):

Suppose, we have a function of three variables—say, the temperature T (x, y,z) in this room. (Start out in one corner, and set up a system of axes; then for each point (x, y,z) in the room, T gives the temperature at that spot.) We want to generalize the notion of "derivative" to functions like T , which depend not on one but on three variables. A derivative is supposed to tell us how fast the function varies, if we move a little distance. But this time the situation is more complicated, because it depends on what direction we move. In fact, the question "how fast does T vary?"

Taking the total derivatives on the temperature (T), we have;

$$dT = \frac{\partial T}{\partial x}dx + \frac{\partial T}{\partial y}dy + \frac{\partial T}{\partial z}dz \qquad (2)$$

This tells us how T changes when we alter all three variables by the infinitesimal amounts dx, dy, dz. Interestingly, the eq. (2) is a reminiscent of the dot product i.e.,

$$dT = \left(\hat{\imath}\frac{\partial T}{\partial x} + \hat{\jmath}\frac{\partial T}{\partial y} + \hat{k}\frac{\partial T}{\partial z}\right)\cdot\left(\hat{\imath}dx + \hat{\jmath}dy + \hat{k}dz\right) \qquad (3)$$

$dT = (\nabla T)\cdot(dr)$ where

$$\boldsymbol{\nabla} T \equiv \hat{\imath}\frac{\partial T}{\partial x} + \hat{\jmath}\frac{\partial T}{\partial y} + \hat{k}\frac{\partial T}{\partial z} \qquad (4)$$

is the **gradient of T**. Note that $\vec{\nabla}T$ is the generalized derivative and is a vector quantity i.e., like any vector, the gradient has magnitude and direction. To determine its geometrical meaning, we can rewrite the total derivative as;

$$dT = (\nabla T)\cdot(dr) = |dT||dr|\cos\theta \qquad (5)$$

If we fix the magnitude |dr| and search around in various directions (that is, vary θ), the maximum change in T evidentally occurs when θ = 0. That is, for a fixed distance |dr|, dT is greatest when I move in the same direction as $\vec{\nabla}T$. Thus,

The gradient points in the direction of maximum increase of the scalar function (f say). In other words, $\vec{\nabla}f$ is normal to the surface of constant f.

Example 1 . Find the gradient of $r = \sqrt{x^2 + y^2 + z^2}$ (the magnitude of the position vector).

Solution

$$\begin{aligned}\nabla r &= \frac{\partial r}{\partial x}\hat{\mathbf{x}} + \frac{\partial r}{\partial y}\hat{\mathbf{y}} + \frac{\partial r}{\partial z}\hat{\mathbf{z}} \\ &= \frac{1}{2}\frac{2x}{\sqrt{x^2+y^2+z^2}}\hat{\mathbf{x}} + \frac{1}{2}\frac{2y}{\sqrt{x^2+y^2+z^2}}\hat{\mathbf{y}} + \frac{1}{2}\frac{2z}{\sqrt{x^2+y^2+z^2}}\hat{\mathbf{z}} \\ &= \frac{x\,\hat{\mathbf{x}} + y\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}}{\sqrt{x^2+y^2+z^2}} = \frac{\mathbf{r}}{r} = \hat{\mathbf{r}}.\end{aligned}$$

Does this make sense? Well, it says that the distance from the origin increases most rapidly in the radial direction, and that its *rate* of increase in that direction is 1. . . just what you'd expect.

Problem 1. Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$.

(b) $f(x, y, z) = x^2y^3z^4$.

The Del (∇) Operator:

The gradient has the formal appearance of a vector, $\vec{\nabla}$, operates a scalar function as seen from the eq (4). Generally, we can represent the **del** operator in vector notation as;

$$\vec{\nabla} = \hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z} \qquad (6)$$

It is also known as the **nabla** operator.

Operation with the Del/Nabla operator:

Just like vector operations, Del operates with scalar and vector functions in three different ways. These are

1. On a scalar function T : ∇T (the gradient);
2. On a vector function $\mathbf{v}$, via the dot product: $\nabla \cdot \mathbf{v}$ (the **divergence**);
3. On a vector function $\mathbf{v}$, via the cross product: $\nabla \times \mathbf{v}$ (the **curl**).

The Divergence ($\vec{\nabla}.$):

From the definition of $\vec{\nabla}$ we construct the divergence:

$$\vec{\nabla}.\vec{f} = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right).\left(\hat{\imath}f_x + \hat{\jmath}f_y + \hat{k}f_z\right)$$

$$= \left(\frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}\right) \qquad (7)$$

Where f_x, f_y and f_z *are the components of the vector function f. The divergence operates on a vector function f yields a* ***scalar quantity****. The divergence of a vector function (F) is, like any dot product of a vector.*

Geometrical Interpretation: The name divergence is well chosen, for $\vec{\nabla} \cdot \vec{f}$ is a measure of how much the vector f spreads out (diverges) from the point in question. For example, the vector function in Fig. 3 has a large (positive) divergence (if the arrows pointed in, it would be a negative divergence).

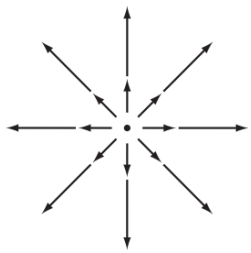


Fig. 3: positive divergence

Example 2: Suppose the functions in Fig. 1.18 are $\mathbf{v}_a = \mathbf{r} = x\,\hat{\mathbf{x}} + y\,\hat{\mathbf{y}} + z\,\hat{\mathbf{z}}$, $\mathbf{v}_b = \hat{\mathbf{z}}$, and $\mathbf{v}_c = z\,\hat{\mathbf{z}}$. Calculate their divergences.

Solution

$$\nabla \cdot \mathbf{v}_a = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3.$$

As anticipated, this function has a positive divergence.

$$\nabla \cdot \mathbf{v}_b = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(1) = 0 + 0 + 0 = 0,$$

as expected.

$$\nabla \cdot \mathbf{v}_c = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(0) + \frac{\partial}{\partial z}(z) = 0 + 0 + 1 = 1.$$

Here, $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ are replaced by $\hat{x}$, $\hat{y}$ and $\hat{z}$ respectively.

Solenoidal Field:

If divergence of a vector field/function (say $\vec{f}$) vanishes ($\vec{\nabla}.\vec{f} = 0$) then the field is known as a solenoidal field.

The Curl $(\vec{\nabla} \times)$:

From the definition of ∇ we construct the curl of a function as;

$$\vec{\nabla} \times \vec{F} = \left(\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right) \times \left(\hat{\imath}F_x + \hat{\jmath}F_y + \hat{k}F_z\right)$$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$\vec{\nabla} \times \vec{F} = \hat{\imath}\left(\frac{\partial F_y}{\partial z} - \frac{\partial F_z}{\partial y}\right) + \hat{\jmath}\left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right) + \hat{k}\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial z}\right) \qquad (8)$$

Notice that the curl of a vector function (F) is, like any cross product of a vector.

Geometrical Interpretation: *The name curl is also well chosen, for $\vec{\nabla} \times \vec{F}$ is a measure of how much the vector F swirls around the point.*

Example:

The curl of $\boldsymbol{F}(x, y, z) = 3x^2\boldsymbol{i} + 2z\boldsymbol{j} - x\boldsymbol{k}$ is:

$$\begin{aligned} \boldsymbol{\nabla} \times \boldsymbol{F} &= \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3x^2 & 2z & -x \end{vmatrix} \\ &= \left(\frac{\partial(-x)}{\partial y} - \frac{\partial(2z)}{\partial z}\right)\boldsymbol{i} - \left(\frac{\partial(-x)}{\partial x} - \frac{\partial(3x^2)}{\partial z}\right)\boldsymbol{j} \\ &\quad + \left(\frac{\partial(2z)}{\partial x} - \frac{\partial(3x^2)}{\partial y}\right)\boldsymbol{k} \\ &= (0-2)\boldsymbol{i} - (-1-0)\boldsymbol{j} + (0-0)\boldsymbol{k} \\ &= -2\boldsymbol{i} + \boldsymbol{j}\,. \end{aligned}$$

Irrotational Field:

The (vector) field (F say) is said to be irrottaional if curl of the field vanishes i.e., $\vec{\nabla} \times \vec{F} = 0$

Operations using the Del:

(i) $$\nabla(fg) = f\nabla g + g\nabla f,$$

(ii) $$\nabla(\mathbf{A}\cdot\mathbf{B}) = \mathbf{A}\times(\nabla\times\mathbf{B}) + \mathbf{B}\times(\nabla\times\mathbf{A}) + (\mathbf{A}\cdot\nabla)\mathbf{B} + (\mathbf{B}\cdot\nabla)\mathbf{A},$$

(iii) $$\nabla\cdot(f\mathbf{A}) = f(\nabla\cdot\mathbf{A}) + \mathbf{A}\cdot(\nabla f),$$

(iv) $$\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B}),$$

(v) $$\nabla\times(f\mathbf{A}) = f(\nabla\times\mathbf{A}) - \mathbf{A}\times(\nabla f),$$

(vi) $$\nabla\times(\mathbf{A}\times\mathbf{B}) = (\mathbf{B}\cdot\nabla)\mathbf{A} - (\mathbf{A}\cdot\nabla)\mathbf{B} + \mathbf{A}(\nabla\cdot\mathbf{B}) - \mathbf{B}(\nabla\cdot\mathbf{A}).$$

(vii) Divergence of gradient of a scalar function (say T):

i.e., $\nabla \cdot (\nabla T) = \nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}$ = ***Laplacian of T***

(viii) Curl of gradient of a scalar function (say T) is always **zero** i.e., $\nabla \times (\nabla T) = 0$

(ix) Gradient of divergence '$\nabla(\nabla \cdot v)$' seldom occurs in physical applications, and it has not been given any special name of its own.

(x) The divergence of a curl, like the curl of a gradient, is always zero:

$$\nabla \cdot (\nabla \times v) = 0$$

(xi) Curl of curl: $\nabla \times (\nabla \times v) = \nabla(\nabla \cdot v) - \nabla^2 v$

(For readers perspective $\vec{\nabla}$ is here replaced by ∇ in the above formulas)

Integrals:

Line, Surface and the Volume Integrals:

In electrodynamics, we encounter several different kinds of integrals, among which the most important are line (or path) integrals, surface integrals (or flux), and volume integrals.

a. Line Integrals of a vector field: A line integral is an expression of the form

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{v} \cdot d\mathbf{l} \qquad (9)$$

where **v** is a vector function, dl is the infinitesimal displacement vector, and the integral is to be carried out along a prescribed path, from point a to point b as shown in the fig. 4.

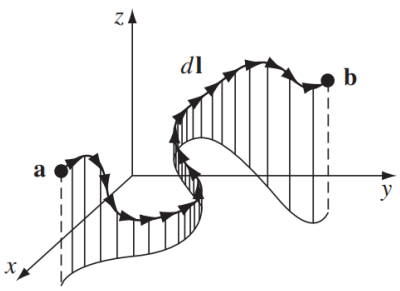


Fig. 4: Line integral between two points a and b.

If the path in question forms a closed loop (that is, if b = a), we shall put a circle on the integral sign then, the line integral becomes;

$$\oint \mathbf{v} \cdot d\mathbf{l}.$$

(10)

Ordinarily, the value of a line integral depends critically on the path taken from a to b, but there is an important special class of vector functions for which the *line integral is independent of path and is determined entirely by the end points.* In this case, the vector field is known as a *conservative field.*

b. Surface Integrals: A surface integral is an expression of the form

$$\int_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}, \tag{11}$$

where **v** is again some vector function, and the integral is over a specified surface (see Fig. 5).

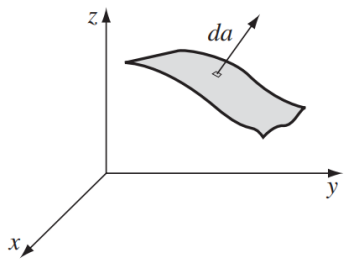


Fig. 5: surface integral

Here **da** is an infinitesimal patch of area, with direction perpendicular to the surface. For a closed surface, the surface integral can be written as;

$$\oint \mathbf{v} \cdot d\mathbf{a},$$

If v describes the flow of a fluid (mass per unit area per unit time) $\oint \mathbf{v} \cdot d\mathbf{a},$ represents the total mass per unit time passing through the surface — hence the alternative name, '**flux**'.

c. Volume Integrals: A volume integral is an expression of the form

(12) $$\int_{\mathcal{V}} T \, d\tau$$

where T is a scalar function and dτ is an infinitesimal volume element. In Cartesian coordinates,

$$d\tau = dx\, dy\, dz$$

For example, if T is the density of a substance (which may vary from point to point), then the volume integral would give the total mass.

Theorems:

Gauss Divergence Theorem: *The volume integral of divergence of a vector field (say **V**) over a given volume is equal to the surface integral of the vector over a closed surface (a) enclosing the volume. Mathematically,*

$$\int_{\mathcal{V}} (\nabla \cdot \mathbf{v})\, d\tau = \oint_{\mathcal{S}} \mathbf{v} \cdot d\mathbf{a}. \qquad (13)$$

Stoke's Theorem: *The surface integral of the curl of a vector field over a given region (say S) is equal to the line integral of the vector along the boundary of the area (P).*

$$\int_{\mathcal{S}} (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} \qquad (14)$$

Maxwell's Equations:

The various laws of electromagnetism were pulled together and were cast into four equations involving time and space derivatives of electric and magnetic fields. These equations are known as Maxwell's electromagnetic equations. The equations are as follows;

1). Gauss' Law in Electrostatics:

Statement: *The total electric flux 'ϕ_E' over a closed surface is equal to $1/\epsilon_0$ times the net charge enclosed by the surface.*

We have, the electric flux (ϕ_E) associated with the charge (as seen from the fig.6) enclosed by a surface's' is given by

$$\phi_E = \frac{q_{enc}}{\epsilon_0} \qquad (15)$$

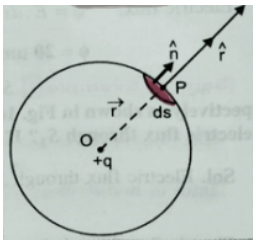


Fig.6: Flux emanating from the elementary surface 'ds'. Here, q_{enc} = +q

where ϵ_0 is permittivity of the free space. Also, the electric flux is defined as the surface integral of the electric field i.e.,

$$\phi_E = \oint_S \vec{E}.\overrightarrow{ds} \qquad (16)$$

Equating both the equations, finally the Gauss' law can be expressed mathematically as;

$$\phi_E = \oint_S \vec{E}.\overrightarrow{ds} = \frac{q_{enc}}{\epsilon_0} \qquad (17)$$

Notes:

- The charge enclosed (q_{enc}) by the surface may be point charges or continuously distributed charges.
- The net enclosed charges may be positive, negative or zero. Accordingly, the net electric flux may be outward, inward or zero.
- The surface enclosing the charges is termed as a Gaussian surface.
- The electric flux doesn't depend on the shape or size of the Gaussian surface as long as the charges are enclosed by the surface.
- Symmetry is crucial w.r.t. application of the Gauss' law.
- The eq. (17) is known as the integral form of the Gauss' law

1.1: Gauss' Law in terms of Electric Displacement (**D**):

We have the electric displacement vector (**D**) in vaccum/free space is defined as;

$$\vec{D} = \epsilon_0 \vec{E}$$

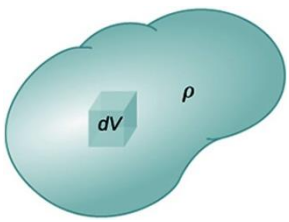
Putting this value in the equation (17), we get

$$\phi_E = \oint_S \vec{D}.\overrightarrow{ds} = q_{enc} \qquad (18)$$

1.2: Gauss' Law in differential form:

Let, ρ is the volume charge distributions (dq/dV). The net enclosed charge by the Gaussian surface will be,

$$\int_V I_{enc} = \rho\, dV \qquad (19)$$



where dV is the elementary volume encloses by the Gaussian surface.

Putting the equation (19) in eq (17), we get;

$$\oint_S \vec{E}.\overrightarrow{ds} = \int_V (\rho\, dV)/\epsilon_0 \qquad (20)$$

Applying the Gauss' divergence theorem to the left hand side of eq.(20), we have

$$\oint_S \vec{E}.\overrightarrow{ds} = \int_V (\vec{\nabla}.\vec{E})dV \qquad (21)$$

Equating eq. (21) and (22), we get

$$\int_V (\vec{\nabla}.\vec{E})dV = \frac{1}{\epsilon_0}\int_V \rho\, dV$$

$$Or \quad \int_V \left(\vec{\nabla}.\vec{E} - \frac{\rho}{\epsilon_0}\right)dV = 0$$

The integrand must vanish, leading to

$$\vec{\nabla}.\vec{E} = \frac{\rho}{\epsilon_0} \qquad (22)$$

This is the *differential form of the Gauss' law*.

In terms of Displacement Vector, eq. (22) can be written as;

$$\vec{\nabla}.\vec{D} = \rho \qquad (23)$$

2. Gauss' law in Magnetostatics:

Statement: *The magnetic flux (ϕ_B) over a closed surface enclosing the volume is always zero.*

Mathematically,

$$\phi_B = \oint_S \vec{B}.\overrightarrow{ds} = 0 \qquad (24)$$

where B, is the magnetic flux density (magnetic flux per unit area) with unit is Weber/m^2. Since the magnetic poles always occur in pairs, isolated magnetic poles don't exist. Therefore, within any macroscopic volume, the net magnetic pole is always zero.

Applying the Gauss' divergence theorem to the equation (24), we have

$$\phi_B = \oint_S \vec{B}.\overrightarrow{ds} = \int_V (\vec{\nabla}.\vec{B})dV = 0$$

$$or \quad (\vec{\nabla}.\vec{B}) = 0 \qquad (25)$$

Equation (25) describes the *differential form of the Gauss' law in Magnetostatics*. *In other words, it shows that non-existence of magnetic monopoles.*

3. Faraday's Law of Electromagnetic Induction:

In 1831 Michael Faraday reported that *A changing magnetic field induces an electric field or simply an emf 'ε'. The magnitude of the emf induced in the loop is equal to the negative rate of change of magnetic flux through the surface enclosed by the loop.*

Mathematically,

$$\varepsilon = -\frac{d\phi_B}{dt} \qquad (28)$$

The *emf* (ε) *is defined as the line integral of the electric field i.e.,*

$$\varepsilon = \oint_C \vec{E}\,.\vec{dl} \qquad (29)$$

Equating eq. (28) and (29) respectively,

$$\oint_C \vec{E}\,.\vec{dl} = -\frac{d\phi_B}{dt}$$

or

$$\oint \vec{E}\,.\vec{dl} = -\frac{d\oint \vec{B}\,.\vec{ds}}{dt} \qquad (\phi_B = \oint \vec{B}.\vec{ds})$$

This is known as the Faraday's law of Electromagnetic Induction.

3.1: Differential form of the Faraday's Law:

Applying Stoke's theorm to the lhs and equating both side, we get

$$\oint (\vec{\nabla} \times \vec{E}).\vec{ds} = -\frac{d\oint \vec{B}\,.\vec{ds}}{dt}$$

or,

$$(\vec{\nabla} \times \vec{E}) = -\frac{d\vec{B}}{dt} \qquad (30)$$

The eq. (30) is known as the *diferential form of the Faraday's Law of Electromagnetic Induction*.

4. Ampere's Circuital Law:

Statement: *The line integral of magnetic field along a closed loop (C) is equal to μ_0 times the net electric current enclosed by the loop.*

$$\oint_C \vec{B} \cdot \vec{dl} = \mu_0 I \qquad (26)$$

Here I = I_{enc}= algebraic sum of the currents enclosed by the loop C (**Amperian loop**).

Let us considered an Amperian loop, which encloses the current I_1, I_2 respectively as shown in fig.7. The Ampere's circuital law becomes

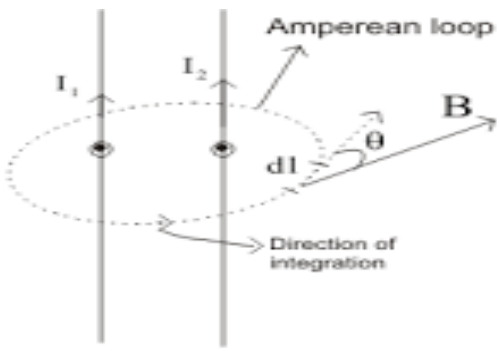


Fig. 7: Amper's loop carrying current in two straight wires.

$$\oint_C \vec{B}.\vec{dl} = \mu_0(I_1 + I_2)$$

Alternatively says, *the integral of magnetic field intensity (H) along an imaginary closed path is equal to the current enclosed by the path.*

$$\oint \vec{B} \cdot \vec{dl} = \mu_0 \, I$$

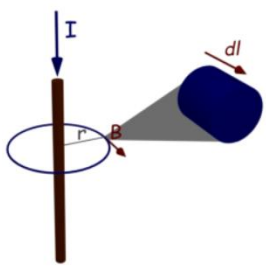
$$\Rightarrow \oint \frac{\vec{B}}{\mu_0} \cdot \vec{dl} = I$$

$$\Rightarrow \oint \vec{H} \cdot \vec{dl} = I$$

$$\left[\because \vec{H} = \frac{\vec{B}}{\mu_0}\right]$$

Example: Use Ampere's circuital law, find the magnitude of the magnetic field **B**, at a point due to a straight, infinitely long conductor carrying a steady current.

Solution: Let us consider an infinitesimal length dl of the Amperian loop at the same point. At each point on the Amperian loop, the value of B is constant since the perpendicular distance of that point from the axis of conductor is fixed, but the direction will be along the tangent on the loop at that point.



Now, according to Ampere's Circuital Law

(**B**, **dl** are in the same direction) $$\oint B \cdot dl = \mu_0 \cdot I$$

Therefore, $$2\pi r\, B = \mu_0\, I$$

or $$\boldsymbol{B} = \mu_0 I/2\pi r$$ (here, **B** indicates a vector quantity)

4.1 Ampere's Law in Differential Form:

We have,

$$\oint B \cdot dl = \mu_0 \cdot I$$

From the current distribution,

$$I = \oint_S \vec{J}.\overrightarrow{ds}$$

where **J** = current density (current per unit area).

$$\oint_S \vec{B}.\overrightarrow{dl} = \mu_0 \oint_S \vec{J}.\overrightarrow{ds}$$

Applying the Stoke's theorm to the lhs,

$$\oint_S (\vec{\nabla} \times \vec{B}).\overrightarrow{ds} = \mu_0 \oint_S \vec{J}.\overrightarrow{ds}$$

Equating both sides, we have

$$(\vec{\nabla} \times \vec{B}) = \mu_0\, \vec{J} \qquad (27)$$

The eq. (27) is known as the **diferential form** of the Ampere's Circuital Law.

5.0: Maxwell's Equations:

5.1: Electrodynamics before Maxwell

So far, we have encountered the following laws, specifying the divergence and curl of electric and magnetic fields:

$$\begin{aligned}
&\text{(i)} \quad \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho && \text{(Gauss's law)},\\
&\text{(ii)} \quad \nabla \cdot \mathbf{B} = 0 && \text{(no name)},\\
&\text{(iii)} \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} && \text{(Faraday's law)},\\
&\text{(iv)} \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} && \text{(Ampère's law)}.
\end{aligned}$$

These equations represent the state of electromagnetic theory, when Maxwell began his work.

5.2: Maxwell's Correction to the Ampere's Law:

Just as *a changing magnetic field induces an electric field (Faraday' law)*, Maxwell proposed that '*a changing electric field induces a magnetic field'.* The real confirmation of Maxwell's theory came in 1888 with Hertz's experiments on electromagnetic waves. Maxwell called his extra term the **displacement current** (**I_d**):

$$I_d = \epsilon_0 A \frac{d\vec{E}}{dt} \qquad (28)$$

The displacement current exists as long as the electric field changes with time. When the electric field attains a constant value (just like the plates of the capacitor gets fully charged), displacement current vanishes and the filed is established by the **conduction current** (I = due to flow of charge particles).

Maxwell introduced this displacement current in the Amper's Circuital Law for a time varying electric field and modified the equation as;

$$\oint \vec{B}.\vec{dl} = \mu_0 \ (I + I_d) \qquad (29)$$

This is known as the Modified Amper's Circuital Law with Maxwell's correction.

The corresponding differential form will be,

$$\left(\vec{\nabla} \times \vec{B}\right) = \mu_0 \left(\vec{J} + \vec{J_d}\right) \qquad (30)$$

Where, $\vec{J_d}$ is known as the **displacement current density (**I_d/A).

or,

$$\left(\vec{\nabla} \times \vec{B}\right) = \mu_0 \left(\vec{J} + \epsilon_0 \frac{d\vec{E}}{dt}\right) \qquad (31)$$

5.3: Distinction between the displacement and the conduction currents.

Displacement Current	*Conduction Current*
The displacement current evolves due to time varying electric field.	*The Conduction current originates from the flow of charges in a conducting medium or in a metal.*
It is a ficticious current which can exist in vaccum or in any medium.	*It is a real current.*
It depends on the electric permitivity of the medium and the rate at which the electric field changes with time.	*It obeys the Ohm's Law and depends on the resistance and potential difference of the conductor.*

Now, the finishing touches on Maxwell's equations are:

$$
\begin{array}{llll}
\text{(i)} & \nabla \cdot \mathbf{E} = \dfrac{1}{\epsilon_0}\rho & & \text{(Gauss's law),} \\
\text{(ii)} & \nabla \cdot \mathbf{B} = 0 & & \text{(no name),} \\
\text{(iii)} & \nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t} & & \text{(Faraday's law),} \\
\text{(iv)} & \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \dfrac{\partial \mathbf{E}}{\partial t} & & \text{(Ampère's law with Maxwell's correction).}
\end{array}
\tag{32}
$$

This is known as the **differential form of Maxwell's Equations**.

The final differential and Integral form of the Maxwell's equations can be qritten as;

Gauss' Law for Electrostatics	$\vec{\nabla} \cdot \vec{E} = \dfrac{\rho}{\epsilon_0}$	$\oiint \vec{E} \cdot \vec{ds} = \dfrac{q_{enc}}{\epsilon_0}$	*Electric flux through a closed surface is proportional to the charged enclosed*
Gauss' Law for Magnetism	$\vec{\nabla} \cdot \vec{B} = 0$	$\oiint \vec{B} \cdot \vec{ds} = 0$	*No magnetic monopole; The total magnetic flux through a closed surface is zero*
Faraday's Law of Induction	$\vec{\nabla} \times \vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot \vec{dl} = -\dfrac{\partial \oiint \vec{B} \cdot \vec{ds}}{\partial t}$	*Changing magnetic flux gives emf.*
Ampere's Law	$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \dfrac{\partial \vec{E}}{\partial t}$	$\oint \vec{B} \cdot \vec{dl} = \mu_0 \oiint \left(\vec{J} + \epsilon_0 \dfrac{\partial \vec{E}}{\partial t} \right) \cdot \vec{dS}$	*Electric current and changing electric flux produces a magnetic field*

For a charge and current free medium (**free space**), the Maxwell's equations become;

$$\nabla \cdot \mathbf{E} = 0, \qquad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$
$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \tag{33}$$

Corresponding the **Integral forms** are;

$$\oint \underline{E} \bullet \underline{dA} = 0$$
$$\oint \underline{B} \bullet \underline{dA} = 0$$
$$\oint \underline{E} \bullet \underline{dl} = -\frac{d\Phi_B}{dt} \tag{34}$$
$$\oint \underline{B} \bullet \underline{dl} = \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

5.4: Physical Significance of Maxwell's Equations:

- *Maxwell's equations incorporate all the laws of electromagnetism, which were developed from experimental observations and were expressed in the form of various emprical laws.*

- *Maxwell's equations lead to the existence of electromagnetic waves, which has been amply confirmed by experimental observations. These equations are consistence with all the observed properties of e.m. waves.*

- *Maxwell's equations are consistence with the special theory of relativity.*

- *Maxwell's equations are used to describe the classical electromagnetic field as well as the quatum theory of interaction of charged particles with electromagnetic field.*

- *Maxwell's equations provided a unified description of the electric and magnetic phenomena which were treated independely.*

Electromagnetic Waves

6: Maxwell's Equations in Free space/vacuum:

In free space, the charge density ρ and the current density $\vec{J}$ are zero. So the four Maxwell's equations in free space are

(i) $\vec{\nabla}.\vec{E} = 0$

(ii) $\vec{\nabla}.\vec{B} = \vec{\nabla}.\vec{H} = 0$

(iii) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(iv) $\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$

(with ρ and σ are zero, $\vec{J} = \sigma\vec{E} = 0$, $\vec{D} = \varepsilon_0\vec{E}$, and $\vec{B} = \mu_0\vec{H}$)

6.1: Electromagnetic Wave Equations in terms of $\vec{E}$

From Maxwell's 3rd equation

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Taking curl on both the sides, we can write

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B}) = -\mu_0 \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{H})$$

Applying Maxwell's 4th equation to the right hand side of the above equation we get;

Or, $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\varepsilon_0\mu_0 \frac{\partial^2 \vec{E}}{\partial t^2}$ .....(1)

But, $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E}$

Now, equation (1) can be written as

$\vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E} = -\varepsilon_0\mu_0 \frac{\partial^2 \vec{E}}{\partial t^2}$ .....(2)

From, Maxwell's 1st equation, $\vec{\nabla}.\vec{E} = 0$

Equation (2) can be written as

$$-\nabla^2\vec{E} = -\varepsilon_0\mu_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

Or, $\boldsymbol{\nabla^2\vec{E} = \varepsilon_0\mu_0 \frac{\partial^2 \vec{E}}{\partial t^2}}$.....(3)

Equation (3) is the required electromagnetic wave equation in free space, in terms of electric vector, $\vec{E}$ (2nd order differential wave equation).

6.2 Electromagnetic Wave Equations in terms of $\vec{H}$

We have Maxwell's 4th equation

$$\vec{\nabla} \times \vec{H} = -\frac{\partial \vec{D}}{\partial t}$$

Taking curl on both the sides, we can write

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{D}) = -\varepsilon_0 \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{E})$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{H}) = -\varepsilon_0 \mu_0 \frac{\partial^2 \vec{H}}{\partial t^2} \quad \text{.....(4)}$$

(Applying Maxwell's 3rd equation)

Or, $\vec{\nabla}(\vec{\nabla}.\vec{H}) - \nabla^2 \vec{H} = -\varepsilon_0 \mu_0 \frac{\partial^2 \vec{H}}{\partial t^2}$

Or, $-\nabla^2 \vec{H} = -\varepsilon_0 \mu_0 \frac{\partial^2 \vec{H}}{\partial t^2}$ (Applying Maxwell's 2nd equation)

Or, $\boldsymbol{\nabla^2 \vec{H} = \varepsilon_0 \mu_0 \frac{\partial^2 \vec{H}}{\partial t^2}}$ .....(5)

*Equation (5) is the required **electromagnetic** wave equation in free space, in terms of magnetic vector, $\vec{H}$ (2nd order differential wave equation).*

6.3 Velocity of electromagnetic wave

The general wave equation (2nd order differential wave equation) is

$$\nabla^2 \vec{\Psi} = \frac{1}{v^2} \frac{\partial^2 \vec{\Psi}}{\partial t^2}$$

Comparing this equation with e.m. wave equation (equation 3 or 5), we can write the expression for velocity, v

Hence, $v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \cong 2.99 \times 10^8$ m/s $\cong c$ (velocity of light)

7.0 Maxwell's equations in a dielectric medium:

Maxwell's equations in a dielectric medium are

(i) $\vec{\nabla}.\vec{E} = 0$

(ii) $\vec{\nabla}.\vec{B} = \vec{\nabla}.\vec{H} = 0$

(iii) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(iv) $\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$

(with ρ and $\vec{J}$ are zero, $\vec{D} = \varepsilon \vec{E}$, and $\vec{B} = \mu \vec{H}$)

7.1 Electromagnetic Wave Equations in terms of $\vec{E}$:

From Maxwell's 3rd equation

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Taking curl on both the sides, we can write

$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B}) = -\mu \frac{\partial}{\partial t}(\vec{\nabla} \times \vec{H})$

Or, $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$ .....(6) (Applying Maxwell's 4th equation)

But, $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E}$

Now, equation (1) can be written as

$\vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E} = -\varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$ ..... (7)

From, Maxwell's 1st equation, $\vec{\nabla}.\vec{E} = 0$

Equation (2) can be written as

$-\nabla^2\vec{E} = -\varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}$

Or, $\boldsymbol{\nabla^2\vec{E} = \varepsilon\mu \frac{\partial^2 \vec{E}}{\partial t^2}}$..... (8)

Equation (8) is the required electromagnetic wave equation in a dielectric medium, in terms of electric vector, $\vec{E}$ (2nd order differential wave equation).

Similarly, we can derive for magnetic component $\vec{H}$ and is represented by

$\boldsymbol{\nabla^2\vec{H} = \varepsilon\mu \frac{\partial^2 \vec{H}}{\partial t^2}}$..... (9)

7.2 Velocity of electromagnetic wave:

The general wave equation (2nd order differential wave equation) is

$$\nabla^2\vec{\Psi} = \frac{1}{v^2}\frac{\partial^2\vec{\Psi}}{\partial t^2}$$

Comparing this equation with em wave equation (equation 8 or 9), we can write the expression for velocity, v

$$v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{1}{\sqrt{\varepsilon_0\varepsilon_r\mu_0\mu_r}} = \frac{c}{\sqrt{\varepsilon_r\mu_r}}$$

If μ_r =1 (non-magnetic medium), then

$$\boldsymbol{v = \frac{c}{\sqrt{\varepsilon_r}} = \frac{c}{n}}$$

n is refractive index of the medium

So, $n = \sqrt{\boldsymbol{\varepsilon_r}} = \sqrt{k}$ (k is dielectric constant of the medium)

8.0 Transverse nature of electromagnetic wave:

Suppose the em wave is propagating in any arbitrary direction defined by the propagation vector k. The electric and magnetic vectors are represented by the equations

$\vec{E}\,(r,t) = \vec{E}_0\, e^{i(wt-k.r)}$
$\vec{H}\,(r,t) = \vec{H}_0\, e^{i(wt-k.r)}$
Here, the vectors E_0 and H_0 are constant in time, k is propagation vector whose magnitude is equal to $\frac{2\pi}{\lambda}$ and its direction is the direction of propagation of wave and $i = \sqrt{-1}$.

8.1 Transverse nature of $\vec{E}$

From Maxwell's 1st equation (in free space)

$\vec{\nabla}.\vec{E} = 0$

or, $\vec{\nabla}.\vec{E}_0\, e^{i(wt-k.r)} = 0$

or, $\left[\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right].\left[\hat{\imath}E_{0x}\, e^{i(wt-k.r)} + \hat{\jmath}E_{0y}\, e^{i(wt-k.r)} + \hat{k}E_{0z}\, e^{i(wt-k.r)}\right] = 0$

or, $iE_{0x}\, e^{i(wt-k.r)}k_x + iE_{0y}\, e^{i(wt-k.r)}k_y + iE_{0z}\, e^{i(wt-k.r)}k_z = 0$

or, $i\left(E_xk_x + E_yk_y + E_zk_z\right) = 0$

or, $\vec{E}.\vec{k} = 0$

Two vectors are perpendicular to each other. That means the electric vector $\vec{E}$ is perpendicular to the direction of propagation.

8.2 Transverse nature of $\vec{H}$

From Maxwell's 2nd equation

$\vec{\nabla}.\vec{H} = 0$

or, or, $\vec{\nabla}.\vec{H}_0\, e^{i(wt-k.r)} = 0$

or, $\left[\hat{\imath}\frac{\partial}{\partial x} + \hat{\jmath}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right].\left[\hat{\imath}H_{0x}\, e^{i(wt-k.r)} + \hat{\jmath}H_{0y}\, e^{i(wt-k.r)} + \hat{k}H_{0z}\, e^{i(wt-k.r)}\right] = 0$

or, $iH_{0x}\, e^{i(wt-k.r)}k_x + iH_{0y}\, e^{i(wt-k.r)}k_y + iH_{0z}\, e^{i(wt-k.r)}k_z = 0$

or, $i\left(H_xk_x + H_yk_y + H_zk_z\right) = 0$

or, $\vec{H}.\vec{k} = 0$

Two vectors are perpendicular to each other. That means the magnetic vector $\vec{H}$ is perpendicular to the direction of propagation.

8.3 Relative orientation of vector $\vec{E}$ and $\vec{H}$

From Maxwell's 3 rd equation,

$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$

or, $\hat{\imath}\left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) + \hat{k}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right) = -\mu \frac{\partial \vec{H}}{\partial t}$

or, $\hat{\imath}\left[iE_{0z}\, e^{i(wt-k.r)} k_y - iE_{0y}\, e^{i(wt-k.r)} k_z\right] + \hat{\jmath}\left[iE_{0x}\, e^{i(wt-k.r)} k_z - iE_{0z}\, e^{i(wt-k.r)} k_x\right] + \hat{k}\left[iE_{0y}\, e^{i(wt-k.r)} k_x - iE_{0x}\, e^{i(wt-k.r)} k_y\right] = -\mu \frac{\partial \vec{H}_0\, e^{i(wt-k.r)}}{\partial t}$

or, $i\left[\hat{\imath}\left(E_z k_y - E_y k_z\right) + \hat{\jmath}(E_x k_z - E_z k_x) + \hat{k}\left(E_y k_x - E_x k_y\right)\right] = -\mu\, i\,(-w)\, \vec{H}_0\, e^{i(wt-k.r)}$

or, $\vec{\boldsymbol{k}} \times \vec{\boldsymbol{E}} = \mu \boldsymbol{w} \vec{\boldsymbol{H}}$ .....(10)

(using the relation of cross product of two vectors)

The equation (10) implies that the vector $\vec{H}$ is perpendicular to both the vectors $\vec{k}$ and $\vec{E}$.

8.4. Similarly, from Maxwell's 4th equation in a source free medium, we have

$\vec{\nabla} \times \vec{H} = -\frac{\partial \vec{D}}{\partial t}$

or, $\hat{\imath}\left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}\right) + \hat{\jmath}\left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}\right) + \hat{k}\left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}\right) = \varepsilon \frac{\partial \vec{E}}{\partial t}$

or, $\hat{\imath}\left[iH_{0z}\, e^{i(wt-k.r)} k_y - iH_{0y}\, e^{i(wt-k.r)} k_z\right] + \hat{\jmath}\left[iH_{0x}\, e^{i(wt-k.r)} k_z - iH_{0z}\, e^{i(wt-k.r)} k_x\right] + \hat{k}\left[iH_{0y}\, e^{i(wt-k.r)} k_x - iH_{0x}\, e^{i(wt-k.r)} k_y\right] = \varepsilon \frac{\partial \vec{E}_0\, e^{i(wt-k.r)}}{\partial t}$

or, $i\left[\hat{\imath}\left(H_z k_y - H_y k_z\right) + \hat{\jmath}(H_x k_z - H_z k_x) + \hat{k}\left(H_y k_x - H_x k_y\right)\right] = \varepsilon i\,(-w)\, \vec{E}_0\, e^{i(wt-k.r)}$

or, $\vec{\boldsymbol{H}} \times \vec{\boldsymbol{k}} = \boldsymbol{\varepsilon w} \vec{\boldsymbol{E}}$.....(11)

The equation (11) implies that the vector $\vec{E}$ is perpendicular to both the vectors $\vec{k}$ and $\vec{H}$

From equation (10) and (11), it is clear that the electric vector $\vec{E}$, magnetic vector $\vec{H}$ and propagation vector $\vec{k}$ are mutually perpendicular to each other. They form a right hand orthogonal set. So, if the em wave propagates along z direction, then the electric vector, $\vec{E}$ and magnetic vector, $\vec{H}$ are in x or y direction. If the electric vector is along x direction then the magnetic component is along y direction. The propagation of em wave in z direction is shown in the figure 8.

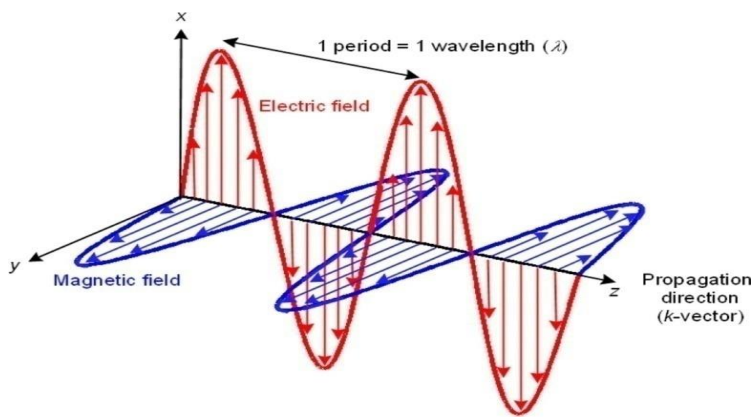


Fig.8: Propagation of em wave in z direction.

Further, the amplitudes of $\vec{E}$ and $\vec{B}$ are also related.

From, equation (10), we have

$\vec{k} \times \vec{E} = \mu w \vec{H}$

If $\vec{E}$ is along x axis then $\vec{H}$ is along y axis (as the wave is propagating along z axis). Above equation can be written as

$\hat{k}k \times \hat{\imath}\, E_x = \hat{\jmath}\mu w H_y$

or, $kE_x = wB_y$

or, $\frac{E_x}{B_y} = \frac{w}{k} = \mathbf{v}$ (velocity of the em wave)

9.0 Poynting's Theorem (The Energy Conservation Theorem):

Energy may be transported through space by means of electromagnetic waves. This energy must be conserved. The conservation of energy in electromagnetic waves is given by Poynting's theorem. Assume that the space is linear, isotropic, homogeneous and characterized by permeability μ, permittivity ε, and conductivity σ.

Maxwell's equations are

(i) $\vec{\nabla}.\vec{E} = 0$

(ii) $\vec{\nabla}.\vec{B} = \vec{\nabla}.\vec{H} = 0$

(iii) $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

(iv) $\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$

Taking the scalar product (dot product) of $\vec{H}$ to the both sides of Maxwell's 3rd equation, we have

$\vec{H}.(\vec{\nabla} \times \vec{E}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t}$.....(12)

Taking the scalar product of $\vec{E}$ to the both sides of Maxwell's 4th equation, we have

$\vec{E}.(\vec{\nabla} \times \vec{H}) = \vec{E}.\vec{J} + \vec{E}.\frac{\partial \vec{D}}{\partial t}$.....(13)

Subtracting equation (13) from equation (12), we get

$$\vec{H}.(\vec{\nabla} \times \vec{E}) - \vec{E}.(\vec{\nabla} \times \vec{H}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E}.\vec{J} - \vec{E}.\frac{\partial \vec{D}}{\partial t}$$

or, $\vec{\nabla}.(\vec{E} \times \vec{H}) = -\vec{H} \cdot \frac{\partial(\mu\vec{H})}{\partial t} - \vec{E}.\frac{\partial(\varepsilon\vec{D})}{\partial t} - \vec{E}.\vec{J}$ (using the vector relation)

$$= -\frac{1}{2}\frac{\partial(\mu\vec{H}^2)}{\partial t} - \frac{1}{2}\frac{\partial(\varepsilon\vec{E}^2)}{\partial t} - \vec{E}.\vec{J}$$

$$= -\frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right] - \vec{E}.\vec{J}$$

or, $-\vec{\nabla}.(\vec{E}\times\vec{H}) = \frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right] + \vec{E}.\vec{J}$ .....(14)

Integrating both sides of above equation over the volume V enclosed by the surface S, we get

$$-\oint_V \vec{\nabla}.(\vec{E}\times\vec{H})dv = \oint_V \frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right]dv + \oint_V \vec{E}.\vec{J}\,dv$$

Applying Gauss divergence theorem to the LHS of the above expression, we get

$$-\oint_S (\vec{E}\times\vec{H}).\vec{ds} = \oint_V \frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right]dv + \oint_V \vec{E}.\vec{J}\,dv \text{.....} \quad (15)$$

Interpretation of RHS of equation (15)

(i) First term $\oint_V \frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right]dv$

$\frac{1}{2}\vec{B}.\vec{H} = \frac{1}{2}\mu H^2$ is the magnetic energy density

$\frac{1}{2}\vec{D}.\vec{E} = \frac{1}{2}\varepsilon E^2$ is the magnetic energy density

So, $\frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right]$ is the time rate of increase of electromagnetic energy density in the space and $\oint_V \frac{\partial}{\partial t}\left[\frac{1}{2}\vec{B}.\vec{H} + \frac{1}{2}\vec{D}.\vec{E}\right]dv$ represents the rate of increase of electromagnetic energy enclosed in volume V.

(ii) Second term $\oint_V \vec{E}.\vec{J}\,dv$

Let dw be the work done by the electromagnetic forces in displacing a charge q through a distance dl, then

$$dw = \vec{F}.\vec{dl} = q\left[\vec{E} + (\vec{v}\times\vec{B})\right].\vec{dl} = q\left[\vec{E} + (\vec{v}\times\vec{B})\right].\vec{v}dt$$

$$= q\vec{E}.\vec{v}dt \qquad (\text{as } \vec{v}\times\vec{v} = 0)$$

The rate at which work is done per unit volume, is given by

$$\frac{dw}{dt} = \frac{q\vec{E}.\vec{v}}{V} = \rho\vec{E}.\vec{v} = \vec{J}.\vec{E}$$

Now, the rate at which work is done on all charges in a volume V

$$\frac{dw}{dt} = \oint_V \vec{E}.\vec{J}\,dv$$

Therefore, $\oint_V \vec{E}.\vec{J}\,dv$ represents the power expended by the em field for the motion of charge carriers.

Since, the RHS of equation (15) represents the sum of the time rate of increase of stored electromagnetic energy in fields and the power expended by the field for the motion of charge carriers, hence LHS must represent the power flow into the volume V through the surface S or power flow out of the volume V across the surface S. This is called Poynting's theorem. It is the law of conservation of energy in the em field.

The vector $\vec{E} \times \vec{H}$ is called Poynting vector $\vec{P}$. It is the power density associated with the field. *The direction of Poynting vector is along the direction of wave propagation (perpendicular to both* $\vec{E}$ and $\vec{H}$).

Dimension of $\vec{P}$

$[P] = [E][H]$

but $E = \frac{F}{q} = \frac{[MLT^{-2}]}{[AT]} = [MLT^{-3}A^{-1}]$

and H = $\frac{[Idl\, sin\theta]}{[r^2]} = [AL^{-1}]$

Now, $[P] = [MLT^{-3}A^{-1}][AL^{-1}] = [MT^{-3}] = \frac{[ML^2T^{-2}]}{[L^2T]}$

or, $[P] = \frac{Energy}{Area \times time} = \frac{Power}{Area}$ = power density

So, dimension of Poynting vector is same as power density.

Q.1 Show that energy is equally shared between the electric component and the magnetic component?

Ans.: We know that the electric energy density in vacuum is

$\frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 (cB)^2$ (as $\frac{E}{B} = c$)

$= \frac{1}{2}\varepsilon_0 (c\mu_0 H)^2 = \frac{1}{2}\mu_0 H^2$ (as $c = \frac{1}{\sqrt{\varepsilon_0\mu_0}}$)

= it is magnetic energy density (proved)

Q.2 Find the average value of Poynting vector over one cycle, also called intensity.

$\langle P \rangle = \langle E\, H \rangle = \langle E\, \frac{B}{\mu_0} \rangle = \langle E\, \frac{E}{c\mu_0} \rangle = \frac{1}{2}\frac{E_0^2}{c\mu_0}$ [as $\langle sin^2\theta \rangle = \frac{1}{2}$, in one cycle]

or, I = $\langle P \rangle = \frac{1}{2}\varepsilon_0 c E_0^2$

Q.3 Show that the average value of Poynting vector for a plane em wave is $\frac{1}{2}\sqrt{\frac{\mu_0}{\varepsilon_0}}\, H_0^2$

$\langle P\rangle = \langle E\ H\rangle = \langle Bc\ \frac{B}{\mu_0}\rangle = \frac{1}{2}\frac{c}{\mu_0}\boldsymbol{B_0^2} = \frac{1}{2}\frac{1}{\mu_0}\frac{1}{\sqrt{\varepsilon_0\mu_0}}\mu_0^2 H_0^2 = \frac{1}{2}\ \sqrt{\frac{\mu_0}{\varepsilon_0}}\ H_0^2$ (proved)

Q.4 Show that Poynting's theorem is according to conservation of energy.

10.0 Propagation of electromagnetic wave through conducting media

Consider the propagation of electromagnetic waves through conducting, linear, isotropic and homogeneous medium characterized by permeability μ, permittivity ε and conductivity σ.

Four Maxwell's equations are

(i) $\vec{\nabla}.\vec{E} = 0$

(ii) $\vec{\nabla}.\vec{B} = \vec{\nabla}.\vec{H} = 0$

(iii) $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$

(iv) $\vec{\nabla}\times\vec{H} = \vec{J} + \frac{\partial\vec{D}}{\partial t}$

With $\rho = 0$ (absence of free charges), $\vec{D} = \varepsilon\vec{E}$, $\vec{B} = \mu\vec{H}$ and $\vec{J} = \sigma\vec{E}$

From Maxwell's 3rd equation

$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$

Taking curl on both the sides, we can write

$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B}) = -\mu\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{H})$

Applying Maxwell's 4th equation

Or, $\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = -\varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2} - \mu\sigma\frac{\partial\vec{E}}{\partial t}$ .....(16)

But, $\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E}$

Now, equation (16) can be written as

$\vec{\nabla}(\vec{\nabla}.\vec{E}) - \nabla^2\vec{E} = -\varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2} - \mu\sigma\frac{\partial\vec{E}}{\partial t}$ .....(17)

From, Maxwell's 1st equation, $\vec{\nabla}.\vec{E} = 0$

Equation (17) can be written as

$-\nabla^2\vec{E} = -\varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2} - \mu\sigma\frac{\partial\vec{E}}{\partial t}$

or, $\boldsymbol{\nabla^2\vec{E} = \varepsilon\mu\frac{\partial^2\vec{E}}{\partial t^2} + \mu\sigma\frac{\partial\vec{E}}{\partial t}}$.....(18)

Equation (18) is the required electromagnetic wave equation, in a conducting medium, in terms of electric vector, $\vec{E}$.

Similarly, in terms of magnetic vector, we can write

$$\nabla^2\vec{H} = \varepsilon\mu\frac{\partial^2\vec{H}}{\partial t^2} + \mu\sigma\frac{\partial\vec{H}}{\partial t}$$.....(19)

Here, $\mu\sigma\frac{\partial\vec{E}}{\partial t}$ and $\mu\sigma\frac{\partial\vec{H}}{\partial t}$ are the dissipative terms.

Let us assume that the field varies as $e^{i(k.r-wt)}$, then the solution of equation (18) may be expressed as $\vec{E}\ (r,t) = \vec{E}_0\ e^{i(k.r-wt)}$

Substituting the above expression in equation (18), we get

$(-k^2 + \varepsilon\mu\omega^2 + i\mu\sigma\omega)\ \vec{E} = 0$

or, $k^2 = \varepsilon\mu\omega^2\left(1 + \frac{i\sigma}{\omega\varepsilon}\right)$.....(20)

k is complex and can be written as

$$k = \alpha + i\beta$$

or, $k^2 = \alpha^2 - \beta^2 + 2i\alpha\beta$.....(21)

Comparing equation (20) and (21), we get

$\alpha^2 - \beta^2 = \varepsilon\mu\omega^2$

$2\alpha\beta = \mu\sigma\omega$

Solving, we get

$$\alpha = \omega\sqrt{\mu\sigma}\left[\frac{1+\sqrt{1+\left(\frac{\sigma}{\mu\omega}\right)}}{2}\right]^{1/2}$$

And $\beta = \omega\sqrt{\mu\sigma}\left[\frac{-1+\sqrt{1+\left(\frac{\sigma}{\mu\omega}\right)}}{2}\right]^{1/2}$

In terms of α and β, $\vec{E}$ can be written as

$\vec{E}\ (r,t) = \vec{E}_0\ e^{i(\alpha+i\beta)\hat{n}.r-iwt} = \vec{E}_0\ e^{-\beta\hat{n}.r}\ \ e^{i(\alpha\hat{n}.r-wt)}$.....(22)

From equation (22), it is clear that the field amplitude is spatially attenuated due to the presence of the term $e^{-\beta\hat{n}.r}$.

Here, β is called attenuation coefficient.

For good conductor,

$\frac{\sigma}{\mu\omega} \gg 1$, then

$\alpha = \beta = \sqrt{\frac{\mu\omega\sigma}{2}}$.....(23)

If $\frac{\sigma}{\mu\omega} \ll 1$, then the medium is called as dielectric.

Skin depth or Penetration depth (d)

The distance, at which, amplitude of electric field of the wave, decreases to $\frac{1}{e}$ times of its initial value is called skin depth.

d = $\frac{1}{\beta} = \sqrt{\frac{2}{\mu\omega\sigma}} = \sqrt{\frac{1}{\mu\pi f\sigma}}$.....(24)

Skin depth measures, how far the wave penetrates into the conductor.

NUMERICAL

1. A nonmagnetic medium is characterized by relative permittivity ε_r = 80. Calculate the speed of electromagnetic wave in the medium and refractive index of the medium.

Sol : $v = \frac{1}{\sqrt{\varepsilon\mu}} = \frac{1}{\sqrt{\varepsilon_0\varepsilon_r\mu_0\mu_r}} = \frac{c}{\sqrt{\varepsilon_r\mu_r}}$

For a nonmagnetic medium, μ_r =1, then

$v = \frac{c}{\sqrt{\varepsilon_r}} = \frac{3\times10^8}{\sqrt{80}}$ = 3.35 $\times$ 10^7 m/s

The refractive index of the medium is given by n = $\sqrt{\varepsilon_r\mu_r} = \sqrt{80}$ = 8.94.

2. Calculate the value of Poynting's vector on the surface of the sun if the power radiated by sun is 4 $\times$ 10^{26} watts and its radius is 7 $\times$ 10^8 meter.

Sol: Poynting's vector (P) is the power passing through unit area (intensity).

So, P = $\frac{Power}{Surface\,area} = \frac{4\times10^{26}}{4\pi r^2} = \frac{4\times10^{26}}{616\times10^{16}} = 6.5 \times 10^7 \frac{watts}{meter^2}$

3. Consider a 100 W monochromatic point source that radiates equally in all directions in free space and is monitored at a distance of 1m. Evaluate the amplitude of electric field at the point of detection.

Sol: $\langle P\rangle = \frac{Power}{Surface\,area} = \frac{100}{4\pi r^2} = \frac{100}{4\times3.17\times1^2}$ = 7.88 W/m^2

Again, $\langle P\rangle = \frac{1}{2}\varepsilon_0 c E_0^2$ = 7.88 W/m^2

or, $E_0^2 = \frac{2\times7.88}{\varepsilon_0 c}$ = 5997 V^2/m^2

or, E_0 = 77.44 V/m

4. A laser beam has a diameter of 2mm. What is the amplitude of the electric and magnetic field in the beam in vacuum if the power of the laser is 1.5 mW?

Sol: Here, diameter d = 2mm, so radius r = 1 mm = 0.001 m

P = $\frac{Power}{Area} = \frac{1.5 \times 10^{-3}}{\pi r^2} = \frac{1.5 \times 10^{-3}}{3.17 \times (0.001)^2}$ = 477.7 W/m^2

But, P is also equal to $\frac{1}{2}\ \varepsilon_0 c E_0^2$. Therefore,

$\frac{1}{2}\ \varepsilon_0 c E_0^2$ = 477.7 W/m^2

Solving, we get

E_0 = 600 V/m

$H_0 = \frac{E_0}{c\mu_0}$ = 1.593 amp-turn/m

5. Calculate the skin depth for radio waves of 3 m wavelength (in free space) in copper, the electrical conductivity of which is 6 × 10^7 S/m.

Sol: Skin depth, d = $\sqrt{\frac{2}{\mu\omega\sigma}}$

Here, $\omega = 2\pi f = \frac{2\pi c}{\lambda} = 2\pi \times 10^8$ rad/s

Now, skin depth, d = $\sqrt{\frac{2}{\mu\omega\sigma}} = \sqrt{\frac{2}{4\pi \times 10^7 \times 2\pi \times 10^8\ 6 \times 10^7}} = 6.5 \times 10^{-6}$ m.

Practice Questions:

Short and Medium Questions

1. Evaluate the divergence of $\vec{A} = 2xy\,\hat{i} + y^2\,\hat{j} - 2yz\,\hat{k}$ at the point (1, 0, 2).
2. Evaluate $\vec{\nabla} \times \vec{r}$, where $\vec{r}$ the position vector is.
3. Evaluate Curl of the vector field $\vec{A} = 2xy\,\hat{i} + y^2\,\hat{j} - 3xyz\,\hat{k}$.
4. Evaluate divergence of the vector field $\vec{F} = xy\,\hat{i} + y\,\hat{j} - 2xyz\,\hat{k}$ at (1, 0, 1).
5. Evaluate gradient of the scalar function f(x, y, z) = 2 xyz + 4 y^{2}z + xz^3 at (1, 2, 3).
6. Write the value of Curl (grad f), where f = x^2 – 2y^2 + 3z^2.
7. If $\phi = 4xy^2 + 2y - z^2$, then find gradient of the above function at (1,2,3).
8. If $\vec{A} = 2x^3 z\hat{\imath} + 3xy^2\hat{\jmath} + 6xyz\ \hat{k}$, find div(curl $\vec{A}$) at (2 , 2, 1).

9. Find the value of 'a' for which the vector field $\vec{v} = ax\hat{i} + 3y\hat{j} - z\hat{k}$ is solenoidal.

10. State the significance of gradient of a scalar function.

11. State the significance of divergence of a vector function.

12. State the significance of curl of a vector function.

13. Show that the position vector $\vec{r}$ is an irrotational vector.

14. State Gauss divergence theorem in vector calculus.

15. Write the mathematical form of Stokes theorem in vector calculus.

16. Show that the surface integral of the position vector over a closed surface is equal to three times the volume enclosed by the surface.

17. Using Gauss divergence theorem prove that the volume of a sphere of radius r is 4/3 πr^3.

18. An electron and a proton is placed inside a sphere of radius 15 cm. Calculate the net flux coming out of surface of sphere.

19. Write the relation between electric displacement vector, electric field intensity and electric polarization vector.

20. Write the S.I unit of electric displacement vector.

21. State the Maxwell's equation, which supports the concept of non existence of magnetic monopole.

22. Write Maxwell's electromagnetic equation in differential form which follows from the concept of displacement current.

23. Distinguish between displacement current and conduction current.

24. The electric field between two parallel metal plates of area $1cm^2$ changes at the rate of 1.2×10^8 v/ms. Calculate the displacement current.

25. The electromagnetic wave is propagating in free space with electric vector

$E(x, t) = e^{i(wt - kx)}\,\hat{k}$ volt/m. Deduce the magnetic vector.

26. A non magnetic medium is characterized by relative permittivity 80. Calculate the speed of electromagnetic wave in that medium.

27. Write the S.I unit and direction of Poynting vector.

28. One of the Maxwell's electromagnetic equations involves the curl of the electric field. Write the equation and mention the law of electromagnetism which is represented by the equation.

29. The magnetic flux ϕ through a coil perpendicular to its plane is varying with time't' obeying the equation $\phi = t^2 + 3t + 5$ Weber. Calculate the emf induced in the coil at time = 2 sec.

30. Write down all four Maxwell's equations in electromagnetism.

31. The electromagnetic wave is propagating in free space with electric vector $E(z,t) = 50\cos(4\times10^7 t - Kz)\hat{i}$ volt/meter. Obtain an expression for magnetic vector.

32. In free space electric field intensity is given by $\vec{E} = \hat{J}\,25\,\cos(wt - 20x)\,{}^{volt}\!/_{m}$. Calculate the displacement current density.

33. A medium is characterized by relative permittivity 45 & relative permeability 5; calculate the speed of an electromagnetic wave in the medium and the refractive index of the medium.

34. Write the electromagnetic wave equation in terms of scalar potential by using Maxwell's field equations in free space.

35. Calculate the value of pointing vector on sun, which radiates 4×10^{26} watts & its radius is 7×10^8m.

36. If an electromagnetic wave is represented by $\vec{E}(y,t) = \hat{Z}\,50\cos(wt + ky)$, with a diagram show the directions of the propagation of the wave, the electric field & magnetic fields of it and calculate the maximum value of $\vec{E}$ & $\vec{H}$.

37. Find the impedance in vacuum, when an electromagnetic wave propagates in it.

38. In an ionosphere there are approximately 10^{11} number of electrons per unit volume. Calculate the plasma frequency of the medium. (ε_0 = 8.85×10^{-12} and $m_e = 9.11 \times 10^{-31}$,in S.I system).

39. Write the physical significance of displacement current.

40. The electric field between two parallel plates of area 2 cm^2 changes at the rate of 1.2×10^8 V/m.s. Calculate the displacement current.

41. Calculate the speed of electromagnetic wave in vacuum. $(\varepsilon_0 = 8.85\times10^{-12}\ and\ \mu_0 = 4\pi\times10^{-7}, in\ S.I\ system)$.

42. If an electromagnetic wave in free space is represented as $\vec{B}(y,t) = \hat{i}30\cos(\omega t - ky)$, show the direction of flow of energy in the electro-magnetic field and find the maximum value of its electric component.

43. A laser beam of 100 watt source is focused on an area of $10^{-8}m^2$. Evaluate the magnitude of Poynting vector of the area.

44. A 500 watt electric bulb is glowing with full capacity illuminating the space around it uniformly. Calculate the amplitude of electric and magnetic field of radiation at a distance of 2 meters.

45. If the electric field varies with time as $\vec{E} = E_0 \sin \omega t$, find the maximum displacement current density, where $E_0 = 0.1$ V/m and $\omega = 2\pi \times 60$ Hz.

Long Questions

1. State Gauss's law in electrostatics. Obtain its differential form in vacuum.
2. State Gauss's law in electrostatic. Obtain its differential form in a dielectric medium.
3. State Faraday's laws of electromagnetism. Obtain its differential form.
4. State Ampere's circuital law. Obtain its differential form.
5. How is ampere's circuital law modified due to the presence of displacement current? Obtain the corresponding Maxwell's equation.
6. Write the four Maxwell's equations in differential form in a charge free non-conducting medium. Mention the significance of Maxwell equations.
7. Derive the electromagnetic wave equation in terms of electric vector/magnetic vector when the wave is propagating in vacuum/dielectric medium. Obtain an expression for its velocity.
8. Starting from Maxwell's electromagnetic equations, obtain the wave equation for $\vec{E}$ in an ionized medium. Identify the dissipative term in the equation.
9. Show that electromagnetic wave is transverse in nature.
10. Starting from Maxwell's equations, derive an expression for Poynting's theorem.
11. State and explain Poynting's theorem. State the direction and S.I unit of Poynting vector.
12. Show that the average value of Poynting's vector for a plane electromagnetic wave is $\frac{1}{2}\sqrt{\frac{\epsilon}{\mu}}\, E_0^2$.
13. Write Poynting's theorem in vector form. For an electro-magnetic wave given by $\vec{H}(z,t) = \hat{y}50\cos(\omega t - kz)$, find the average energy crossing through a rectangular area of length 3cm. and width 1.5cm. in xy plane in one minute.

Module – 3: INTRODUCTION TO QUANTUM PHYSICS

Classical physics is dominated by two fundamental concepts. The first is the concept of a particle, a discrete entity with definite position and momentum which moves in accordance with Newton's laws of motion. The second is the concept of an electromagnetic wave, an extended physical entity with a presence at every point in space that is provided by electric and magnetic fields which change in accordance with Maxwell's laws of electromagnetism. The classical world picture is neat and tidy: the laws of particle motion account for the material world around us and the laws of electromagnetic fields account for the light waves which illuminate this world. This classical picture began to crumble in 1900 when Max Planck published a theory of black-body radiation; i.e. a theory of thermal radiation in equilibrium with a perfectly absorbing body. Planck provided an explanation of the observed properties of black-body radiation by assuming that atoms emit and absorb discrete quanta of radiation with energy $E = h\nu$, where ν is the frequency of the radiation and h is a fundamental constant of nature with value

$$h = 6:626 \times 10^{-34} \text{ Js.}$$

This constant is now called Planck's constant. In this chapter we shall see that Planck's constant has a strange role of linking wave-like and particle-like properties. In so doing it reveals that physics cannot be based on two distinct, unrelated concepts, the concept of a particle and the concept of a wave. These classical concepts, it seems, are at best approximate descriptions of reality.

There are a few phenomenon which the classical mechanics fail to explain.

1. Stability of an atom
2. Spectral series of Hydrogen atom
3. Non-relativistic motion of proton, electron, neutron, atom etc (particles of atomic dimension).
4. Discrete energy spectra
5. Radioactive decay
6. Black body radiation, photoelectric effect, Compton effect etc.

The most outstanding development in modern science was the conception of Quantum Mechanics in 1925. This new approach was highly successful in explaining about the behavior of atoms, molecules and nuclei. Max Planck in 1900 at a meeting of German Physical Society read his paper “On the theory of the Energy distribution law of the Normal Spectrum”. This was the start of the revolution of Physics i.e. the start of **Quantum Mechanics.** Quantum

physics is a generalization of Classical Physics that includes classical laws as special cases. Quantum Physics extends that range to the region of small dimensions. Just as 'c' the velocity of light signifies universal constant, the Planck's constant characterizes Quantum Physics.

Photon:

A packet or bundle of light energy is called a photon. Energy of a photon is $E = h\nu = hc/\lambda$, where h is the Planck's constant, ν is the frequency of the radiation or photon, c is the speed of light (e.m. wave) and λ is the wavelength.

Properties of photons:

i) Photons are the carriers of electromagnetic interaction(force)
ii) A photon travels at a speed of light c in vacuum. (i.e. 3×10^8 m/s)
iii) It has zero rest mass. i.e. the photon cannot exist at rest.
iv) The kinetic mass of a photon is, $m = E/c^2 = h/c\lambda$. The momentum of a photon is, $p = h/\lambda = E/c$
v) Photons travel in a straight line.
vi) Energy of a photon depends upon frequency of the photon; so the energy of the photon does not change when photon travels from one medium to another.
vii) Wavelength of the photon changes in different media; so, velocity of a photon is different in different media.
viii) Photons are electrically neutral.
ix) Photons may show diffraction under given conditions.
x) Photons are not deviated by magnetic and electric fields.
xi) Photons can be created (pair annihilation) or destroyed (pair production).

Particle aspects of radiation:
Everything in the world is a wave; everything in the world is a particle. They are the manifestation of same thing in different forms.
The particle aspects of radiation are exhibited in the phenomena of blackbody radiation, photoelectric effect, Compton scattering and pair production

Black body radiation:
A black body is an idealized physical body that absorbs all incident electromagnetic radiation, regardless of frequency or angle of incidence. A simple **example** of a **black body** radiator is the furnace. If there is a small hole in the door of the furnace heat energy can enter from the outside. The name "black body" is given because it absorbs radiation in all frequencies, not because it *only* absorbs. Indeed, a black body can also *emit* radiation. On the contrary, a white body is one with a rough surface that reflects all incident rays completely and uniformly in all directions.

A black body in thermal equilibrium (that is, at a constant temperature) emits electromagnetic radiation called black-body radiation. The distribution of radiant energy

among the various frequencies components of the black body radiation depends on its temperature. The energy distribution curve for black body radiation shows the following characteristics

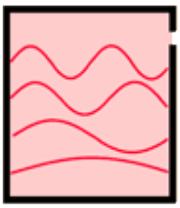
- Non-uniform distribution of energy density
- At a given temperature the energy density has maximum value corresponding to a value of frequency or wavelength.
- The frequency corresponds to maximum energy density increases with increase of temperature (wavelength decreases with increasing temperature).
- The energy density decreases to zero for both higher and lower values of frequency or wavelength.
- The energy density corresponding to a given frequency or wavelength increases with increase of temperature.

Many formulations are formulated to explain the above experimental observations like Stefan-Boltzmann law, Wein's displacement law and Planck's radiation formula. Out of which Planck's radiation formula successfully explains the theory of black body radiation. Planck's theory is based on quantum physics whereas other two theories are classical physics.

Planck's law of black body radiations:

"Blackbody radiation" or "cavity radiation" refers to an object or system which absorbs all radiation incidents upon it and re-radiates energy which is characteristic of this radiating system only, not dependent upon the type of radiation which is incident upon it. The radiated energy can be considered to be produced by standing wave or resonant modes of the cavity which is radiating.

Radiation modes in a hot cavity provide a test of quantum theory



	#Modes per unit frequency per unit volume	Probability of occupying modes	Average energy per mode
CLASSICAL	$\frac{8\pi\nu^2}{c^3}$	Equal for all modes	kT
QUANTUM	$\frac{8\pi\nu^2}{c^3}$	Quantized modes: require hν energy to excite upper modes, less probable	$\frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$

Energy per unit volume per unit frequency	$S_\upsilon = \frac{8\pi h}{c^3} \frac{\upsilon^3}{e^{h\upsilon/kT} - 1}$

Energy per unit volume per unit wavelength	$S_\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$

The Planck radiation formula is an example of the distribution of energy according to Bose-Einstein statistics. The above expressions are obtained by multiplying the density of states in terms of frequency or wavelength times the photon energy times the Bose-Einstein distribution function with normalization constant A=1.

At low wavelength (high frequency), Plank's law reduces to Wien's law and at high wavelength (low frequency) Planck's theory reduces to Rayleigh-Jeans law

(i) At low wavelength, hc/λkT is very large, 1 can be neglected as compared to the exponential term.

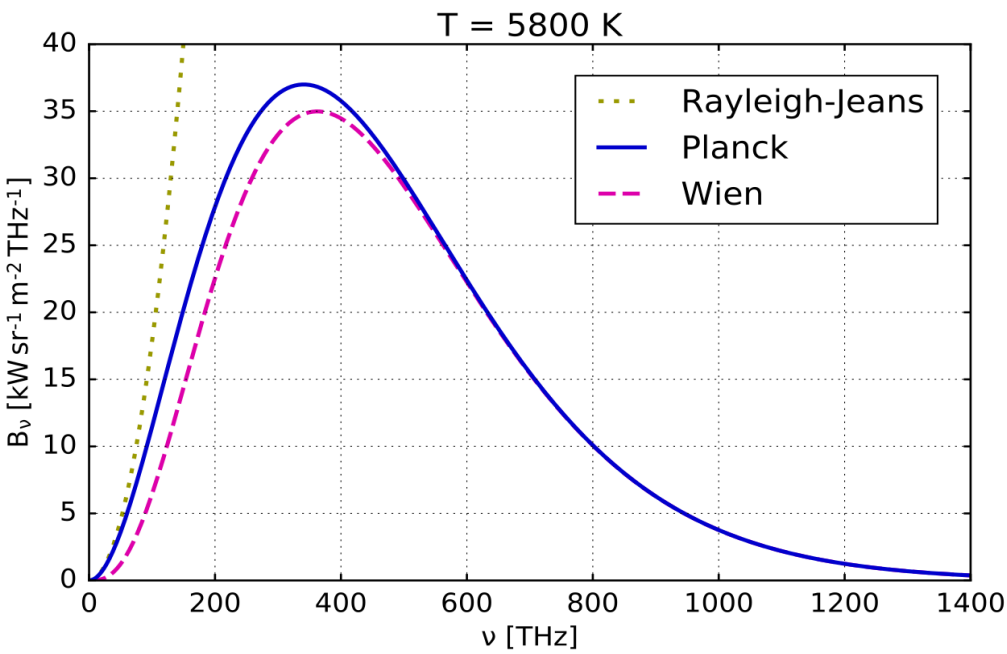
$S_\lambda = \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda kT}$ dλ

It is Wien's law

(ii) At high wavelength, hc/λkT is very small

$e^{hc/\lambda kT}$ = 1 + hc/λkT

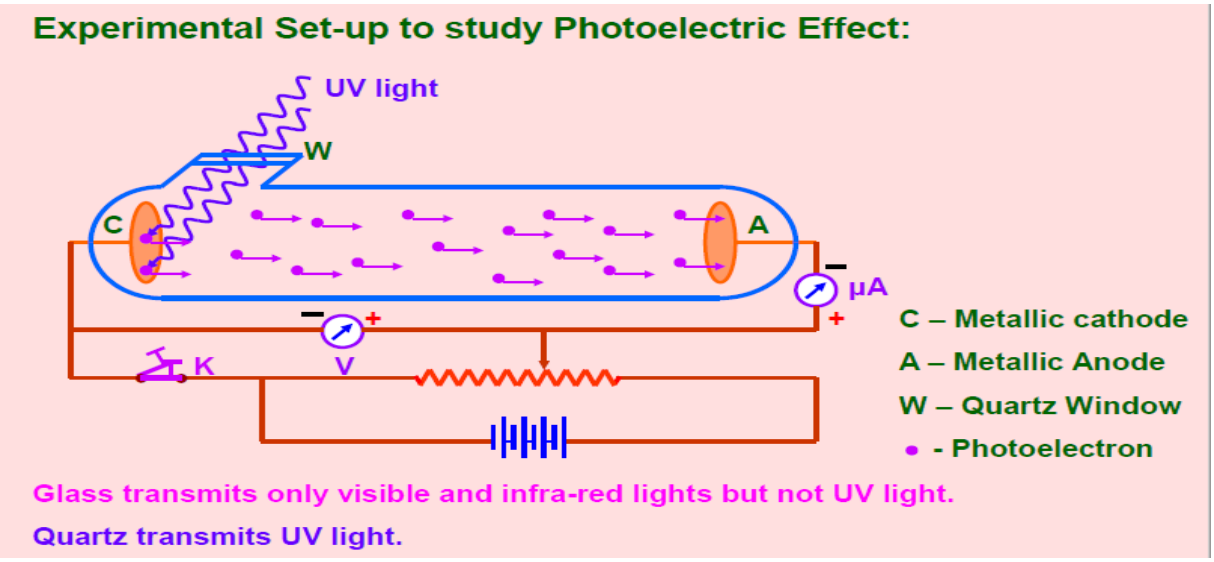
Planck's formula becomes $\frac{8\pi kT}{\lambda^4}$ dλ, which is Rayleigh-Jeans law.



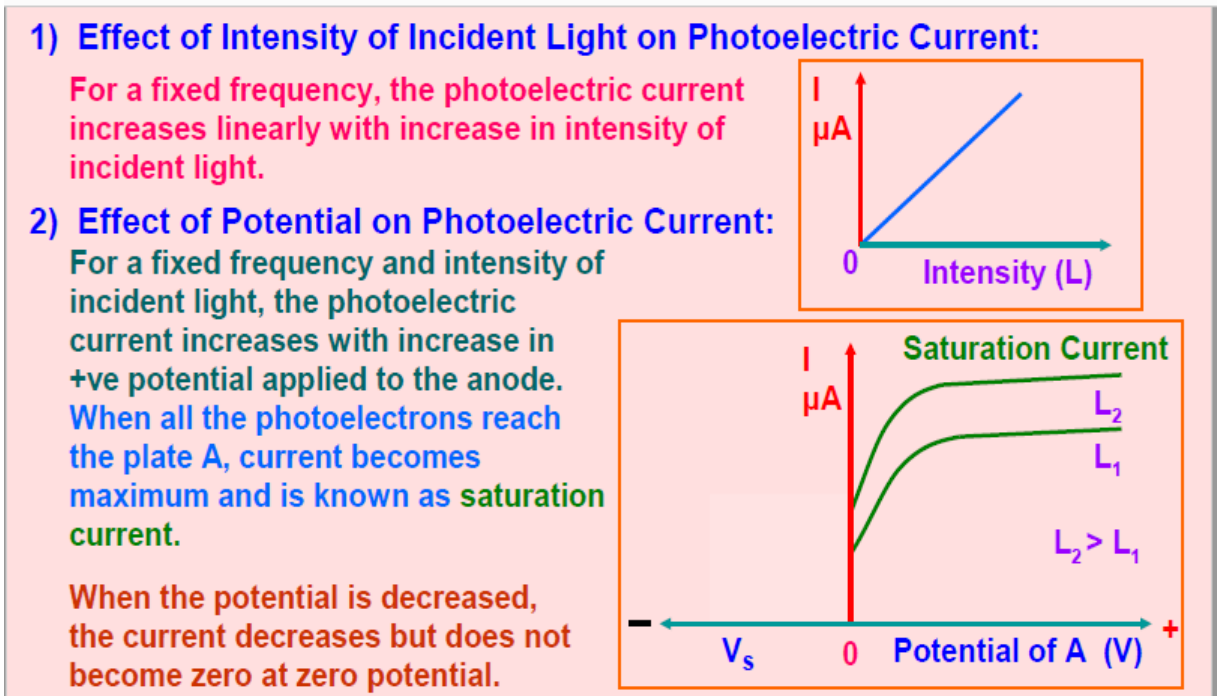
Photoelectric Effect:

The phenomenon of emission of electrons from mainly metal surfaces exposed to light energy (X – rays, γ – rays, UV rays, Visible light and even Infra Red rays) of suitable frequency is known as photoelectric effect. The electrons emitted by this effect are called photoelectrons. The current constituted by photoelectrons is known as photoelectric current.

Note: Nonmetals also show photoelectric effect. Liquids and gases also show this effect but to limited extent.



Laws/characteristics of photoelectric effect:



This shows that even in the absence of accelerating potential, a few photoelectrons manage to reach the plate on their own due to their K.E.

When –ve potential is applied to the plate A w.r.t. C, photoelectric current becomes zero at a particular value of –ve potential called stopping potential or cut-off potential.

Intensity of incident light does not affect the stopping potential.

3) Effect of Frequency of Incident Light on Photoelectric Current:

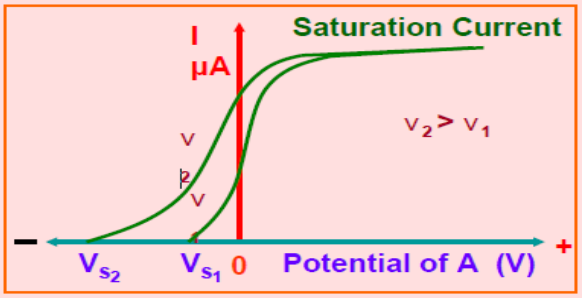
For a fixed intensity of incident light, the photoelectric current does not depend on the frequency of the incident light. Because, the photoelectric current simply depends on the number of photoelectrons emitted and in turn on the number of photons incident and not on the energy of photons.

4) Effect of Frequency of Incident Light on Stopping Potential:

For a fixed intensity of incident light, the photoelectric current increases and is saturated with increase in +ve potential applied to the anode.

However, the saturation current is same for different frequencies of the incident lights.

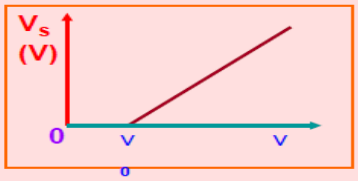
When potential is decreased and taken below zero, photoelectric current decreases to zero but at different stopping potentials for different frequencies.



Higher the frequency, higher the stopping potential. i.e. $V_s \alpha \nu$

5) Threshold Frequency:

The graph between stopping potential and frequency does not pass through the origin. It shows that there is a minimum value of frequency called threshold frequency below which photoelectric emission is not possible however high the intensity of incident light may be. It depends on the nature of the metal emitting photoelectrons.



Laws of Photoelectric Emission:

i) For a given substance, there is a minimum value of frequency of incident light called threshold frequency below which no photoelectric emission is possible, howsoever, the intensity of incident light may be.

ii) The number of photoelectrons emitted per second (i.e. photoelectric current) is directly proportional to the intensity of incident light provided the frequency is above the threshold frequency.

iii) The maximum kinetic energy of the photoelectrons is directly proportional to the frequency provided the frequency is above the threshold frequency.

iv) The maximum kinetic energy of the photoelectrons is independent of the intensity of the incident light.

v) The process of photoelectric emission is instantaneous. i.e. as soon as the photon of suitable frequency falls on the substance, it emits photoelectrons.

vi) The photoelectric emission is one-to-one. i.e. for every photon of suitable frequency one electron is emitted.

Einstein's Photoelectric Equation:

When a photon of energy $h\nu$ falls on a metal surface, the energy of the photon is absorbed by the electron and is used in two ways:

i) A part of energy is used to overcome the surface barrier and come out of the metal surface. This part of the energy is called 'work function' ($\Phi = h\nu_0$).

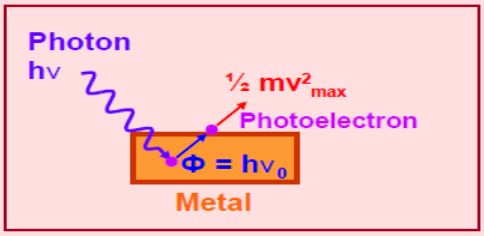
ii) The remaining part of the energy is used in giving a velocity 'v' to the emitted photoelectron. This is equal to the maximum kinetic energy of the photoelectrons ($\frac{1}{2} mv^2_{max}$) where 'm' is mass of the photoelectron.

According to law of conservation of energy,

$$h\nu = \Phi + \frac{1}{2} mv^2_{max}$$

$$= h\nu_0 + \frac{1}{2} mv^2_{max}$$

$$\boxed{\frac{1}{2} mv^2_{max} = h (\nu - \nu_0)}$$



Verification of Laws of Photoelectric Emission based on Einstein's Photoelectric Equation:

$$\boxed{\frac{1}{2} mv^2_{max} = h (\nu - \nu_0)}$$

i) If $\nu < \nu_0$, then $\frac{1}{2} mv^2_{max}$ is negative, which is not possible. Therefore, for photoelectric emission to take place $\nu > \nu_0$.

ii) Since one photon emits one electron, so the number photoelectrons emitted per second is directly proportional to the intensity of incident light.

iii) It is clear that $\frac{1}{2} mv^2_{max} \alpha \nu$ as h and ν_0 are constant. This shows that K.E. of the photoelectrons is directly proportional to the frequency of the incident light.

iv) Photoelectric emission is due to collision between a photon and an electron. As such there can not be any significant time lag between the incidence of photon and emission of photoelectron. i.e. the process is instantaneous. It is found that delay is only 10^{-8} seconds.

Application of Photoelectric Effect:

1. Automatic fire alarm
2. Automatic burglar alarm
3. Scanners in Television transmission
4. Reproduction of sound in cinema film
5. In paper industry to measure the thickness of paper
6. To locate flaws or holes in the finished goods
7. In astronomy
8. To determine opacity of solids and liquids
9. Automatic switching of street lights
10. To control the temperature of furnace

Compton Effect

When a monochromatic beam of X-ray is scattered by a light element, the scattered radiation contains two wavelengths- (i) Same as incident wavelength (Thomson component) (ii) Greater than incident wavelength (Compton component). This is called Compton effect.

The phenomenon of change in wavelength of incident radiation upon scattering from a material is called Compton effect.

If the wavelengths of incident and scattered radiation is λ and λ' respectively, then the change in wavelength is given by

$\lambda' - \lambda = \frac{h}{m_0 c}(1 - Cos\phi)$; h is Planck's constant, m_0 is mass of an electron, and c is speed of light in vacuum

ϕ is angle of scattering

From above it is seen that the compton shift is independent of wavelength of the incident photon

According to classical physics, when an electromagnetic radiation of a particular frequency and wavelength is incident on atoms, the atomic electrons oscillate with the frequency of the incident radiation. Due to this oscillation, the electrons radiate EM waves of same frequency and wavelength. Thus, classical electromagnetic theory cannot explain the change in wavelength of the scattered radiation observed in Compton effect.

Compton made the following assumptions (**quantum assumptions**)

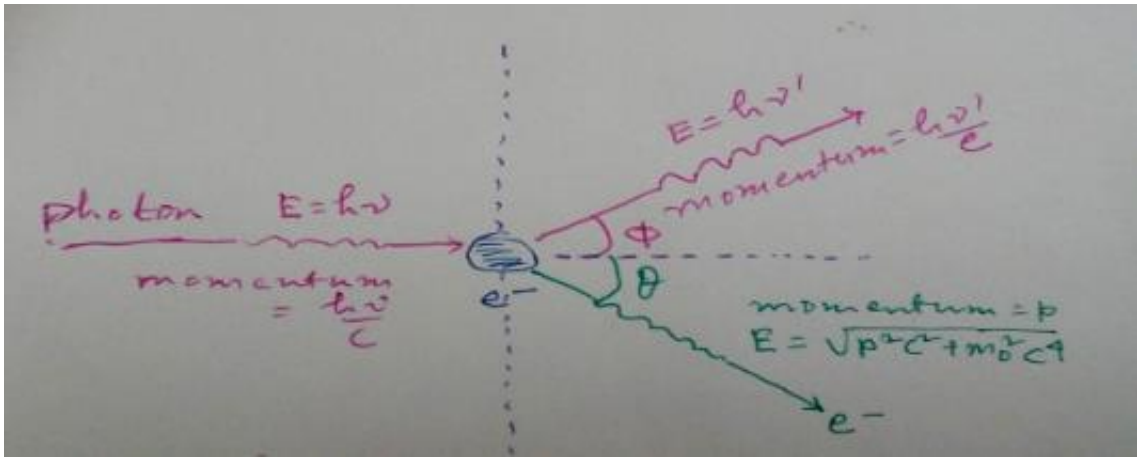
(i) Electromagnetic radiation is quantized. It consists of discrete packets of energy hv, where v is the frequency of the wave. These packets are called photons which behave as particles.

(ii) When an electromagnetic wave is incident on a medium, the photons collide with the target electrons

(iii) In these collisions, photons lose energy to target electrons. So, energy of photons decreases and wavelength of the photons

Derivation of Compton shift:

A photon of energy $h\nu$ collides with an electron at rest. After collission the photon and the electron is scattered at angles ϕ and θ respectively. Energy and momenta are conserved in the collission process



Energy of the photon before collission= $h\nu$

Momentum of the photon before collission= $\frac{h\nu}{c}$ [along x dirction]

Momentum of the electron before collission= 0

Energy of the electron before collission= m_0c^2

Energy of the photon after collission= $h\nu'$

Momentum of the photon after collission along the x and y directions= $\frac{h\nu'}{c}Cos\phi$ and $\frac{h\nu'}{c}Sin\phi$

Energy of the electron after collission= $E = mc^2 = \sqrt{p^2c^2 + m_0^2c^4}$; here p is electron momentum

Momentum of the electron after collission along the x and y directions= $pCos\theta$ and $pSin\theta$

From conservation of energy:

$h\nu + m_0c^2 = h\nu' + mc^2$ or, $mc^2 = h(\nu - \nu') + m_0c^2$

or, $\sqrt{p^2c^2 + m_0^2c^4} = h(\nu - \nu') + m_0c^2$

or, $p^2c^2 + m_0^2c^4 = h^2(\nu - \nu')^2 + m_0^2c^4 + 2hm_0c^2(\nu - \nu')$ [squring both sides]

or, $p^2c^2 = h^2(\nu - \nu')^2 + 2hm_0c^2(\nu - \nu')$.....(1)

<u>From momentum conservation along x axis :</u>

$\frac{h\nu}{c} = \frac{h\nu'}{c}Cos\phi + pCos\theta$ or, $pcCos\theta = h\nu - h\nu'Cos\phi$.....(2)

<u>From momentum conservation along y axis :</u>

$0 = \frac{h\nu'}{c}Sin\phi - pSin\theta$ or, $pcSin\theta = h\nu'Sin\phi$.....(3)

Squaring and adding Eq (2) and (3) -

$p^2c^2 = h^2\nu^2 + h^2\nu'^2Cos^2\phi - 2h^2\nu\nu'Cos\phi + h^2\nu'^2Sin^2\phi$

or, $p^2c^2 = h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'Cos\phi$.....(4)

From Eq (1) and (4), we can write -

$h^2(\nu^2 + \nu'^2 - 2\nu\nu') + 2hm_0c(\nu - \nu') = h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'Cos\phi$

$$or, \quad 2hm_0c(\nu - \nu') = 2h^2\nu\nu'(1 - Cos\phi)$$

$$or, \quad \frac{c}{\nu'} - \frac{c}{\nu} = \frac{h}{m_0c}(1 - Cos\phi)$$

$$or, \quad \lambda' - \lambda = \frac{h}{m_0c}(1 - Cos\phi)$$

$$or, \quad \Delta\lambda = \frac{h}{m_0c}(1 - Cos\phi).....(5)$$

$Eq.(5)$ describes Compton shift, i.e., change in wavelength of the incident radiation

$\lambda_c = \frac{h}{m_0c}$ is called Compton wavelength

For electron, $\lambda_c = 0.02426\overset{0}{A}$

$\because \Delta\lambda = 0.02426(1 - Cos\phi)$

Features:

(i) When $\phi = 0, \Delta\lambda = 0$, so $\lambda' = \lambda$; No change of wavelength takes place along the direction of incident radiation

(ii) $\Delta\lambda$ incresaes with increase in ϕ, when $\phi = 90^o, \Delta\lambda = 0.02426\overset{o}{A}$

(iii) Change in wavelength is independent of wavelength of incident radiation

(iv) When $\phi = 180^o, \Delta\lambda = 0.04852\overset{0}{A}$ (the maximum value of Compton shift, in this case electron recoils back after striking the electron)

Pair Production:

When high energy photons (γ- rays) pass near an atomic nucleus, it converts fully into an electron, e^- and positron, e^+. This phenomenon is called pair production.

In pair production, electromagnetic energy is converted into matter (e^+ and e^-). In this process, charge, mass and momentum are conserved.

<u>Why γ- ray is suitable for pair production-</u>

Suppose electron and positron are at rest. The rest mass is m_0. The rest energy of electron and positron are $E = m_0c^2 + m_0c^2 = 2\ m_0c^2 = 1.02$ MeV . The energy comes from the incident ray. The frequency of the ray of energy 1.02 MeV energy is

$$\nu = E/h \sim 2.5 \times 10^{20}\ Hz$$

γ rays have frequency in this range. So, g ray is used in pair production.

Pair Annihilation:

In this case an electron, e^- and positron, e^+ comes together due to Coulomb attraction and they are destroyed. The total mass is converted into energy. This phenomenon is called pair annihilation.

In pair annihilation, matter (e^+ and e^-) is converted to electromagnetic energy

In this process, charge, mass and momentum are conserved.

<u>Why pair production is not possible in free space:</u>

In free space $h\nu = mc^2 + mc^2 = 2mc^2$

Considering the momentum conservation-

$h\nu/c = 2p\cos\theta,$

Or, $h\nu = 2pc\cos\theta = 2m(v/c)c^2\cos\theta < 2mc^2$ *[as* $v/c < 1$ *and* $\cos\theta < 1$*]*

This violates the conservation of energy, so pair production is not possible in free space.

Wave aspect of particles

<u>De-Broglie theory of matter waves:</u>

A particle in motion has wave nature. The wave associated with the particle is called matter wave or de-Broglie wave.

For a photon, $E = h\nu$

Also, the relativistic energy is given by $E = [p^2c^2 + m_0^2c^4]^{1/2}$, the rest mass of photon =0

Hence, $E = pc$, Comparing we get,

$P = h/\lambda$

The wavelength of the matter wave is given by

$\lambda = \frac{h}{p}$; h is Planck's constant and p is momentum of the particle

So, $\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$; Where E is kinetic energy $= \frac{p^2}{2m}$

Case I: For an electric charge passing through a voltage difference V

$$\lambda = \frac{h}{\sqrt{2mqV}} = \frac{12.3}{\sqrt{V}}\,\text{Å} \quad \text{.....(1)}$$

Case II: For thermal neutron, the linetic energy is ~ kT

$$\lambda = \frac{h}{\sqrt{2mkT}} \quad \text{.....(2)}$$

For relativistic particles:

Kinetic Energy $E_k = E - m_0c^2 = \sqrt{m_0^2c^4 + p^2c^2} - m_0c^2$

or, $E_k^2 + m_0^2c^4 + 2E_km_0c^2 = m_0^2c^4 + p^2c^2$

So, $p = \dfrac{\sqrt{E_k(E_k + 2m_0c^2)}}{c}$

So, $\lambda = \dfrac{h}{p} = \dfrac{hc}{\sqrt{E_k(E_k + 2m_0c^2)}}$ .....(3)

Phase velocity and Group velocity of De-Broglie waves

A wave is associated with a moving particle. The velocity of the wave is called phase velocity. When a few wave of nearly equal wavelength is superposed, a wave packet is formed. The velocity of the wave packet is called group velocity. The group velocity of a wave is given by vg=*dw/dk,* where *w* is the angular frequency and *k* is wave number.

It is observed that the phase velocity of matter wave associated with a particle of mass m and velocity v is different from particle velocity. The group velocity associated with the particle is same as particle velocity.

From De - Broglie theory of matter waves: $p = \dfrac{h}{\lambda} = \dfrac{h}{2\pi} \times \dfrac{2\pi}{\lambda} = \hbar k$

Energy of a matter wave $E = h\nu = \dfrac{h}{2\pi} \times 2\pi\nu = \hbar\omega$

Therefore, phase velocity of matter waves: $v_p = \nu\lambda = \dfrac{\omega}{k} = \dfrac{\hbar\omega}{\hbar k} = \dfrac{E}{p}$

Nonrelativistic particles:

Phase velocity of a matter wave associated with a free particle: $v_p = \dfrac{E}{p} = \dfrac{p^2/2m}{mv} = \dfrac{p^2}{2m^2v} = \dfrac{m^2v^2}{2m^2v} = \dfrac{v}{2}$

Group velocity of matter wave associated with a free particle $v_g = \dfrac{d\omega}{dk} = \dfrac{dE}{dp} = \dfrac{d}{dp}\left[\dfrac{p^2}{2m}\right] = \dfrac{2p}{2m} = \dfrac{p}{m} = v$

Relativistic particle

Phase velocity of relativistic free particle $v_p = \dfrac{E}{p} = \dfrac{mc^2}{mv} = \dfrac{c^2}{v}$ [impossible because $v_p > c$]

Group velocity of relativistic free particle $v_g = \dfrac{dE}{dp} = \dfrac{d\sqrt{p^2c^2 + m_0^2c^4}}{dp} = \dfrac{pc^2}{mc^2} = \dfrac{p}{m} = v$

Characteristics of matter waves:

(i) A matter wave is associated with a particle in motion. The wave nature becomes apparent only for very small particles (atomic dimension). For large bodies, wavelength is so small that wave properties are not visible.

(ii) Wavelength of matter wave ranges from zero to infinity.

(iii) Matter waves can be observed experimentally.

(iv) Matter waves are independent of the charge of the particle.

(v) Velocity of the matter wave is variable.

(vi) Matter waves show phenomenon like interference, diffraction, etc.

Q: why matter waves are not observed in daily life?

Heisenberg's uncertainty principle (1928):

It states that for a particle of atomic dimension in motion, it is impossible to simultaneously determine the position and momentum of the particle with perfect accuracy.

The principle of uncertainty follows from wave particle duality.

If Δx is uncertainty in position and Δp_x is uncertainty in momentum for motion along x direction, then

$$\Delta x.\Delta p_x \geq \frac{\hbar}{2} \quad ; \; \hbar = \frac{h}{2\pi}$$

Similarly,

$$\Delta y.\Delta p_y \geq \frac{\hbar}{2} \quad \text{and} \quad \Delta z.\Delta p_z \geq \frac{\hbar}{2}$$ for motion along y and z direction, respectively.

A similar principle of uncertainty is valid between

(i) energy and time -

$$\Delta E.\Delta t \geq \frac{\hbar}{2}$$

(ii) angular momentum and angle of rotation

$$\Delta L.\Delta \theta \geq \frac{\hbar}{2}$$

Significance of uncertainty relation:

(i) When $\Delta x \to 0$, $\Delta p_x \to \infty$ and when $\Delta x \to \infty$, $\Delta p_x \to 0$

This means that if position is determined with high accuracy then the accuracy in momentum become very less accurate and vice - versa

(ii) From the uncertainty relation in position and momentum we get

$$\Delta x.\Delta v_x \geq \frac{\hbar}{2m}$$ (m is mass of the particle)

For heavy particle $\Delta x.\Delta v_x \to 0$ because m is large. So the position and momentum of heavy particle can be determined with perfect accuracy and in this limit quantum mechanics tends to classical mechanics

Applications of uncertainty principle:

(i) Non existence of electrons in the atomic nucleus

(ii) Ground state energy of harmonic oscillator

(iii) Ground state energy of hydrogen atom

(i) Non existence of electrons in the atomic nucleus

Suppose electrons exists inside the nucleus. Then maximum uncertainty in finding the position of the electron inside the nucleus is nuclear diameter $\Delta x_{\max} \approx 2\times10^{-15}\,\mathrm{m}$
The minimum uncertainty in determining the momentum is

$$\Delta \mathrm{p}_{\min} \cong \frac{\hbar}{\Delta x_{\max}}$$

The minimum kinetic energy of the electron is $\mathrm{E} = \dfrac{(\Delta \mathrm{p}_{\min})^2}{2m}$

$$= \frac{\hbar^2}{(\Delta x_{\max})^2 2\mathrm{m}} \cong 20MeV$$

The maximum energy of electron emitted from the nucleus during β decay is found to be $\approx$ 4MeV. Hence electrons can not reside inside the nucleus.

(ii) Ground state energy of harmonic oscillator

The ground state is the lowest or minimum energy state of a harmonic oscillator.
According to classical physics, the ground state energy of a harmonic oscillator= 0.
In quantum mechanics, the ground state energy has a minimum value due to uncertainty principle

The minimum value of position x and momentum p_x in the ground state is of the same order as the uncertainty in position and momentum of the oscillator Δx and Δp_x
From the uncertainty principle

$$x.p_x \approx \frac{\hbar}{2} \quad \text{or,} \quad p_x = \frac{\hbar}{2x} .....(1)$$

Energy of the harmonic oscillator $E = \dfrac{p_x^2}{2m} + \dfrac{1}{2}kx^2 = \dfrac{p_x^2}{2m} + \dfrac{1}{2}m\omega^2 x^2$

The ground state energy of the harmonic oscillator is obtained using Eq (1) as

$$E = \frac{\hbar^2}{8mx^2} + \frac{1}{2}m\omega^2 x^2 .....(2)$$

at ground state E is minimum, so $\frac{\partial E}{\partial x}=0$,

or, from (2), $\quad -\frac{2\hbar^2}{8mx^3}+m\omega^2 x=0$

or, $\quad m\omega^2 x=\frac{2\hbar^2}{8mx^3}$

or, $\quad x^4=\frac{\hbar^2}{4m^2\omega^2} \quad$ or, $x=\left(\frac{\hbar}{2m\omega}\right)^{\frac{1}{2}}$.....(3)

The ground state energy E of the harmonic oscillatoris obtained from Eq (2) and (3) as

$$E=\frac{\hbar^2}{8m.\frac{\hbar}{2m\omega}}+\frac{1}{2}m\omega^2\frac{\hbar}{2m\omega}=\frac{1}{4}\hbar\omega+\frac{1}{4}\hbar\omega=\frac{1}{2}\hbar\omega.....(4)$$

This is the ground state energy or minimum energy of a quantum harmonic oscillator.

The magnitude of this energy is very small ($\sim 10^{-34}$ Joule).So this energy can not be measured for macroscopic oscillators. In quantum oscillators like electronin an atom or two atoms bound as molecule,this energy is significant.

(iii) Ground state energy of hydrogen atom

The energy of a hydrogen atom $E=\frac{p^2}{2m}-\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$.....(1);

where p is the momentum of the electronin the Bohr orbit and the second term represent the potenrial energy of the electron. r is the radius of the Bohr orbit.

We assume that in the ground state where the energy is minimum, the uncertainty is position and momentum is of the order of their magnitude. Then, according to uncertainty relation-

$r.p=\hbar$ or, $p=\frac{\hbar}{r}$.....(2)

So, the enrgy of the Hydrogen atom is

$$E=\frac{\hbar^2}{2mr^2}-\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}.....(3)$$

At ground state, $\frac{\partial E}{\partial r} = 0$, From (1) and (2) above,

$$\frac{\partial E}{\partial r} = -\frac{2\hbar^2}{2mr^3} + \frac{e^2}{4\pi\varepsilon_0 r^2} = 0$$

or, $\frac{e^2}{4\pi\varepsilon_0 r^2} = \frac{2\hbar^2}{2mr^3}$ or, $r = \frac{\hbar^2 4\pi\varepsilon_0}{me^2}$.....(4)

From (1), the ground state energy $E = \frac{\hbar^2}{2m\hbar^4\left(\frac{4\pi\varepsilon_0}{me^2}\right)^2} - \frac{e^2}{4\pi\varepsilon_0\hbar^2\left(\frac{4\pi\varepsilon_0}{me^2}\right)}$

$$= \frac{m^2e^4}{2m\hbar^2(4\pi\varepsilon_0)^2} - \frac{me^4}{\hbar^2(4\pi\varepsilon_0)^2}$$

$$= \frac{me^4}{32\pi^2\hbar^2\varepsilon_0^2} - \frac{me^4}{16\pi^2\hbar^2\varepsilon_0^2}$$

$$= -\frac{me^4}{32\pi^2\hbar^2\varepsilon_0^2} \text{.....(5)}$$

Equation (4) and (5) is the expression for Bohr radius and Ground state energy of Hydrogen atom

Questions

Short answer type

1. How does the kinetic energy of the emitted photoelectron depend on intensity of the incident radiation?
2. Define work function in photoelectric effect.
3. Define stopping potential. How does it depend on frequency of incident radiation?
4. What are the laws of photoelectric effect?
5. Explain graphically the variation of stopping potential with frequency of incident light in case of photoelectric effect.
6. Define Compton effect
7. Under what condition is the Compton shift a minimum, in the Compton effect?
8. What is Compton wavelength? What is the difference between Compton shift and Compton wavelength?
9. How does the Compton shift depend on wavelength of incident radiation?
10. Define pair production.
11. What physical quantities are conserved in pair production?
12. Why gamma ray photon is suitable for pair production?
13. Why is pair production not possible in vacuum?
14. Why a radiation of 1 Å wavelength cannot produce pair production?
15. What is de-Broglie wave?

16. Write wave particle duality in case of matter wave.
17. How matter waves differ from electromagnetic wave?
18. Find the de-Broglie wavelength for a particle of mass m moving with kinetic energy E.
19. What is Heisenberg's uncertainty principle?
20. Explain how wave particle duality leads to uncertainty.

Long answer type

1. What is photoelectric effect? Write the laws of photoelectric effect. Graphically show the variation of photocurrent with intensity, kinetic energy of photoelectron with frequency and stopping potential with frequency of incident radiation.
2. What is photoelectric effect? Derive Einstein's photoelectric equation. Explain the laws of photoelectric effect from it.
3. (i) Define stopping potential, threshold frequency and work function in the phenomenon of photoelectric effect. (ii) The work function of aluminum is 4.2 eV. Calculate the kinetic energy of the fastest and slowest photoelectrons, stopping potential and the cutoff wavelength when light of wavelength 4000 Å falls on the clean aluminum surface.
4. What is Compton scattering? Derive an expression for Compton shift with suitable diagram.
5. (i) State and explain Compton effect with a neat diagram. Discuss how it differs from the photoelectric effect. (ii) Can X-rays of wavelength 1 Å undergo pair production?
6. (i) State and explain Compton effect. (ii) In an experiment on Compton scattering, the wavelength of the incident radiation is 1.8 Å. Calculate the wavelength of the radiation scattered at 45^0. Calculate the speed and the direction of the recoiled electron.
7. What do you mean by pair production? Calculate the threshold energy for pair production. Show that pair production is not possible in vacuum.
8. (i) What is matter wave? What are the properties of matter wave? Obtain an expression for de-Broglie wavelength of matter wave for an electron of mass m moving in an electric field of potential V volt.
9. State Heisenberg's uncertainty principle. Using this prove that the ground state energy of hydrogen atom is non-zero.
10. State Heisenberg's uncertainty principle. Using this prove that the ground state energy of harmonic oscillator is non-zero.
11. State Heisenberg's uncertainty principle. Using this prove that the non-existence of electron inside the nucleus.

Numerical

1. Calculate the momentum of an X-ray having a wavelength of 0.7×10^{-10} m.
2. Find the energy of a gamma ray photon having wavelength of 1 A^o , given that

3. $h=6.62X10^{-27}$ erg-s.
4. Define work function of a metal. How it can be expressed in terms of threshold frequency? Find the work function of a metal in eV, whose threshold wavelength is 6000 Å.
5. In an experiment tungsten cathode which has a threshold wavelength 2000 Å is irradiated by ultra-violet light of wavelength 1800 Å. Calculate the maximum energy of emitted photoelectrons and work function of the metal in electron volts.
6. Find the Compton shift for X-rays of wavelength 1.3 A^0, scattered by 60^0. Find the wavelength of scattered X-rays ($m_0= 9.1\times10^{-31}$ kg).
7. X-rays of wavelength 1.0 A^0 are scattered from a carbon block. Find the wavelength of the scattered beam in a direction making 90^0 with incident beam. How much kinetic energy is imparted to the recoiling electron?
8. Under what condition the Compton shift is equal to i) Zero ii) 0.0242Å iii) Twice the Compton wavelength.
9. Calculate the de-Broglie wavelength of a particle of mass 3gm moving with a speed of 400 m/s.
10. The De-Brogile wavelength associated with a neutron is $1.4 X 10^{-10}$ m whose mass is $1.675X10^{-27}$ kg. Estimate its kinetic energy in eV. Also calculate its velocity.
11. Estimate the de-Broglie wavelength for a beam of electrons whose energy is 20 eV.
12.
13. An electron is accelerated by a potential difference V volts, has a de Broglie wavelength λ. If the electron is accelerated by a potential difference of 4V volts, find its de-Broglie wavelength.
14. An electron has a speed of 400 m/sec, with an accuracy of 0.005 %. Find the uncertainty of the position of the electron.
15. The energy width of an excited state in a system is 1.5 eV. What is the average lifetime of that state?
16. For an electron moving with a velocity $2X10^7$ m/s, estimate the smallest possible uncertainty in the position of the electron.
17. Calculate the minimum electron energy in the hydrogen atom using uncertainty principle (Atomic radius is 0.53 Å).

Module – 4: QUANTUM MECHANICS-II

Transition from the deterministic to probabilistic description:

In classical physics the physical quantities like position, momentum etc can be determined exactly at any time and the value of these quantities at a later time can be predicted through equation of motion. In quantum mechanics, uncertainty principle imposes a restriction on the exact value of a physical quantity. So, in quantum physics, instead of having exact values, there are a number of allowed values with different probabilities. In classical physics, intensity of a wave at a point depends on square of amplitude of the wave at that point, whereas in quantum physics, probability of finding a particle at a point depends on square of amplitude of the wave at that point.

For a system, instead of a single sharply defined value, a physical quantity can take a large number of different values with different probability. Each of these values corresponds to a possible state of the system. The General state of the system is a linear superposition of all such possible states. The observed value of the physical quantity is the weighted average of the different values. Example: if a particle can have energy E_1, E_2, E_3 with probability 0.7, 0.2, and 0.1, respectively, then the observed energy will be the weighted average $E_{av}=0.7E_1+0.2E_2+0.1E_3$. The general state of a physical system is described by wave function Y (x,y,z,t). The state of a system changes with time. The time variation in Y is described by Schrodinger equation.

Wave function of matter wave and characteristics of wave function:

(i) Wave function ψ of a particle is a mathematical representation which contains all the information about a system that can be determined.

(ii) ψ is a function of position and time $\psi = \psi(x, y, z, t)$, it is single valued function.

(iii) ψ is a complex quantity; $\psi = a + ib$

(iv) ψ and it's derivative $\frac{\partial \psi}{\partial x}$ are continuous at all places

(v) The quantity $\psi^* \psi = |\psi|^2$ represents the probability per unit volume (probability density) of finding the system in the state ψ

(vi) $\int_{V=\text{all space}} \psi^* \psi dV = \int_{V=\text{all space}} |\psi|^2 dV = 1$ is called the normalization condition

(vii) ψ *must be finite and it has no direct physical meaning.*

(Here ψ^* is the complex conjugate of ψ, and $\psi\,\psi^* = a^2 + b^2$ i.e. real & has physical meaning. Ψ is called probability amplitude but $\psi^*\,\psi$ is called probability density)

The time dependent Schrodinger equation:

We consider a free particle of mass m is moving along x direction under the action of a force F. Let motion of the particle is non - relativistic.

we know $p_x = \hbar k$ and $E = \hbar\omega$

The free particle wave function is $\psi = Ae^{i(kx-wt)} = Ae^{i\left(\frac{px}{\hbar}-\frac{Et}{\hbar}\right)} = Ae^{\frac{i}{\hbar}(px-Et)}$ ------(1)

$$\frac{\partial\psi}{\partial x} = \frac{i}{\hbar}p_x Ae^{\frac{i}{\hbar}(px-Et)} = \frac{i}{\hbar}p_x\psi$$

$$\frac{\partial^2\psi}{\partial x^2} = \frac{i}{\hbar}p_x.\frac{i}{\hbar}p_x Ae^{\frac{i}{\hbar}(px-Et)} = -\frac{p_x^2}{\hbar^2}\psi$$

or, $p_x^2\psi = -\hbar^2\frac{\partial^2\psi}{\partial x^2}$.....(2)

$$\frac{\partial\psi}{\partial t} = -\frac{iE}{\hbar}Ae^{\frac{i}{\hbar}(px-Et)} = -\frac{iE}{\hbar}\psi.$$

or, $E\psi = i\hbar\frac{\partial\psi}{\partial t}. =$ .....(3)

We know that for a free particle $\frac{p_x^2}{2m} = E,$

In quantum mechanics, we consider both the term in the above equation as operators.

So the equation is $\frac{p_x^2}{2m}\psi = E\psi$.....(4)

we put the values of p_x and E from eq. (2) and (3) to get

$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} = i\hbar\frac{\partial\psi}{\partial t}$.....(5) ; This is free particle Schrodinger equation in one dimension

If the particle is moving in a potential V(x), the energy of the particle is

$$\text{E} = \text{Kinetic energy} + \text{Potential Energy} = \frac{\text{p}_\text{x}^2}{2m} + V$$

Thus the Schoredinger equation is

$$\frac{\text{p}_\text{x}^2}{2m}\psi + V\psi = E\psi.....(6)$$

or, from eq (2) and (3)-

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi = i\hbar\frac{\partial\psi}{\partial t}.....(7)$$

Equation (7) and equation (5) are the required one dimensional Schrodinger's time dependent wave equation for a free particle and bound particle respectively.

(E = hν = $\frac{h}{2\pi}$ x 2π ν = ħω and p = $\frac{h}{\lambda}$ **=** $\frac{h}{2\pi}$ **x** $\frac{2\pi}{\lambda}$ **= ħk, k is called wave number or propagation constant)**

In three dimension, Schrodinger equation becomes-

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t}; \qquad \left[\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right]$$

For a free particle in 3D, V = 0, So the equation becomes

$$-\frac{\hbar^2}{2m}\nabla^2\psi = i\hbar\frac{\partial\psi}{\partial t}$$

The time independent Schrodinger equation:

$\psi = Ae^{\frac{i}{\hbar}(px-Et)} = Ae^{\frac{i}{\hbar}p_x x}e^{-\frac{i}{\hbar}Et} = \psi_0 e^{-\frac{i}{\hbar}Et}$; here ψ_0 is the space part of the wave function.

$$\frac{\partial^2\psi}{\partial x^2} = e^{-\frac{i}{\hbar}Et}\frac{\partial^2\psi_0}{\partial x^2} \text{.....(8)}$$

$$\frac{\partial\psi}{\partial t} = \psi_0.\left(-\frac{i}{\hbar}\right).Ee^{-\frac{i}{\hbar}Et} \text{.....(9)}$$

we substitute (8) and (9) in Schrodinger equation $i\hbar\frac{\partial\psi}{\partial t}. = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$ to get

$$-\frac{\hbar^2}{2m}e^{-\frac{i}{\hbar}Et}\frac{\partial^2\psi_0}{\partial x^2} + V\psi_0 e^{-\frac{i}{\hbar}Et}. = i\hbar\psi_0.\left(-\frac{i}{\hbar}\right).Ee^{-\frac{i}{\hbar}Et}$$

$$or, \qquad -\frac{\hbar^2}{2m}\frac{\partial^2\psi_0}{\partial x^2} + V\psi_0 = E\psi_0$$

$$or, \qquad \frac{\partial^2\psi_0}{\partial x^2} + \frac{2m}{\hbar^2}(E-V)\psi_0 = 0\text{.....(10)}$$

For a free particle V = 0 so eq (10) becomes-

$$\frac{\partial^2\psi_0}{\partial x^2} + \frac{2m}{\hbar^2}E\psi_0 = 0$$

Equation (10) is the required time independent Schrodinger equation for a bound particle. Here ψ_o is only function of position.

Principle of superposition:

The wave function ψ representing a general state of the system is a linear superposition of different possible states $\psi_1, \psi_2, \ldots\ldots\ldots\ldots \psi_n$ in which the system can exist

It $\psi_1, \psi_2, \ldots\ldots\ldots\ldots \psi_n$ are the solutions of Schrodinger equation, then the linear combination of the solutions $\psi = c_1\psi_1 + c_2\psi_2 + \ldots\ldots\ldots + c_n\psi_n = \sum_n c_n\psi_n$ is also a solution of Schrodinger equation.

The square of the coefficients $|c_i|^2$ is the probability that the system will be in state ψ_i

For a given system, the allowed states are obtained by solving the Schrodinger equation.

Probability density:

Probability density : $P = |\psi|^2 = \psi^*\psi$ is called the probability density. It is desbcribed as the probability of finding the the system in state ψ per unit volume.

The probability of finding the particle in a volume dV is given by PdV

The probability of finding the particle in a volume V is $\int_V PdV = \int_V |\psi|^2 dV$

In one dimension, such as motion along x - axis, the probability of finding the particle between x and x + dx is $P = \int_x^{x+dx} |\psi|^2 dx$

The probability that the particle is somewhere within the entire space is $\int_{V=\text{all space}} |\psi|^2 dV = 1$

In this case ψ is said to be normalized

Dimension of Ψ:

Dimension of ψ

$$3D: |\psi|^2 = \frac{probability}{Volume} \sim L^{-3}; \quad \psi \sim L^{-\frac{3}{2}}$$

$$2D: |\psi|^2 = \frac{probability}{Area} \sim L^{-2}; \quad \psi \sim L^{-1}$$

$$1D: |\psi|^2 = \frac{probability}{length} \sim L^{-1}; \quad \psi \sim L^{-\frac{1}{2}}$$

Normalized wave function:

If $\int_V \psi^* \psi dV = \int_{V=\text{all space}} |\psi|^2 dV = 1$ then the wave function is said to be normalized

If $\int_{V=\text{all space}} |\psi|^2 dV = N$, then what is the normalized wavefunction?

$$\int_{V=\text{all space}} |\psi|^2 dV = N \quad \text{or,} \quad \int_{V=\text{all space}} \psi^* \psi dV = N \quad \text{or} \quad \int_{V=\text{all space}} \frac{\psi^*}{\sqrt{N}} \frac{\psi}{\sqrt{N}} dV = 1$$

In this case, $\frac{\psi}{\sqrt{N}}$ is the normalized wavefunction

Eigen states, Eigen function and Eigen values:

In quantum mechanics, a general state of a ψ system is a superposition of a number of states with different probabilities.

$$\psi = c_1\psi_1 + c_2\psi_2 + \dots\dots\dots + c_n\psi_n = \sum_n c_n\psi_n$$

These states are called eigenstates and the wavefunctions ψ_i representing these states are called eigen functions or eigenstates.

Number

In quantum mechanics, each physical quantities like momentum, energy etc are represented by operators. The value of the quanity in an eigenstate can be determined by operating the corresponding operator on the eigenfunction. This value is called eigenvalue.

$$\hat{A}\psi_i = a_i\psi_i \dots\dots\dots\dots (1)$$

The above equation is called eigenvalue equation. $\hat{A}$ is an operator and a_i is the eigenvalue.

A operates on ψ_i to give back ψ_i multiplied by a quantity a_i

ψ_i are in general complex numbers. But eigenvalues are real quantities. They correspons to real physical quantities called observables.

(**Numbers of definite states which are allowed for the system are called eigen states. The wave function associated with eigen state is called eigen function. The eigen values of a physical quantity of a given system is defined as the set of permitted values of the physical quantity of the given system.)**

Momentum, Kinetic energy, and energy operators

We consider a free particle wave function moving along x - direction

The free particle wave function is $\psi = Ae^{\frac{i}{\hbar}(p_x x - Et)}$

$$\frac{\partial \psi}{\partial x} = \frac{i}{\hbar} p_x A e^{\frac{i}{\hbar}(px - Et)} = \frac{i}{\hbar} p_x \psi$$

or, $-i\hbar \frac{\partial \psi}{\partial x} = p_x \psi$

$-i\hbar \frac{\partial}{\partial x}$ operates on ψ to give a value p_x

Therefore, $-i\hbar \frac{\partial}{\partial x}$ is called momentum operator ; in 3D the momentum operator is $-i\hbar\nabla$

Kinetic energy $= \frac{p_x^2}{2m}$, Therefore Kinetic energy operator is $\frac{1}{2m} p_x p_x = \frac{1}{2m} p_x p_x = \frac{1}{2m} -i\hbar \frac{\partial}{\partial x} -i\hbar \frac{\partial}{\partial x} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$

$$\frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar} A e^{\frac{i}{\hbar}(px - Et)} = -\frac{iE}{\hbar} \psi \quad ; \text{or}, i\hbar \frac{\partial \psi}{\partial t} = E\psi$$

Therefore, $i\hbar \frac{\partial}{\partial t}$ is called energy operator

Expectation Value:

In quantum mechanics we talk of probable values of the physical quantities instead of actual values (due to inherent uncertainty). If we make several measurements of a physical parameter of a system under similar situations then all the measured values do not come out same. The average of all these measured values of a physical quantity is called its expectation value.

For any physical quantity defined by an operator O, the expectation value is given by

$< O > = \int_{-\infty}^{\infty} \Psi^* \, \hat{O} \, \Psi \, dx$, in one dimension.

Here, Ψ^* is the complex conjugate of Ψ.

For position, $< x > = \int_{-\infty}^{\infty} \Psi^* \, \hat{x} \, \Psi \, dx$

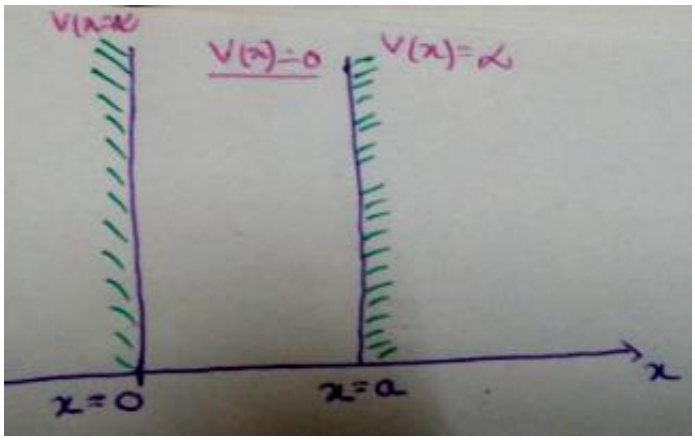
For momentum, $< p > = \int_{-\infty}^{\infty} \Psi^* \, \hat{p} \, \Psi \, dx$

Applications of Quantum Mechanics:

(1) Particle in a one-dimensional box (infinite deep potential well):

We consider one dimensional motion of a particle of mass m confined between two rigid walls of a box of width *a*. The potential in the box is zero and the potential at the rigid walls is infinity. The particle is confined in the box due to infinite potential of the walls. **This is one-dimensional particle in a box problem.**

This is the example of a bound system. Classically, the energy of the particle is continuous. Quantum Mechanics dictates that the energy of the particle will be discrete integral values.



The form of the potential is

$V(x) = 0 \quad \text{for } 0 < x < a$

$V(x) = \infty \quad \text{for } x < 0 \text{ and } x > a$

Figure looks like a potential well of infinite height.

The probability of finding the particle in the region outside the box is zero due to infinite potential

$P = |\psi(x)|^2 = 0 \quad \text{for } x < 0 \text{ and } x > a$

or, $\psi(x) = 0 \quad \text{for } x < 0 \text{ and } x > a$.....(1)

The time independent Schrodinger equation inside the box is -

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

or, $$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}E\psi = 0 \quad [\because \mathrm{V} = 0]$$

or, $$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \text{.....(2)},$$

$$k^2 = \frac{2mE}{\hbar^2} \text{.....(3)}$$

The general solution of eq(2) is

$\psi = \mathrm{C}_1 e^{ikx} + \mathrm{C}_2 e^{-ikx}$, which can be written as

$\psi = A\,Sinkx + B\,Coskx$ .....(4)

The coefficients A and B are determined from the boundary constant that

ψ and $\frac{d\psi}{dx}$ should be finite and constant at the boundaries $x = 0$ and $x = a$

At $x = 0, \psi = 0$ for $x < 0$ and $\psi = ASinkx + BCoskx$ for $x > 0$

So from the condition of continuity of ψ we get -

$ASinkx + BCoskx\big|_{x=0} = 0$

or, $B = 0$

So, form eq(4),

$\psi = ASinkx$.....(5)

<u>Energy eigenvalues</u>

From the continuity condition of ψ at $x = a$, we get $\psi(x) = 0$ at $x = a$

$\psi = ASinkx\big|_{x=a} = 0$

$A \neq 0$, because this makes ψ(x) = 0 at all values of x

So, $Sinka = 0$ or $ka = n\pi$, where $n = 1,2,3,4.....$ [n = 0 is excluded because $k \neq 0$ and $a \neq 0$]

or, $$k = \frac{n\pi}{a} .....(6)$$

From eq(3), $k^2 = \frac{2mE}{\hbar^2}$

Therefore, <u>The energy of the particle</u> $E = \frac{k^2\hbar^2}{2m} = \frac{\pi^2\hbar^2}{2ma^2}n^2$.....(7)

Equation (7) shows that the energy of a particle in an one - dimensional box is discrete or quantized. Different values of the integer n corresponds to different state of the system.

<u>Features</u>

(i) Energy eigen values are discrete. The energy of the particle is lowest for $n = 1$, $E_1 = \frac{\pi^2\hbar^2}{2ma^2}$.

This is called the ground state. The energy E is called zero point energy. Ground state energy/zero pont energy is non - zero. The states n = 2,3,.... are called excited states. $n = 2$ is 1st excited state, $n = 3$ is called 2nd excited state etc.

(ii) The energy of the higher states is given by $E_1 \times$ (integer number)^2 *i.e.* En = n^2 E_1.

(iii) The spacing between two consicutive energy level is $(2n+1)E_1$, non - uniform/not - equispaced.

(iv) The spacing between the energy states increases with increasing n.

Eigenfunctions of particle in a box :

From the normalization condition we know $\int_{-\infty}^{\infty} |\psi|^2 dx = 1$

In the present case, $\psi = 0$ for $x < 0$ and $x > a$, therefore

$$\int_0^a |\psi|^2 dx = 1 \quad \text{or} \, |A|^2 \int_0^a Sin^2 kx dx = 1, \quad \text{or} \, |A|^2 \int_0^a \frac{1 - Cos2kx}{2} dx = 1,$$

$$\text{or} \quad \frac{|A|^2}{2}\left[x\Big|_0^a - \frac{Sin2kx}{2k}\Big|_0^a \right] = 1$$

$$\text{or,} \quad \frac{|A|^2}{2}\left[a - 0 - \{Sin2ka - o\}\right] = 1,$$

From eq (5), $ka = n\pi, \therefore Sin2ka = Sin2n\pi = 0$

Therefore, $\frac{|A|^2 a}{2} = 1, \quad \text{or} \, A = \sqrt{\frac{2}{a}}$

The eigenfunctions of particle in a box are therefore

$$\psi_n(x) = ASink_n x = \sqrt{\frac{2}{a}} Sin\frac{n\pi x}{a} \quad , n = 1,2,3.....$$

This is the normalized wave function.

First few eigenfunctions are

$$For \;\; n = 1, \quad \psi_1(x) = \sqrt{\frac{2}{a}} Sin\frac{\pi x}{a} \;, \qquad E_1 = \frac{\pi^2 \hbar^2}{2ma^2}$$

$$For \;\; n = 2, \quad \psi_2(x) = \sqrt{\frac{2}{a}} Sin\frac{2\pi x}{a} \;, \qquad E_2 = \frac{4\pi^2 \hbar^2}{2ma^2}$$

$$For \;\; n = 3, \quad \psi_3(x) = \sqrt{\frac{2}{a}} Sin\frac{3\pi x}{a} \;, \qquad E_3 = \frac{9\pi^2 \hbar^2}{2ma^2}$$

De-Broglie wavelength for particle in a box:

K = $\frac{n\pi}{a}$ = $\frac{2\pi}{\lambda}$

Hence, λ = $\frac{2a}{n}$

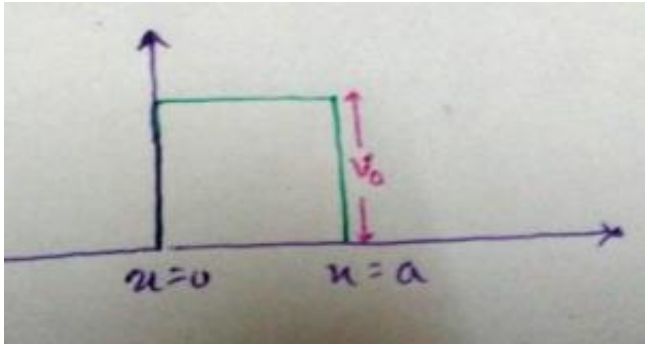
Ground state, n=1, So, λ_{ground} = 2a (maximum wavelength)

For first excited state, n=2, $\lambda_{1st\ excited}$ = a

Similarly, for other excited states de-Broglie wavelength can be calculated.

(2) Potential Barrier:

The sharpest increase of potential energy to certain value at a point and a sharpest fall of the potential energy to zero at another point and remaining constant continuously over the interval constitute a potential barrier.



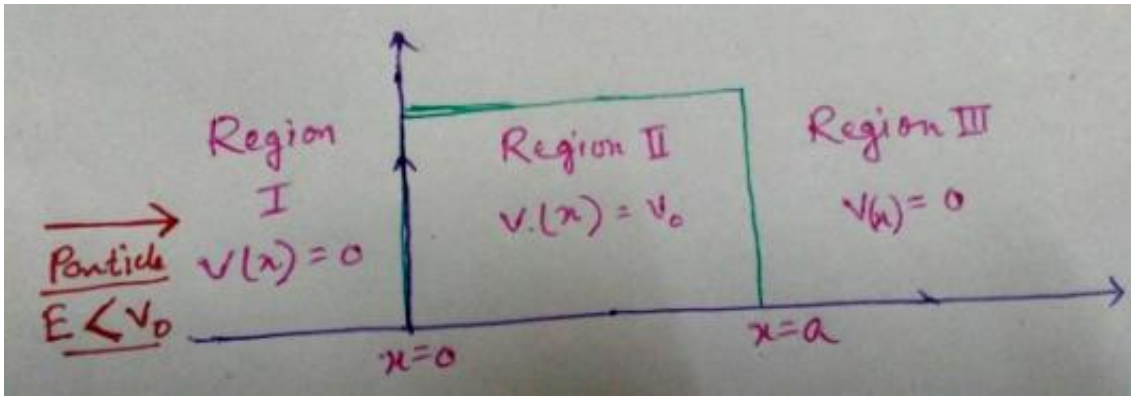
A potential barrier of width a is represented by

$V(x) = V_o$ for $0 \le x \le a$

$= 0$ for $x < 0$ and $x > a$

We consider a particle of mass m is incident on the potential barrier from the left
The energy of the particle is $E < V_0$
According to classical physics, the particle can not go across the barrier
According to quantum mechanics, there is a finite probability that the particle will go across the barrier.

Setting up the Schrodinger equation for one dimensional pitential barrier
We divide the one dimensional region into Region I, Region II, and Region III
Schrodinger equation is solved in these three regions with the boundary condition that the wave function ψ and $\frac{d\psi}{dx}$ are finite and continuous at the boundaries



Region I : $x < 0, \quad V_0 = 0$

$$\frac{d^2\psi_1}{dx^2} + \frac{2mE}{\hbar^2}\psi_1 = 0$$

$$\text{or, } \frac{d^2\psi_1}{dx^2} + k^2\psi_1 = 0 \text{.....(1)}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

Region II : $0 < x < a, \quad V_0 = 0$

$$\frac{d^2\psi_2}{dx^2} + \frac{2m(E - V_0)}{\hbar^2}\psi_2 = 0$$

$$\frac{d^2\psi_2}{dx^2} - \alpha^2\psi_2 = 0 \text{.....(2)}$$

$$\alpha^2 = \frac{2m(V_0 - E)}{\hbar^2}$$

Region III : $x > a, \quad V_0 = 0$

$$\frac{d^2\psi_3}{dx^2} + \frac{2mE}{\hbar^2}\psi_3 = 0$$

$$\text{or, } \frac{d^2\psi_3}{dx^2} + k^2\psi_3 = 0 \text{.....(3)}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

The general solution of Equation (1), (2), and (3) are

$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$ in Region I

$\psi_2(x) = Ce^{\alpha x} + De^{-\alpha x}$ in region II

$\psi_3(x) = Fe^{ikx} + Ge^{-ikx}$ in Region III

Ge^{-ikx} represents a particle wave going from right to left. But we assumed the particle is incident from left. So, there can not be any matter wave coming from right in Region III. So $G = 0$. Therefore the solutions are -

$\psi_1(x) = Ae^{ikx} + Be^{-ikx}$ in Region I

$\psi_2(x) = Ce^{\alpha x} + De^{-\alpha x}$ in region II

$\psi_3(x) = Fe^{ikx}$ in Region III

Classically, A particle of energy $E < V_0$ can not penetrate the barrier of height V_0. Using Quantum mechanics we see that the wave function ψ_3 is nonzero in region III, which shows that the probability of finding the particle in region III is fintite. So a particle incident from left on the barrier with $E < V_0$ can penetrate through the barrier and appear on the other side of the barrier. This phenomenon is called quantum mechanical tunneling or simply tunneling.

The coefficients A, B, C, D, and F are determined from the boundary condition that ψ and $\frac{d\psi}{dx}$ should be finite and continuous at the boundaries $x = 0$ and $x = a$.

$$\psi_1\big|_{x=0} = \psi_2\big|_{x=0} \quad \text{and} \quad \left.\frac{d\psi_1}{dx}\right|_{x=0} = \left.\frac{d\psi_2}{dx}\right|_{x=0} \quad .....(4)$$

$$\psi_2\big|_{x=a} = \psi_3\big|_{x=a} \quad \text{and} \quad \left.\frac{d\psi_2}{dx}\right|_{x=a} = \left.\frac{d\psi_3}{dx}\right|_{x=a} \quad .....(5)$$

Transmission Coefficient :

The probability of transmission of a particle through the potential energy barrier is defined by transmission coefficient T

For a very wide and high potential barrier, The transmission probability can be approximated as

$$T = \frac{16}{4+\alpha^2/k^2} e^{-2\alpha a} \approx e^{-2\alpha a} \quad .....(6) \quad \left[\because \frac{16}{4+\alpha^2/k^2} \approx 1\right]$$

Examples of Quantum mechanical Tunneling

(i) Emission of a particle from radioactive nuclei: The a particle inside the nucleus is under

attractive nuclear force. It is under a potential well. The energy of the alpha particle is less than the potential well. But the a particle comes out of the nucleus through tunneling.

(ii) Nuclear Fusion: Two light nuclei comes close to each other and undergo fusion. The coulomb repulsion acts as a barrier as the particles come close to each other. The particles can undergo fusion at lower energies because they tunnel through the coulomb barrier.

Devices developed using tunneling effect: Tunnel diode, Josephson Junction, Field emission devices, Scanning tunneling microscope etc.

Questions discussions

Short Questions

1. Mention two differences between classical physics and quantum physics.
2. What is wave function in quantum physics? Write its physical significance.
3. Write the Schrodinger's time independent wave equation for a free particle moving along z- axis with low speed.
4. Write the Schrodinger's time dependent wave equation for a particle moving along x- axis with low speed.
5. State and explain the superposition principle for a wave function.
6. What do you mean by normalization of wave function?
7. What is probability density? Derive an expression for it.
8. Write the dimension of wave function of matter wave in 1-d, 2-d and 3-d.
9. What do you mean by eigenfunctions of physical system?
10. What do you mean by eigenvalues of physical quantity?
11. Derive an expression for energy operator in quantum physics.
12. Write the expression for energy and momentum operator in 3D.
13. Derive an expression for momentum operator in quantum physics.
14. What do you mean by expectation value of a physical quantity in quantum physics? Write an expression for position expectation value.
15. Prove that the momentum of a particle in a one-dimensional well of infinite height is quantized.
16. Prove that the de-Broglie wavelength of a particle in an infinite deep potential well is quantized.
17. A particle of mass 'm' is enclosed inside a potential well of infinite height. Show that de-Broglie wave length of the particle in the first excited state is equal to the width of the well.
18. A particle of mass 'm' is enclosed inside a potential well of infinite height. Show that maximum de-Broglie wave length of the particle is twice the width of the well.
19. A particle of mass 'm' is enclosed inside a potential well of infinite height. Find the maximum de-Broglie wave length of the particle?
20. If a particle trapped in infinitely deep potential well has de-Broglie's wavelength 'λ' in the ground state then what is its wavelength in the first excited state?

21. Why the ground state energy of a free particle moving in a one dimensional potential box is non zero?
22. What is the eigenfunction of a particle trapped in an infinite deep potential well in its second excited state?
23. An electron confined in a one dimensional box of width L is known to be in its first excited state. Determine the probability density of electron in the central half.
24. Define a potential barrier.
25. Write the potential function of a potential barrier of width 'a' and height V_0.
26. What is quantum mechanical tunneling? Give two examples of it.

Long Questions

1. A particle of mass 'm' and energy 'E' is moving non-relativistically along X-axis, under the action of potential energy 'V'. Develop the Schrodinger's time dependent wave equation for the particle. Discuss the above equation for a free particle.
2. What do you mean by wave function of a matter wave? Write its important characteristics. A particle trapped in a one-dimensional box of length L is described by the wave function $\Psi = x$. Normalize the wave function between a and b.
3. (i) Define probability density. What is the total probability of a system in different states? (ii) The normalized wave function for certain particle is $\psi(x)= 3/\pi \cos x$, in the region $-\pi/2<x<\pi/2$. Calculate the probability of finding the particle in the region $0<x<\pi/4$.
4. Describe operators, eigen functions and igen values in relation to quantum mechanics. Write down the expression for linear momentum operator and energy operator in quantum mechanics.
5. Write the Schrodinger's equation for a particle confined in a one dimensional box of infinite height. Show that the energy of the particle is quantized.
6. Derive an expression for the normalized wave function of a particle trapped in an infinite deep potential well.
7. Define a potential barrier? Set up Schrodinger's time independent equation in different regions of a potential barrier, when energy of the incident particle is less than the height of the potential barrier. Explain how quantum physics differs from classical physics in this problem.
8. What is quantum mechanical tunneling? Write some examples of it.
   (ii) A beam of electrons are incident on a barrier of height 6.0 eV and 0.2 nm wide. Find the energy they have, if 1.0% of them are tunnel through the barrier.

Numerical

1. A particle can exist in the states ψ_1, ψ_2 and ψ_3 with probabilities 1/2, 1/3 and 1/6 respectively. Write the wave function of the particle. If the energy eigen values in the three states are E_1 = 2eV, E_2 = 4 eV and E_3 = 8 eV respectively, then find the expectation value of the energy of the particle.
2. Calculate the probability of finding a particle in the region $2\leq x\leq 4$, if the wave function for the particle is given by $\psi(x)=0.25\ e^{2ix}$.
3. The wave function for certain particle is $\Psi = A \cos^2 x$ for $-\pi/2 < x < \pi/2$. Find the normalized wave function for the particle.

4. Normalize the wave function in one-dimension

$$\psi(x) = A\, e^{-\alpha x}, \quad \text{for } x > 0$$

$$\psi(x) = A\, e^{\alpha x}, \quad \text{for } x < 0$$

here, α is a positive constant.

5. If the wave function of a quantum particle $\psi(x) = \sqrt{\frac{2}{L}}\ \sin\frac{n\pi x}{L}$., find the probability of the particle between x=0 and x= $\frac{L}{2}$ in the first excited state.
6. A particle trapped in a one dimensional box of length 1 cm, is described by the normalized wave function $\psi = x$. What is the expectation value of the particle's position $< x >$?
7. A particle of mass m is trapped in a one-dimensional infinite potential well of length L is described the wave function $\psi(x) = A \sin\frac{n\pi x}{L}$. Normalize the wave function.
8. Energy of an electron confined in a one dimensional potential box of 0.2 nm size is 151 eV. What is the order of the state in which the electron exists? h= 6.6 X 10^{-34} Js.
9. The ground state energy of a particle in an infinite one-dimensional potential well is 8 eV. If the width of the well is halved, what is the new ground state energy?
10. The ground state energy of a particle, trapped in a one-dimensional infinite potential well is 4 x 10^{-19} J. Find the energy of the particle in second excited state.
11. An electron is constrained to a one-dimensional potential box of 0.1nm side. Find the energy eigenvalue for 2nd excited state.
12. The energy of an electron in a one dimensional potential box of length 4 Å is 9.664 X 10^{-17} J. Find the order of the state and the momentum of the electron in this state.
13. If the energy in the second excited state of an electron trapped in a 1-D box is 77.4 eV, find the width of the box. h= 6.6 X 10^{-34} Js.
14. An electron is trapped in a one-dimensional box (with perfectly rigid walls) of length 2 Å. How much energy is required to excite the electron from the first excited state to the second excited state?
15. The ground state energy of a particle, trapped in a one-dimensional infinite potential well is 3 x 10^{-19} J. Find the energy of the particle in second excited state.
16. Find out the energy gap between the 2nd and 3rd excited states of an electron confined in a potential box of length 4Å.
17. 1.2 million electrons with energy 3.0 eV are incident on a potential barrier of 9.0 eV high and 0.5 nm width. How many electrons can tunnel through the barrier?
18. A beam of electrons are incident on a barrier of height 6.0 eV and 0.2 nm wide. Find the energy they should have if 1% of them are to tunnel through the barrier.
19. Electrons with energy 3 eV are incident on a potential barrier of 10 eV high and 4 Å wide. Find the transmission probability.

Module – 5: LASER AND OPTICAL FIBER

Introduction:

The term laser is the acronym for Light amplification by stimulated emission of radiation

The principle of laser was developed by the American physicist Charles Townes and, independently by Soviet physicist A Basov and N Prokhorov in 1950. These three physicists shared the Novel Prize in 1964.

Here are the important steps that led to the discovery of the laser:

- 1887: Heinrich Hertz accidentally discovers the photoelectric effect. This lucky breakthrough will allow Albert Einstein to introduce the notion of photons.
- 1901: The scientific conundrum known as "the ultraviolet catastrophe" (spectral energy densities diverge at high frequencies) is solved by Planck. He hypothesises that the energy of a type of frequency v is not a random continuous variable but a random discrete set of variables, represented by the values nh v. Interestingly, Planck and his contemporaries at first find it very difficult to accept this idea of discrete leaps in energy. However, subsequent experiments prove that the theory is entirely correct.
- 1905: Einstein introduces a means to quantify electromagnetic energy. The photon is born. Unfortunately, the arrival of the photon cannot take into account the phenomenon of black body radiation (the spectral density of electromagnetic energy emitted by an enclosed area at a temperature T and at thermal equilibrium). However, shortly afterwards, Born devises a means to quantify the energy levels of electrons (1913). This in turn allows Einstein to prove that photons and black body radiation are in fact compatible thanks to the notion of stimulated emission.
- 1949: Kastler and Brossel develop the first optical pumping and the first population inversion. By 1950, the first MASERs appear (Microwave Amplification by Stimulated Emission Radiation), devices that are capable of amplifying an electromagnetic wave in the microwave region (Weber, Townes and Basov).
- 1954: The first MASER is built (an ammonia maser with a 13 mm wavelength). The electromagnetic wave is confined in three dimensions by a "box" and is reflected off its sides. However, this is still in the microwave rather than the optical domain. In fact, scientists at the time thought it was impossible to make an optical laser because the cavity would have to be incredibly small (of the order of magnitude of a wavelength i.e. only tens of µm at the most!).
- 1958: Schawlow and Townes decide to use an open Fabry-Pérot cavity for their experiments. The idea is to confine the electromagnetic field like in a closed box but in only one dimension: the main axis of light propagation in the cavity. This means that

only certain specific electromagnetic waves are amplified, but the resulting beam is much more powerful than when using a closed cavity.

- 16th May 1960: Maiman demonstrates the first ever optical laser effect. The amplifying medium is a ruby, the crystal most used in early lasers because it was already well known from its application in MASERs. This is a pulsed operation laser with a wavelength of 694.3 nm.
- 1961: Javan, Bennet and Herriot build the first gas helium-neon laser operating continuously at 1.15 μm . In fact, this laser can emit over a whole range of discrete wavelengths, from green to infrared via orange and red (633 nm).
- 1962: First red helium-neon laser.
- 1965: First semiconductor lasers.
- 1966: First coloured pulsed lasers (red, orange, yellow).

Light emission process (atomic transition/electronic transition):

When an electron in an atom, transits (drops) from a high energy state to a lower energy state, the energy lost by electrons is emitted as a photon. In a solid, when an electron drops from higher energy band to a lower energy band through a band gap, the energy lost by electron is emitted as a photon.

Light emission process involves

(1) absorption of photons by electrons and transition to higher energy state

(2) Transition of electron from higher to lower energy state and emission of the energy difference between two states as photons-this can happen in two ways –

(a) Spontaneous emission without external disturbance/energy

(b) Stimulated emission which occurs due to interaction of electrons in higher energy state with another external photon. ***This type of emission happens in laser***.

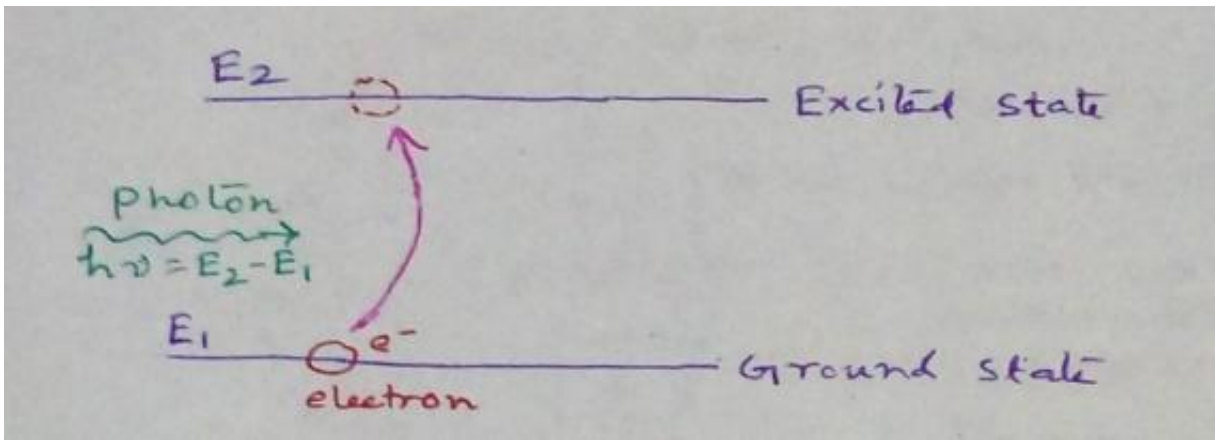
The details of the above mentioned processes are discussed below.

Stimulated Absorption: We consider a group of atoms of any material on which light is falling, i.e. the material atoms are subjected to photon irradiation. We also consider two energy levels in the atoms, one lower and one higher energy level. The lower energy level E_1 is occupied by electrons and the higher energy level E_2 is empty. ***We call the lower energy level ground state and higher energy level as excited state.***

If the photon energy *hv* is equal to the energy difference ($E_2 – E_1$), then electrons can absorb the photon and go to higher energy state E_2.

The transition of an electron from lower energy level E_1 to higher energy level E_2 by absorbing a photon of suitable energy ($h\nu = E_2 - E_1$), is called stimulated absorption. After absorption of a photon by an electron of an atom, the atom is said to be in excited state. The rate of stimulated absorption depends on population of electrons at the lower energy level, N_1 and the incident energy density.

atom+photon → atom* (atom in excited state)

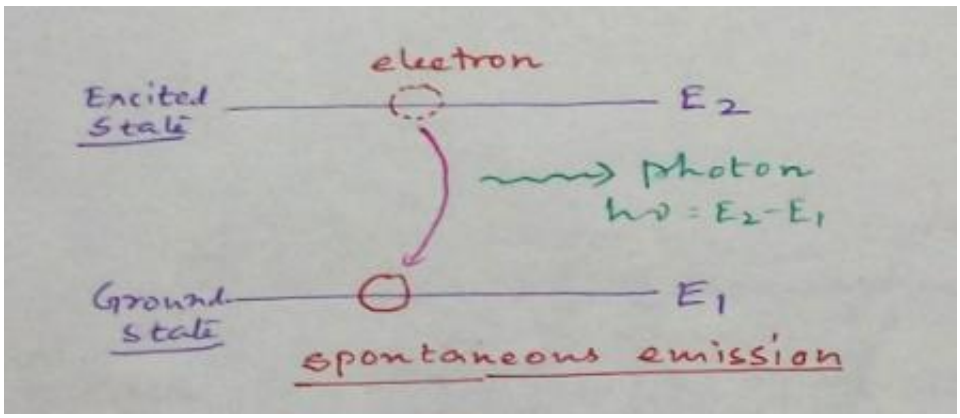


Spontaneous emission:

The lifetime of electron in the higher energy level is very small ~10^{-8} second, on the average. After this time, electron drops to lower energy state. The energy lost by the electron ($E_2 - E_1$), is emitted as a photon having the same energy ($h\nu = E_2 - E_1$). This is called spontaneous emission. The rate of spontaneous emission depends on population of electrons at the higher energy level, N_2.

Electron at the excited state E_2 may drop to the ground state E_1 spontaneously without any external radiation is called spontaneous emission

The photons are emitted at different direction. The excited atoms emit photons at different times. So the emitted photons are incoherent.



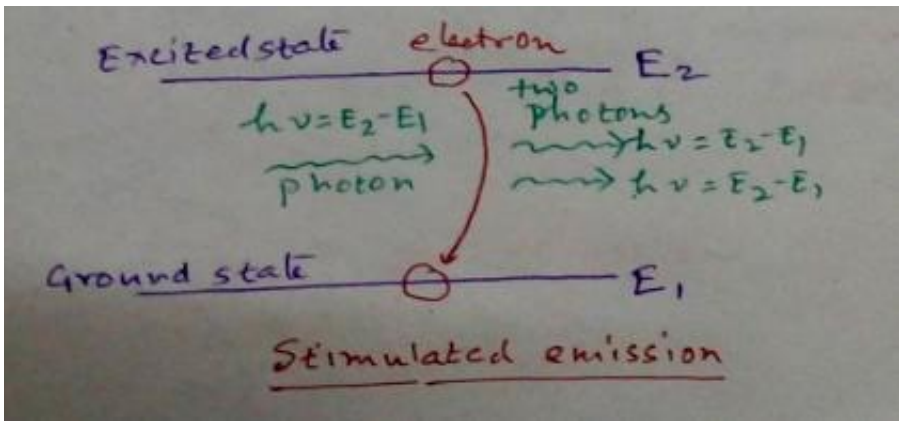
Stimulated emission:

We consider an electron in a higher energy level E_2 level (excited state). If photon of energy $h\nu = E_2 - E_1$ is irradiated on an electron in higher energy then the electron may interact with the photon and drops to the ground state. In this process, two photons are emitted. This emission is called the stimulated emission or lasing action. The rate of stimulated emission depends on population of electrons at the higher energy level, N_2 and the incident energy density.

The process of emission of an electron from excited state E_2 to ground state E_1 by absorbing a photon of suitable energy ($h\nu = E_2 - E_1$) is called stimulated emission. In this process, two identical photons are emitted.

Characteristics of emitted photons:

- The photons are emitted in the same direction (direction of the incoming photon).
- The photons are in phase: they are coherent.
- photons are monochromatic (frequency $n = E_2 - E_1/h$)



The three processes occur simultaneously when light is incident on the system. At steady state, a balance is achieved between absorption and emission. The process stimulated emission is of interest in lasers.

Electron distribution in a two state system and principle of laser:

Let the total number of atom in the system be N_o.

According to Maxwell-Boltzmann Statistics, the number of atoms in state E_1 (i.e. electron in the atoms are in state E_1) is

$$N_1 = N_o e^{(-E_1/kT)}$$; k is Boltzmann constant and T is temperature

Similarly, no. of atoms in state E_2 is

$$N_2 = N_o e^{(-E_2/kT)}$$

So, $N_2/N_1 = e^{-(E_2 - E_1)/kT}$ .....(1)

We know $E_2 > E_1$, so we get from eq. (1) : $N_2 < N_1$

This means that the number of atoms in excited state is lower than the number of atoms in the ground state. Thus number of electrons in excited state E_2 is small and they have small life time (~10^{-8} sec) in the excited state. They quickly drop to the ground state. So, the probability of interaction of electrons in the excited state with incident photon, which is required for stimulated emission, is less. Thus, spontaneous emission of photons is dominating process in general.

[If T<0 (not possible in reality) then $N_2>N_1$, that is why the situation of is called a –ve temperature state]

To get laser emission, we need dominating stimulated emission of photons. It requires two conditions: (i) larger number of electrons in the excited state ($N_2>N_1$) (ii) larger lifetime of electrons in the excited state.

The condition/situation of larger number of electrons being in the excited state compared to the ground state (($N_2>N_1$) in a material is called population inversion. The system in which population inversion is achieved is called an active system and the method by which population inversion is achieved is called pumping.

Special situation and techniques are necessary to create population inversion, where larger numbers of electrons are there in the excited state than the ground state. This is achieved in a material/atomic arrangement where there are 3 or 4 energy levels with a **metastable energy level or state**, in which electrons have larger lifetime (10^{-3} sec) that any unstable excited state (lifetime ~10^{-8} sec) .

Metastable states are the longer lived higher energy states where life time of an electron is ~ 10^{-3} seconds.

Pumping:

The process of exciting electrons to higher energy state is called pumping. This is done by supplying energy to electrons in a material from outside.

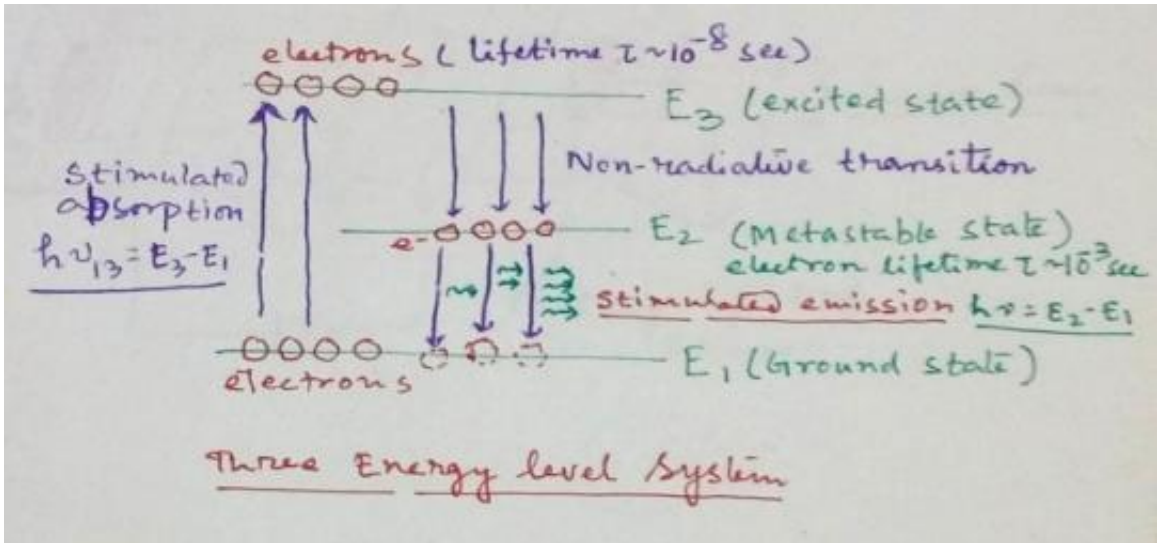
Methods of pumping:

- **(i)** **Optical pumping**: light energy is used for excitations. Suppose in a system (atom), there are two energy states E_1 and E_2. Light of suitable frequency ($h\nu = E_2 - E_1$) is used to excite electrons from E_1 to E_2. Example is ruby laser.
- **(ii)** **Electrical discharge pumping**: Energy is supplied to electrons of a system by passing electrical discharge through it. Example is gas laser like, He-Ne laser, CO_2 laser etc.

(iii) **Chemical pumping**: Heat generated in exothermic reaction is used to excite electrons.

Three Energy Level systems for laser emission:

A three level system consists of a ground state of energy E_1, a high energy state of energy E_3 and a metastable state of energy E_2 in between E_3 and E_1. Electrons are initially in the ground state. During pumping photons of energy $h\nu_{13} = E_3 - E_1$ are irradiated. Electrons absorb the photons and get excited to E_3 level. The lifetime of electrons in the excited level is ~10^{-8} second. In this time, the electrons drop to the metastable state of energy E_2 through non-radiative transition. The energy of the electrons (E_3-E_2) is given to the crystal lattice. The lifetime of electrons in the metastable state E_2 is ~10^{-3} second, much higher than electron lifetime at E_3.



By continuous pumping the metastable state is populated and the ground state E_1 is depopulated. So, population inversion occurs between the states E_2 and E_1. A spontaneous emission of photon ($h\nu = E_2 - E_1$) takes place by electron transition from E_2 to E_1. The photon emitted by spontaneous transition interacts with on electron in state E_2 and induce stimulated emission. Two photons ($h\nu = E_2 - E_1$) are emitted in this process. These two photons further interact with other electrons in state E_2 and four photons are emitted. This process continues and a laser beam is produced.

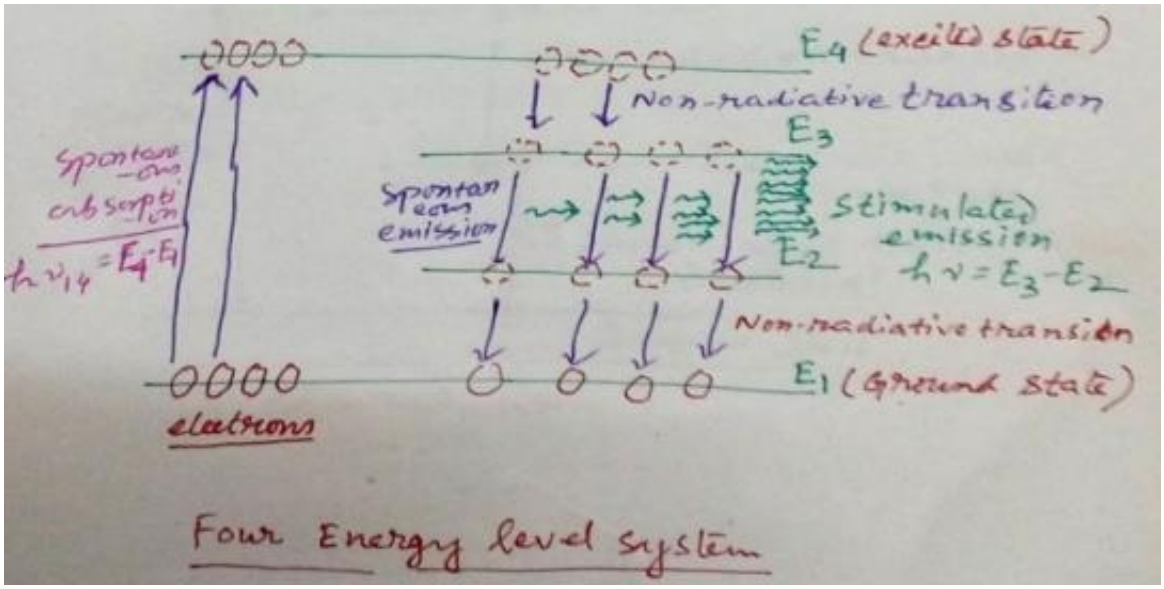
Limitations of three level systems:

(i) The laser is produced in **pulsed manner**. When lasing transition takes place, E_2 is depopulated and E_1 is populated. To achieve population inversion again, one need to wait for some time. So, a material with three levels system emits pulsed laser.

(ii) It requires very high pumping energy, as the lower lasing transition level is ground state.

Four energy level system:

The system is shown in the figure. There are four states involved in laser emission process. In between the ground state E_1 and highest energy state E_4, there are two states, a metastable state E_3 and an intermediate state E_2. During pumping, photons ($h\nu_{14} = E_4-E_1$) are absorbed by the electrons in the ground state E_1 and electrons go to state E_4. From E_4, electrons drop to metastable state E_3 through non-radiative transition. The energy is given to the crystal lattice. Thus, state E_3 is populated and population inversion takes place between state E_3 and E_2.



In the LHS of the fig, it is stimulated absorption, not spontaneous absorption (E_1 to E_3).

A spontaneous emission of photon ($h\nu = E_3-E_2$) takes place by electron transition from E_3 to E_2. The photon emitted by spontaneous transition interacts with one electron in state E_3 and induce stimulated emission. Two photons ($h\nu = E_3-E_2$) are emitted in this process. These two photons further interact with other electrons in state E_3 and four photons are emitted. This process continues and a laser beam is produced. Electrons dropped to state E_2 comes back to ground state E_1 by radiative or non-radiative transition

In this process, the state E_3 remains populated and the state E_2 remains depopulated. So, a four level system emits continuous laser. Here the lower laser transition level is not ground state.

Population inversion cannot be achieved in two energy level system hence no laser action.

According to Maxwell-Boltzmann Statistics, the number of atoms/electrons in state E_1 is

$$N_1 = N_o e^{(-E_1/kT)}$$; k is Boltzmann constant and T is temperature

Similarly, no. of atoms/electrons in state E_2 is

$$N_2 = N_o e^{(-E_2/kT)}$$

So, $$N_2/N_1 = e^{-(E_2 - E_1)/kT} \text{ .....(1)}$$

As $E_2 > E_1$, so we get from eq. (1) : $N_2 < N_1$

Hence population inversion is not achieved.

Characteristics of LASER:

(i) A Laser beam is highly monochromatic
(ii) It is Coherent
(iii) Laser beam is unidirectional
(iv) A laser beam is of high intensity

Structure of Laser:

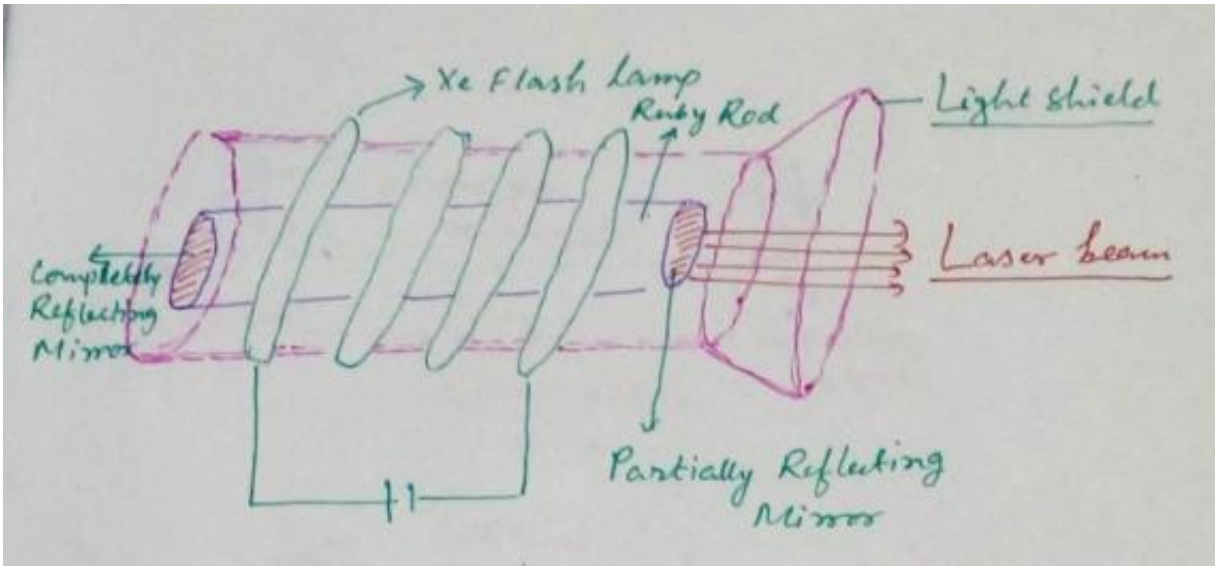
Any laser consists of three essential parts:

(i) Active material: in this material, populations' inversion is achieved and laser emission takes place. Example: Ruby in solid state laser, He-Ne mixture in gas laser.

(ii) Excitation source for Pumping: Energy is supplied by a pumping method to the electrons in the active material for excitation of electron to a higher energy state.

(iii) A resonant cavity: It is the Laser Cavity consisting of an active medium bound between two mirrors. The mirrors reflect the light to and fro through the active medium. So more and more stimulated emission takes place by interaction of electron with light photons and laser Intensity increases.

Ruby laser: (Solid state laser, three energy level system)



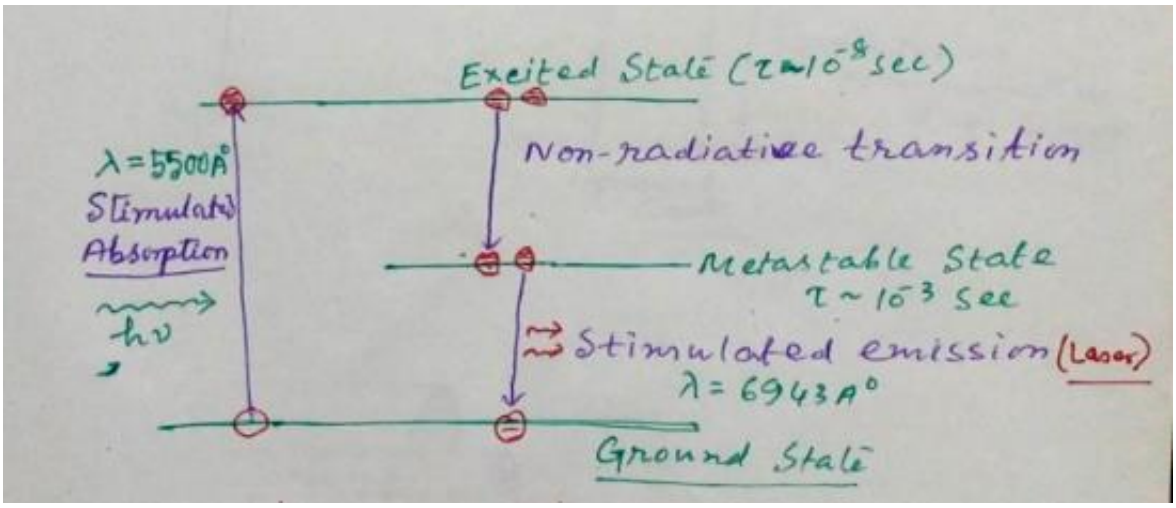
Construction:

Ruby Laser consists of three parts

(i) **Active material**: A rod of Ruby crystal (Al_2O_3 doped with 0.05 % of Cr). Cr ions emit red color laser.

(ii) **A resonant cavity** is formed by two optical mirrors. One is completely reflecting and the other is partially reflecting. Two mirrors are attached to two ends of the ruby rod. Laser is emitted through the partially reflecting mirror.

(iii) **A Xe flash lamp** is used as excitation source for **pumping**.



Working Principle

The ruby rod is illuminated by an intense flash from the Xe lamp. The impulse of light excites Cr ions from ground state to E_1 to excited state E_3. Electrons in the state E_3 are unstable (life time ~ 10^{-8} sec) and they quickly drop from E_3 to a metastable state E_2 by non-radiative transition. The life time of electrons in E_2 is longer (~ 10^{-3} sec), So population inversion is achieved between the states E_2 and E_1. **Laser transition takes place when electron drops from E_2 and E_1.** Emission Wavelength is 6943 A°.

Due to the presence of resonant cavity, laser radiation is intensified. The radiation comes out from the partially reflecting mirror.

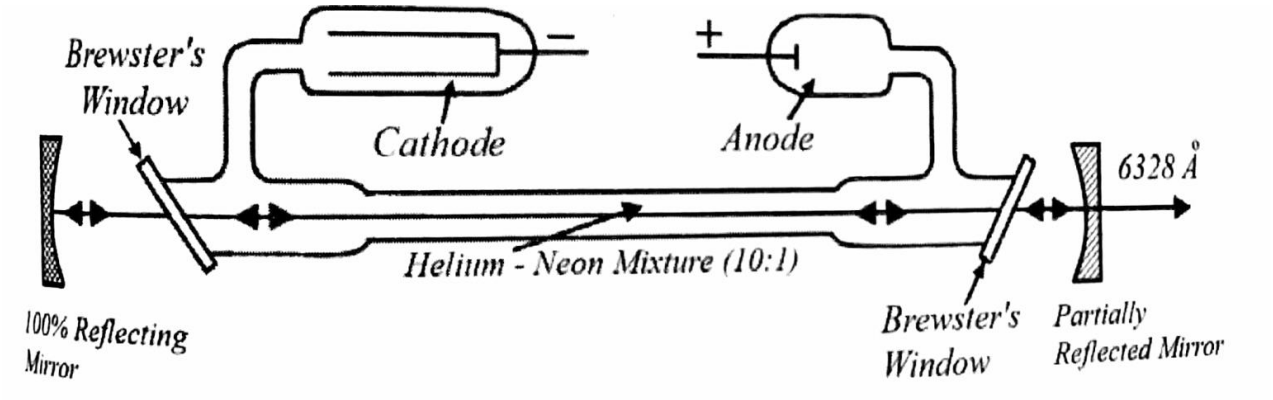
Limitations: same as 3 level laser system.

He-Ne Laser (Gas Laser: 4 level laser system)

Construction:

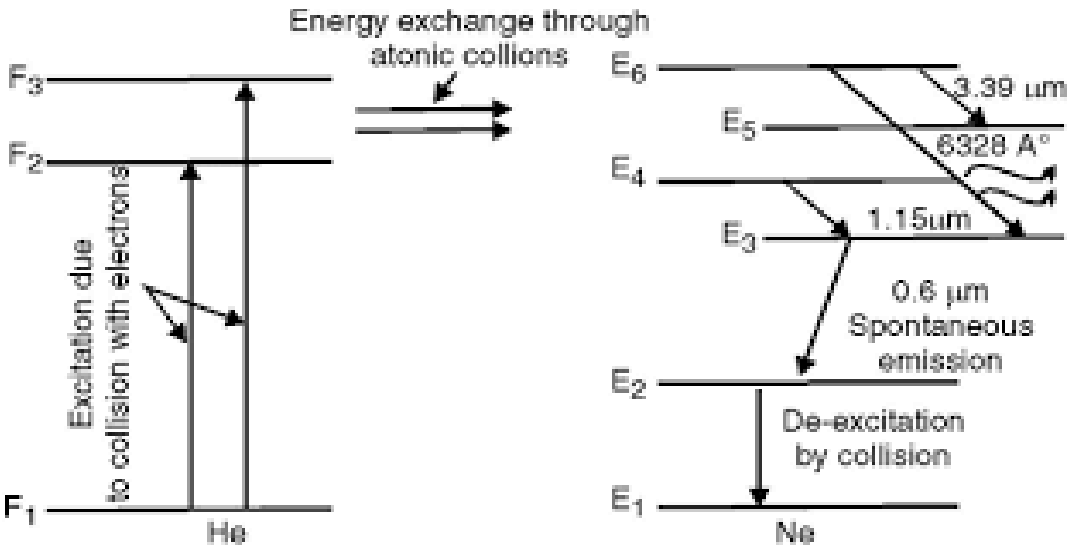
It consists of a long discharge tube of length 50 cm and diameter 1 cm. The tube is filled with a mixture of He and Ne in the ratio 10:1. Electrodes are provided to produce a discharge in the gas and they are connected to high voltage power supply. The tube is sealed by inclined

windows arranged at its end. On the axis of tube two reflectors are fixed which forms resonator. He atoms help to archive population inversion by imparting their energy to the neon atom. Pumping is done by a DC electrical discharge in the low pressure gas in the tube. The wavelength of the laser is 632.8 nm.



Working principle:

He - Ne gas laser employ four level pumping schemes. When the power is switched on the electric field ionizes some of the atoms in the mixture of He and Ne gases. Due to electric field, the electrons and ions will be accelerated. Since electrons have smaller mass they acquire higher velocity and He atoms are lighter in weight and therefore readily excitable.



The energetic electrons excite He atoms to excited states F2 and F3 which lies at 19 eV and 20 eV above the ground state. These are metastable states for helium. Though the radiative transitions are forbidden, the excited He atom can return to the ground state by transferring their energy to Ne atoms through collision. Such an energy transfer can take place only when the two colliding atoms have identical energy states.

E6 and E4 level of Ne atom nearly coincides with F3 and F2 of Helium. Ne atoms acquires energy and goes to excited state and helium atoms return to ground state by transferring

their energy to Ne atoms. This is main pumping mechanism. Ne atoms are active centers and Helium plays the role of pumping agent. The probability of energy transfer from Ne to He atom is less as there are 10 Helium atoms to 1 Neon atom.

E6 and E4 states are metastable states, as collision goes on neon atoms accumulate in these states whereas E5 and E3 level of neon are sparsely populated. Therefore, a state of population inversion is achieved between E6 and E5, E6 and E3 and E4 and E3. Consequently, three laser transitions take place.
E_6-----E_5 33900 A° (far IR region) E_6-------E_3 6328 A° (visible) E_4----E_3 11500 A° (IR region)

As the terminal levels of lasing transitions are sparsely populated the fraction of Ne atom that must be excited to upper level can be much less. As such the power required for pumping is low. Random photons emitted spontaneously sets stimulated emission and coherent radiation is produced.

From E5 and E3 level neon atom can make downward transition to E_2 level. Incoherent light is emitted due to spontaneous transition. As lower levels depopulate faster than upper levels it is easier to maintain population inversion throughout laser operation. E2 is again a metastable state.
Therefore, Ne atoms tends to accumulate at this level again. However, they are made to collide with the walls of discharge tube and they give up their energy and returns to ground state.

Role of Helium atoms:

Being a good conductor of heat, He acts as a coolant and no separate cooling system is required.

He atoms being lighter than Ne atoms absorbs the energy from the high energy electrons easily and very fast.

The ratio He : Ne=10:1 makes the probability of energy transfer for He atoms to Ne atoms much higher than that of the reverse.

Merits:

- Continuous output laser source
- Highly stable
- No separate cooling is required

Demerits:

- Low efficiency and low power output
- Gases are novel medium for laser as gases are found in the purest form so their optical properties are well defined.

Applications of Laser

(i) Laser is used in industry for the purpose of cutting, drilling and welding (due to narrow angular spread and high intensity)

(ii) Medical Science: microsurgery, treatment of retina, cancer treatment, laser angioplasty for clearing the blocked arteries, microsurgery (due to narrow angular spread).

(iii) Communication: micro-communication, large amount of data is transferred

(iv) Hologram: 3D imaging, both phase and amplitude are considered

(v) Distance measurement: LIDAR (light detection and ranging), distance between earth and moon can be measured by pulse echo technique

(vi) Defense: Ranging and guiding weapons. Also it is used as death ray (high antensity and highly directional)

(vii) Computer and printers (laser diode is used). Storage capacity for information in computer is improved due to narrow band width of laser light. Laser printer

(viii) Spectroscopy: Raman spectroscopy and Photoluminescence

(ix) Environmental study, electronics etc

Question discussion:

1. How is laser different from ordinary light?
2. Explain briefly the terms (i) stimulated emission (ii) spontaneous emission (iii) population inversion (iv) metastable state.
3. What does the acronym LASER stand for?
4. Write two important characteristics of it.
5. Draw the population of atoms in different energy levels in atomic systems in equilibrium.
6. Population Inversion is the necessary condition for lasing action. Explain.
7. What is population inversion? Explain why population inversion cannot be achieved in two energy level system.
8. What are the necessary and sufficient conditions for lasing action?
9. Why is a narrow tube used in He-Ne laser?
10. Why the end is faces of a ruby rod silvered?
11. What is the function of He atoms in He-Ne laser?
12. What is the function of Cr^{+3} ions in ruby laser?
13. Write down the ratio of number of He atoms to number of Ne atoms in a typical He:Ne laser. Why such a ratio is maintained?
14. What is positive feedback in laser? Why it is called optical feedback?

15. Explain why a four-level laser is more efficient than a three-level laser.
16. Explain briefly different uses to which laser beams are put.
17. Explain construction and working of ruby laser with the help of a suitable energy level diagram. Write its limitations.
18. Explain the construction and working of He-Ne laser with the help of an energy level diagram. Write its advantage over ruby laser.
19. What does LASER stand for? Write some characteristics of laser. Describe the working of a three energy level Laser system. Mention its limitations.
20. Describe the components of a Laser system.
21. (i) Distinguish between spontaneous emission and stimulated emission. (ii) Why is population inversion referred to as a negative temperature state?
22. Explain the working of a four-level laser system with a suitable energy level diagram.
23. Draw a neat diagram to represent the component of a ruby laser. Explain the operation.

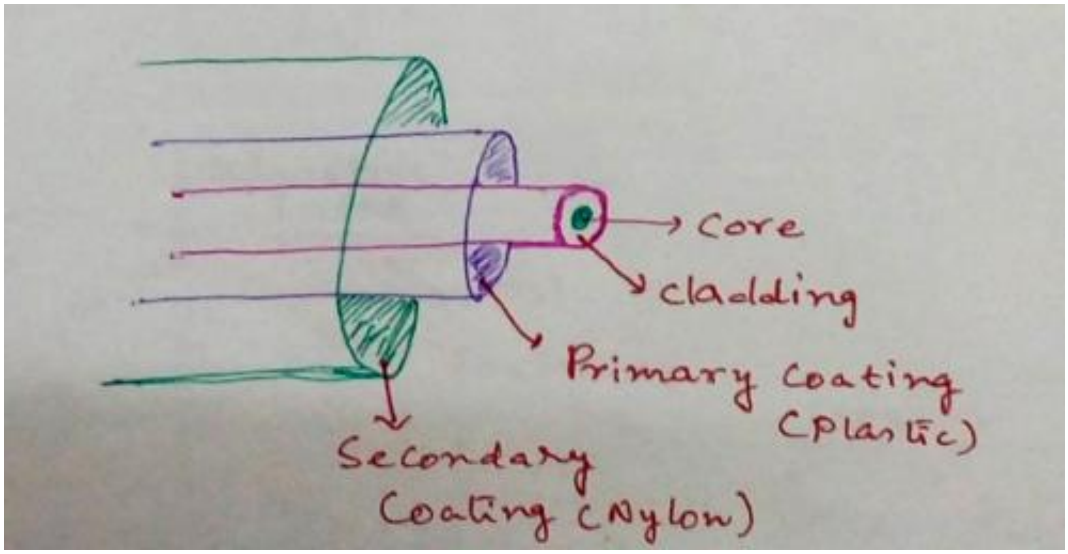
Numerical

1. A Ruby laser emits light of 693.95 nm wavelength. If 1 mole of Cr+3 ions are involved in the lasing process, calculate the pulse energy in eV.
2. If the lasing wavelength in Ruby laser is 694.3 nm, what the ratio of populations of the corresponding energy levels?
3. Calculate the energy of the metastable state in a material where stimulated emission to an energy state at 0.25 eV causes emission of radiation of wavelength 1100 nm.
4. If the wavelength of laser is 632.8 nm, then find the intensity of the laser if the power delivered is 10^3 Watt.
5. A ruby laser has its metastable state at 1.79 eV from which stimulated emission produces laser light. At room temperature when the population inversion is not achieved, calculate the ratio of population of the atoms in the metastable state to that in the ground state.
6. In a material, transition occurs between a metastable state and an energy level of 0.45 eV and the wavelength of the radiation emitted is 2000 nm. Calculate the energy of the metastable state.
7. A laser beam of wavelength 7400Å has coherence time of 4×10^{-5} s. Determine the temporal coherence length.
8. A ruby laser has its metastable state at 1.79 eV from which stimulated emission produces laser light. Calculate the wavelength of laser light.
9. The wavelength of a He-Ne laser generating 3.14 mW power is 632.8 nm. What is the number of photons emitted per minute when it is in operation?

Optical Fiber

Optical Fibers are thin fibers which act as waveguides to transport electromagnetic waves at optical frequencies over very long distance with very low loss of signal. Transmission of data through optical fiber is called photonic devices (metallic wire is electronic devices)

Structure



Core:

The core is the *innermost* part that is made out of glass or transparent plastic. It is extremely thin, flexible, and has a cylindrical shape. Its diameter ranges from 1-100 µm. The core material has high refractive index.

Its sole purpose is to keep all the light within itself. And also to guide the light in a direction parallel to its axis.

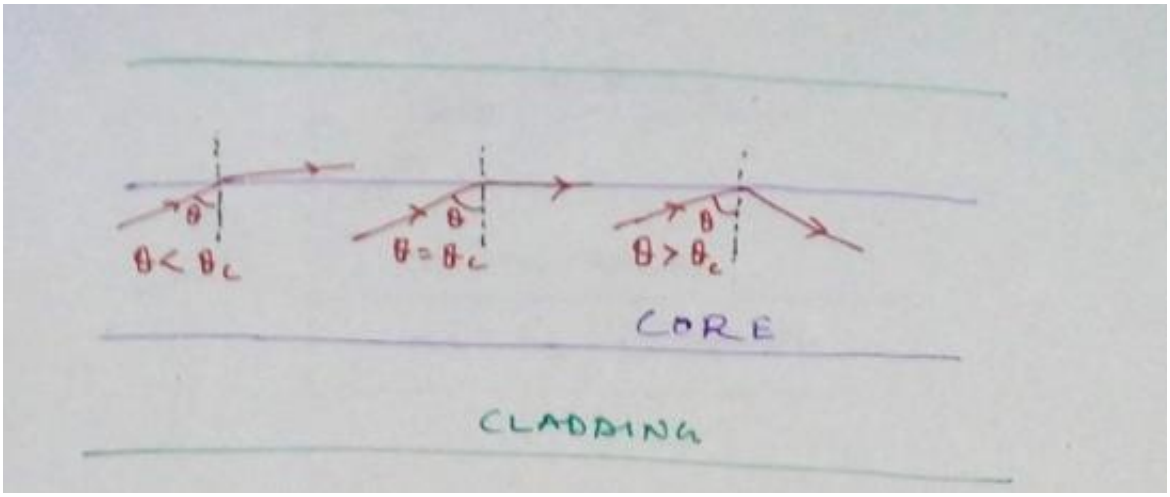
Since it is the primary carrier and guide of the light waves, it can be called an optical waveguide. By the same token, its structure affects the transmission of the light. Hence, all the data that is being transferred will have its transmission parameters or properties based on the structure of this segment of the fiber optic.

Cladding: It surrounds the core. The diameter ranges from 100-300 µm. It is *also* made of glass or transparent plastic. But with a different material, so the refractive index of the cladding is lower than that of the core.

Coating/Jacket: Primary (plastic) and secondary (nylon) coating

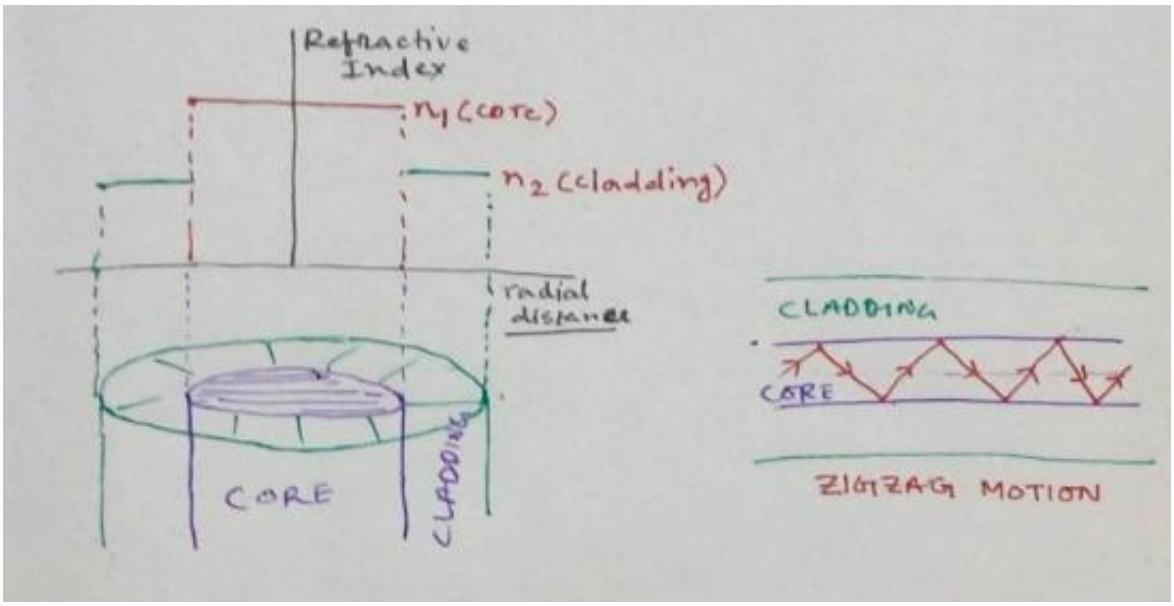
Principle of optical fiber:

When light travels from a high refractive index (r.i) material to a low r.i material, the light rays can get completely reflected when angle of incident $\theta > \theta_c$, *the* critical angle of incidence. The rays travel from one to another end of the optical fiber without much attenuation (loss) through multiple reflections.



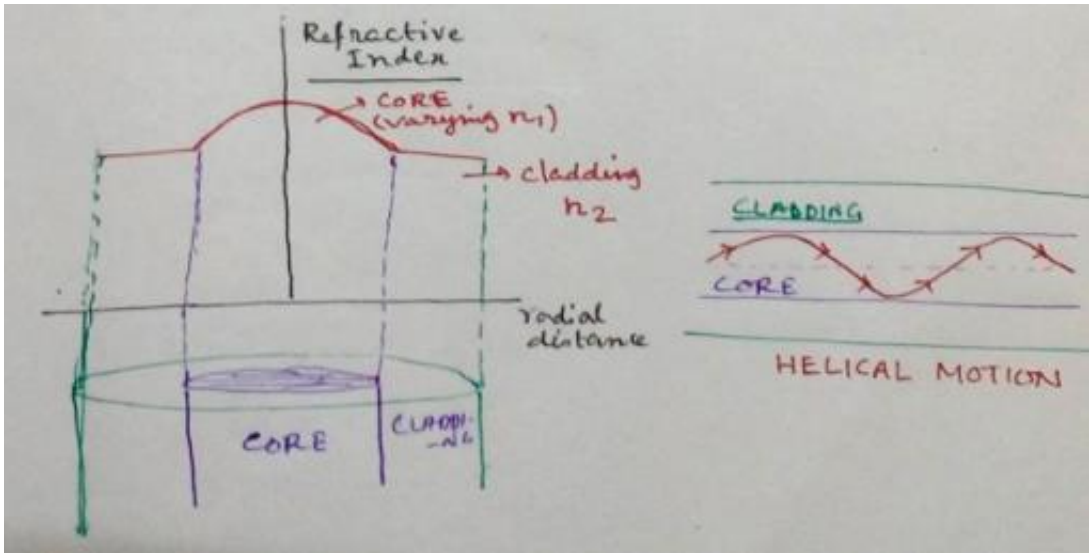
Types of optical fibers

Step index optical fiber: The refractive index of the core is constant. The refractive index of the cladding is also constant but less than the core. It cannot be used for long distance multimode transmission due to energy loss.



Graded Index optical fiber: The refractive index of the core is maximum along the axis of the optical fiber and gradually decreases towards the core/cladding interface. Refractive index of the cladding is uniform.

It is used in long distance communication



Difference between step index and graded index optical fiber

	Step index	**Graded index**
(i)	Structure (written above)	
(ii)	Schematic diagram (drawn above)	
(iii)	Light follow zigzag path	Light follow helical path
(iv)	Use in short distance communication	Use in long distance communication
(v)	More dispersion of light rays	Less dispersion

Types of optical fiber based upon use: (i) Single mode (ii) multimode optical fiber

Single mode optical fiber can carry only one electromagnetic field configuration (signal). Multimode optical fibers can carry more than one electromagnetic field configuration (signal).

Single mode fiber	**Multimode fiber**
Properties: • In single mode fiber only one mode can be propagate. • The single mode fiber has a smaller core diameter and difference in refractive index of core and cladding is small.	• This fiber it allows large number of modes for light to pass through it. • Here, both the core and cladding in refractive indices difference is large as the core diameter is large
Advantages: • No dispersion (i.e. there is no degradation of signal during propagation). Since the information transmission capacity is inversely proportional to dispersion.	Disadvantages: • Dispersion is more due to degradation of signal due to multimode. • Information can be carried to shorter distances only.

• The fiber can carry information to longer distances.	
Disadvantages: • Launching of light and connecting two fibers are difficult. • Installation (fabrication) is difficult as it is more costly.	Advantages: • Launching of light and also connecting two fibers is easy. • Fabrication is easy and the installation cost is low.

Single mode step index fiber: A single mode step index fiber consists of a very thin core of uniform refractive index (R.I) surrounded by Cladding of R.I lower than that of Core. Since the core size is small the numerical aperture is also small. They accept light from laser source. They are used in long distance communications (telephony, TV broadcast systems). Low/no dispersion possibility. Product cost is high.

Multimode step index fiber:

This is similar to single mode step index fiber with the exception that it has a larger core diameter. The core diameter is very large as compared to wavelength of light transmitted. The numerical aperture is large because of large core size. They accept light from both laser as well as from LED. They are used in data links, Local area network (LAN). Dispersion possibility. Product cost is low.

Graded index multimode fiber (GRIN fiber):

In this type of optical fibers, R. I. of the Core varies with distance from the fiber axis. It has high R.I. at center and R.I. falls rapidly as radial distance increases from the axis. In GRIN fibers the acceptance angle and numerical aperture diminishes with radial distance. They accept light from both laser as well as from LED. They are used for medium distance communication for example telephone link between central offices in a small geographic area.

Numerical aperture:

In order to understand the propagation of light through an optical fibre, consider the figure (2). Consider a light ray (i) entering the core at a point A , travelling through the core until it reaches the core cladding boundary at point B. As long as the light ray intersects the core-cladding boundary at a small angles, the ray will be reflected back in to the core to travel on to point C where the process of reflection is repeated .ie., total internal reflection takes place. Total internal reflection occurs only when the angle of incidence is greater than the critical angle. If a ray enters an optic fiber at a steep angle(ii), when this ray intersects the core-

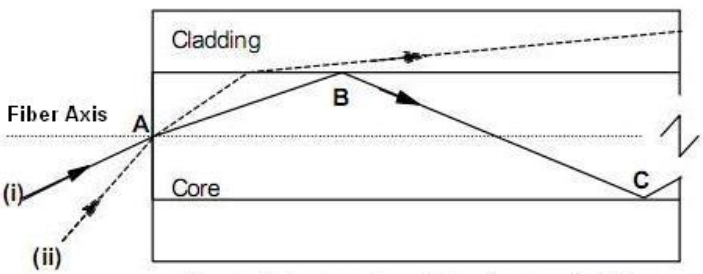


Figure 2 Propagation of light in an optical fibre

cladding boundary, the angle of intersection is too large. So, reflection back in to the core does not take place and the light ray is lost in the cladding. This means that to be guided through an optic fibre, a light ray must enter the core with an angle less than a particular angle called the acceptance angle of the fibre. A ray which enters the fiber with an angle greater than the acceptance angle will be lost in the cladding.

Derivation of numerical aperture:

Consider an optical fibre having a core of refractive index n_1 and cladding of refractive index n_2. let the incident light makes an angle i with the core axis as shown in figure (3). Then the light gets refracted at an angle ϑ and fall on the core-cladding interface at an angle where,

$$\theta' = (90 - \theta) \quad \text{----------} (1)$$

By Snell's law at the point of entrance of light in to the optical fiber we get,

$$n_0 \sin i = n_1 \sin \theta \quad \text{----------} (2)$$

Where n_0 is refractive index of medium outside the fiber. For air n_0 =1.

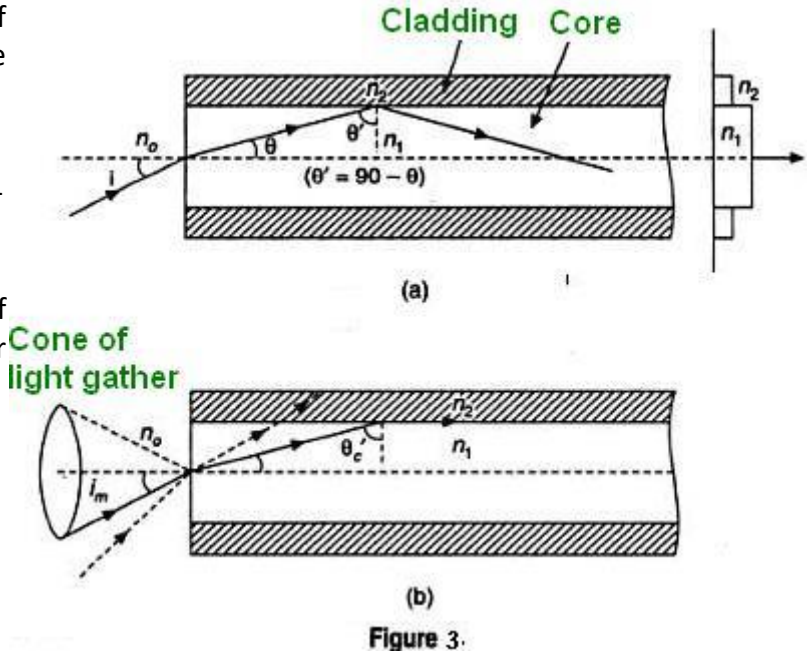


Figure 3.

When light travels from core to cladding it moves from denser to rarer medium and so it may be totally reflected back to the core medium if ϑ' exceeds the critical angle ϑ'_c. The critical angle is that angle of incidence in denser medium (n1) for which angle of refraction become 90°. Using Snell's laws at core cladding interface,

$$n_1 \sin \theta'_c = n_2 \sin 90$$

or

$$Sin\, \theta'_c = \frac{n_2}{n_1} \quad \text{----------} (3)$$

Therefore, for light to be propagated within the core of optical fiber as guided wave, the angle of incidence at core-cladding interface should be greater than ϑ'_c. As i increases, ϑ increases

and so ϑ' decreases. Therefore, there is maximum value of angle of incidence beyond which, it does not propagate rather it is refracted in to cladding medium (fig: 3(b)). This maximum value of i say i_m is called maximum angle of acceptance and $n_0 \sin i_m$ is termed as the numerical aperture (NA).

From equation(2),

$$NA = n_0 \sin i_m = n_1 \sin\theta$$

$$= n_1 \sin(90 - \theta_c)$$

$$Or\ NA = n_1 Cos\ \theta'_c$$

$$= n_1 \sqrt{1 - Sin^2\theta'_c}$$

From equation (2) $\sin\theta'_c = \frac{n_2}{n_1}$

Therefore, $NA = n1 \sqrt{1 - \frac{n_2^2}{n_1^2}}$

$$NA = \sqrt{n_1^2 - n_2^2}$$

The significance of NA is that light entering in the cone of semi vertical angle i_m only propagate through the fibre. The higher the value of i_m or NA more is the light collected for propagation in the fibre. **Numerical aperture is thus considered as a light gathering capacity of an optical fibre**. It is determined by **acceptance angle** i_m (It is the maximum angle for which incident ray undergo total internal reflection at the core-cladding interface and stays in the core of the optical fiber).

Numerical Aperture is defined as the sine of the acceptance angle of a waveguide or fiber (the Sine of half of the angle of fibre's light acceptance cone. i.e. NA=Sin θ_a where θ_a, is called acceptance cone angle)

Source of Attenuation in optical fibers:

(i) Energy absorbed in lattice vibration.

(ii) Energy absorbed by impurities.

(iii) Scattering of light due to local variation in refractive index.

Application of optical fiber:

(i) Communication: High speed, high intensity and low loss data transmission.
(ii) Medical field: Diagnostic instruments- retinal treatment.
(iii) Industry: Laser is carried through optical fibers for cutting, drilling, and welding.
(iv) Fiber Optic Communication Link (FOCL): used to send signal over long distance. Comprises of transmitter, fiber optic cable and receiver.

Fiber optics communication:

Fiber-optic communication is a method of transmitting information from one place to another by sending pulses of infrared light[1] through an optical fiber. The light forms an electromagnetic carrier wave that is modulated to carry information.[2] Fiber is preferred over electrical cabling when high bandwidth, long distance, or immunity to electromagnetic interference are required. This type of communication can transmit voice, video, and telemetry through local area networks, computer networks, or across long distances.

Figure shows the schematic diagram of a fiber optic communication system. The major components of an optical fiber communication system are

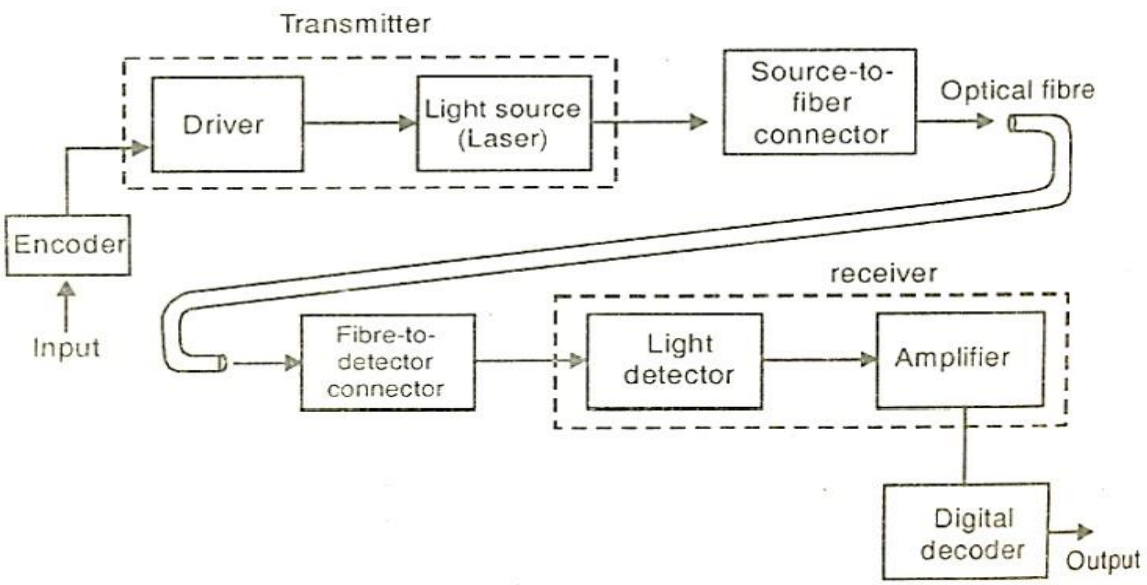
i. The optical transmitter

ii. The optical fiber

iii. The optical receiver

Principle:

Basically, a fiber optic system converts an electrical signal to an infrared light signal. This signal is transmitted through an optical fiber. At the end of the optical fiber, it is reconverted into an electric signal



Working:

1. Encoder encodes the information in the binary sequence zeros and ones.

a. Encoder is an electric circuit where in the information is encoded into binary sequences of zeros and one. In the light wave transmitter each 'one' corresponds to an electrical pulse and 'zero' corresponds to an absence of a pulse. These electrical pulses are used to turn a light source on and off very rapidly. The driver converts the incoming electrical signal into a form that will operate with the light source.

2. These electrical pulses are used to turn a light source on and off rapidly.

3. The optical fiber acts as a wave guide and transmits the optical pulses towards the receiver, by the principle of total internal reflection.

4. The light detector receives the optical pulses and converts them into electrical pulses. These signals are amplified by the amplifier.
5. The amplified signals are decoded by the decoder.

Advantages of FOCL (Fiber optics communication link):

(i) High information density (Fibre optic cabling provides a much higher bandwidth)

(ii) Smaller in size, light weight and cheaper

(iii) Much lower levels of signal attenuation or loss

(iv) High speed data transmission

(v) No leakage: secured communication(Fibre optics do not suffer from stray interference pickup that occurs with coaxial cabling)

(vi) There is no short circuiting as in metal wires

(vii) There is no need to ground and hence no voltage problem occurs.

(viii) It can withstand to any range of temperature and moisture condition

Question Discussion:

1. What is an optical fiber? Sketch the different parts of an optical fiber.
2. State the principle of an optical fiber.
3. Why should clad glass have less refractive index than the core glass?
4. Mention the role of cladding in an optical fiber.
5. What is a step index optical fiber? Draw the refractive index profile of it.
6. What is a graded index optical fiber? Draw the refractive index profile of it.

7. Write down differences between step index optical fiber and graded index optical fiber.
8. Write down the differences between single mode optical fiber and multimode optical fiber.
9. Which of the optical fibers, among single mode and multimode optical fibers, is used for long distance communication and why?
10. Draw the path of rays in the graded index optical fiber.
11. Define numerical aperture of an optical fiber.
12. What are the disadvantages of using fiber optic communication system?
13. Mention the important applications of optical fiber.
14. Schematically show the basic elements of FOCL.
15. What is the principle of optical fibre? Draw a neat ray diagram showing the principle of optical fiber. Obtain an expression for critical angle of incidence at core-cladding interface in terms of their refractive indices.
16. What are the advantages of using fibre optic communication system?
17. What is numerical aperture of an optical fiber? Derive an expression for numerical aperture of a step index optical fiber.

Numerical

1. If the acceptance angle of an optical fiber is 68.16^0, find the numerical aperture. If cladding has refractive index of 1.52, what is the refractive index of the core?
2. What is the critical angle of a ray of light in a step index optical fiber for which refractive index of the core is 1.53 and that of cladding is 2.5 % less than that of the core.
3. The refractive index of the core and cladding of a step index optical fibre are 1.52 and 1.41 respectively. Calculate its critical angle, Numerical aperture and the acceptance angle.
4. If the acceptance angle for a given fiber is 68.16^0, calculate the maximum entrance angle and numerical aperture. If the cladding glass has a refractive index of 1.52, calculate the refractive index of the core glass.
5. If the acceptance angle for a given fiber is 63^0, calculate the maximum entrance angle and numerical aperture.

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